



Digital Image Processing and Computer Vision

Block-1_Unit-1_Part-1

Introduction To Digital Image

Topics to be Covered

Introduction

Introduction to an image

Image Acquisition

Digitization of Images

Important Topics

Summary

Introduction

After going through this session, you should be able to

- Define and represent an image
- Define the various terms used in digital image processing
- Apply sampling and quantization for image digitization



Introduction to an Image

- An image is nothing more than a two dimensional signal. It is defined by the mathematical function $f(x,y)$ where x and y are the two co-ordinates horizontally and vertically.
- The value of $f(x,y)$ at any point is gives the pixel value at that point of an image.
- The shown figure is an example of digital image that you are now viewing on your screen. But actually, this image is nothing but a two dimensional array of numbers ranging between 0 and 255.

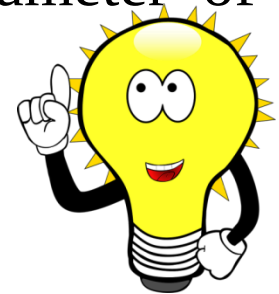


Introduction to an Image

- An image is a two-dimensional function $f(x, y)$ where for each position (x, y) in the projection plane, the values of the function $f(x, y)$ represent the amplitude or light intensity of the image.
- Images are of two types, analog and digital.
- Analog image can be mathematically represented as a continuous range of values representing position and intensity.
- A digital image is composed of picture elements called pixels.

Introduction to an Image

- An image is a projection of the environment under consideration, if the environment is 3D then its projection will be in 2D. If we are having an n-Dimensional space then its projection will be of n-1 dimensions.
- Magnetic resonance images and computerized tomography (CT) images, which are 3D images, are mathematically represented by 3D function $f(x, y, z)$, where x, y, z are spatial coordinates which may represent the amplitude or intensity or any other parameter of the image.



Introduction to an Image

- A digital image is defined as a 2D discrete signal that varies over the spatial coordinates x and y , and can be written mathematically as $f(x,y)$. It is also an $n \times n$ array of elements, and each element represents the sampled intensity.
- In general, the image $f(x, y)$ is divided into X rows and Y columns. The coordinate ranges are $\{X = 0, 1, \dots, m - 1\}$ and $\{Y = 0, 1, 2, \dots, n - 1\}$.



Introduction to an Image

- The image can be written as a mathematical function $f(x, y)$ as given below:

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & f(0,2) & \dots & f(0,y-1) \\ f(1,0) & f(1,1) & f(1,2) & \dots & f(1,y-1) \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ f(m-1,0) & f(m-1,1) & f(m-1,2) & \dots & f(m-1,n-1) \end{bmatrix}$$

Introduction to an Image

- A sample digital image is shown in Fig. 1(a) and its equivalent matrix is shown in Fig.1(b).

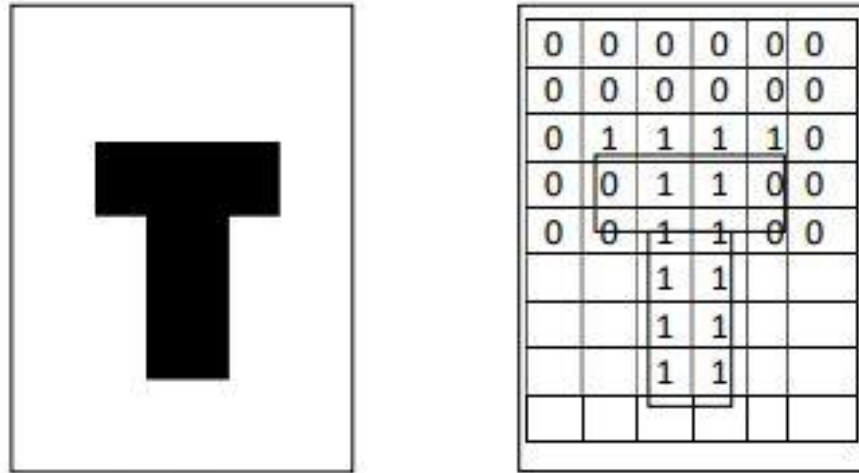


Fig. 1: (a)Sample Digital Image and (b) Matrix of the Image

Introduction to an Image

- In Fig. 1(b), the elements are shown as either 0 or 1, which represents the value of the function $f(x, y)$, each element is known as **pixels** i.e. 'picture element'.
- They represent the discrete data of any digital image, and serve as the actual building block of digital images.
- The value of the function $f(x, y)$ at every point indexed by a row and a column is a number and has no units, and it is known as the **grey value** of intensity of the image at that point.

Introduction to an Image

- You may again refer to Fig. 1(b) , the number of rows in a digital image is called **vertical resolution**. The number of columns is called **horizontal resolution**.
- The number of rows and columns describes the **dimensions** of the image. The image size is often expressed in terms of the rectangular pixel dimensions of the array. Images can be of various sizes.
- Some examples of image size are 256 x 256, 512 x 512, etc.

Image Acquisition

- According to the fundamental concept of optics, when two or more rays of light meet at a point due to reflection or refraction or any other phenomenon, an image is formed. The information of happened phenomenon is recorded or acquired by the respective sensors as an image of that event. The process of capturing real world images and storing them into a computer is called **image acquisition**.
- Generally, the data is gathered in the analog form, which is digitized through a digitizer, and this digital signal is finally utilized by the digital computer for its processing.

Image Acquisition

This leads to three types of image processing :—

- Optical Image Processing
- Analog Image Processing
- Digital Image Processing

We define these types briefly:

- **Optical Image Processing:** An optical image of 3D object is in fact its 2D projection, which is the continuous distribution of light on a 2D surface. This 2D projection holds various types of information like 3D objects that are in focus.

Image Acquisition

- It also includes the study of the radiation source, and other optical processes. This optical image is available in the optical form until it is converted into analog form, leading to the area of analog image processing.
- **Analog Image Processing:** This relates to the processing of analog signals using analog circuits. Analog signals are the time varying continuous signals, often referred to as pictures. These analog signals are transformed into digital image through the process of sampling and quantization, performed by using a digitizer and the process is known as digitization.

Image Acquisition

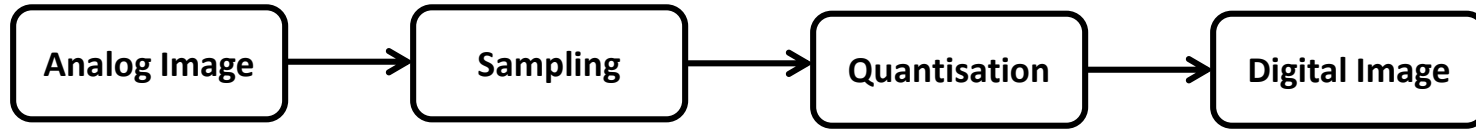


Fig: Analog Image Processing

- **Digital Image Processing:** It relates to the field of using digital circuits, systems, and software algorithms to carry out the image processing operations, which include quality enhancement of an image, counting of objects, image analysis, and many more.

Image Acquisition

- Thus, a typical digital imaging system or digital imaging workstation involves components for the image acquisition, storage, processing, and transmission. Types of imaging systems and acquisition systems are shown in Fig.

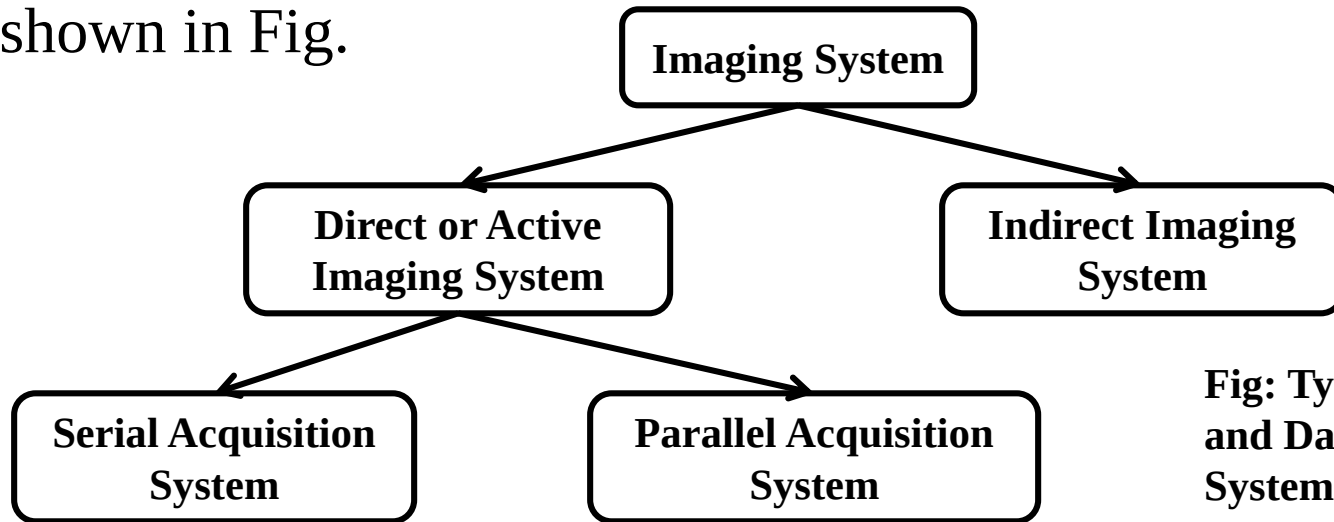


Fig: Types of Imaging and Data Acquisition Systems

Image Acquisition

- **Direct or Active Imaging Systems:** These are the systems where no temporary storage on film or chemical processes is required. In such systems the acquired images are in a recognizable form. For example, human eye, digital camera, scanning confocal microscopes, etc.
- The direct or active imaging systems are divided in to two sub categories i.e. serial and parallel acquisition systems. The human eye and the digital camera are example of parallel active imaging systems, where as scanning microdensitometer and scanning confocal microscopes are examples of serial acquisition imaging systems. The advantage of direct imaging systems is that the final image is a digital image.

Image Acquisition

- **Indirect Imaging Systems:** In comparison to Direct imaging systems, the Indirect imaging systems involves data processing or reconstruction before producing an image for observation. In indirect imaging systems, the image is stored in a film, which is rendered observable by a chemical process. Therefore, there is a delay in the production of images from films.



Digitization of Images

- Image digitization refers to the process of sampling and quantization of the analog signals, which is required to convert the analog signal into digital form. The digital image processing is performed using Digitizers, and this digitization process consists of two steps. One is called sampling and the other is called quantization.
- **Sampling** refers to considering the image only at a finite number of points, each image sample is called a pixel and **Quantization** refers to the representation of the grey level value at the sampling point using finite number of bits. For example 256 level quantization results in image storage of 8 bits per pixel.

Digitization of Images

Sampling

- Sampling refers to the process of converting a continuous image into a discrete representation by selecting a finite number of discrete points or pixels from the continuous image.
- During the process of digitization, an image is captured using a digital device such as a scanner, camera, or sensor. The continuous analog signal from the device is converted into a digital signal consisting of discrete values representing the intensity of light at each pixel in the image.

Digitization of Images

- The sampling process involves selecting a fixed grid of pixels from the continuous image, with each pixel representing a discrete sample of the original signal. The number of pixels in the grid and the distance between them determines the resolution of the resulting digital image.
- Sampling is an important step in the digitization process, as it directly affects the quality and accuracy of the resulting digital image. If the sampling rate is too low, the resulting digital image may appear pixelated or blurred, while a high sampling rate may result in a very large file size.

Digitization of Images

- To sample the signal, the signal should be frozen, as sampling cannot be applied to moving signals. If the signal is frozen for T seconds, T is called the sampling period.
- The **sampling period** is measured in seconds, milliseconds, or microseconds. For example, if the sampling rate is 1000Hz, it means that the signal is sampled every millisecond.
- The **sampling rate** or **sampling frequency** is the reciprocal of the sampling period and is measured in samples per second or Hertz. The sampling process is the multiplication of a continuous signal $f(t)$ by a railing function $r(t)$. $f(n) = f(t) \times r(t) = f(nT)$

Digitization of Images

Example 1: What should be the physical size of a 2D image of a document with dimensions is 2400 x 2400, when scanned at 300dpi. Here dpi stands for dots per inch.

Solution: The physical size of image is

$$\begin{aligned} &= \frac{\text{Number of pixels in width}}{\text{Resolution}} \times \frac{\text{Number of pixels in height}}{\text{Resolution}} \\ &= \frac{2400}{300} \times \frac{2400}{300} \\ &= 8 \text{ inches} \times 8 \text{ inches} \end{aligned}$$

Digitization of Images

Example 2: If the physical size of a medical image is 8×8 inches and the sampling resolution is 5 cycles/mm, then how many pixels per cycle are required to have a better quality image? Will an image of size 256×256 be enough?

Solution: It is given that the sampling resolution is 5 cycles/mm.

Therefore, for better quality, 2 pixels per cycle are required. This is because sampling theorem states that the sampling frequency should be greater than twice the maximum signal frequency. This means that 10 pixels per mm are required. So, the pixel size is 0.1 mm.

You may note here that this double rates in both directions are called Nyquist rates.

The given image size (since 1inch = 2.54cm) is

$$= 8 \times 2.54 \times 8 \times 2.54 \text{ cm}^2$$

$$= 20.32 \times 20.32 \text{ cm}^2$$

Therefore, the minimum number of required pixels = 2032×2032 pixels.

So, an image of size 256×256 is not enough.

Digitization of Images

Quantization

- Image quantization is the process of converting the sampled analog value of the function $f(x, y)$ into a discrete-valued integer. An analog signal has an infinitely larger number of distinct values. So, it is necessary to convert the continuous values into a smaller set through the process of quantization. The sampled analog image i.e. a natural image has continuously varying shades and colours and is known as a continuous tone image. This continuous tone image is required to be converted into a discrete image. The image quantizer maps a continuous value x into a discrete variable x' , where discrete points of grey-tone or brightness are used.

Digitization of Images

- The process of quantization involves the partitioning of input values into equally spaced intervals. The end points of the interval are called decision boundaries.
- Let the decision boundaries be given by $B = \{b_0, b_1, \dots, b_m\}$, and let the input values be in the range $-X_{\max}$ to $+X_{\max}$. The length of the interval between successive decision boundaries i.e. the step size (Δ) is given by $2X_{\max}/m$. The midpoint between successive decision boundaries is called output or reconstruction level and is given by $(R) = \{y_1, y_2, \dots, y_m\}$. Then quantization is in the range $[-\Delta/2, \Delta/2]$, and the number of bits necessary to encode the output levels is given by $R = \lceil \log_2 m \rceil$.

Important Topics

Introduction to an image

Image Acquisition

Digitization of Images

Summary

In this session, we discussed:

- About an image and Digital images.
- Image acquisition can be done using various ways and the concepts required to understand the acquisition of any image.
- Image digitization and models for representing an image.
- The two steps of image digitization, sampling and quantization.

*Thank
You*



Digital Image Processing and Computer Vision

Block-1_Unit-2

Image Transformations

Topics to be Covered

Introduction

Signals

Image as a 2-D signal

Transformations of 2-D Signals

Orthogonal Transformations of 2-D Signals

Unitary Transformations of 2-D Signals

Fundamental Properties of Unitary Transformations

Important Topics

Summary

Introduction

After studying this session, you should be able to:

- Define the 1-dimensional (1-D) and 2-dimensional (2-D) signals.
- Relate how an image is also a 2-dimensional signal
- Define and apply the unitary and orthogonal transforms of signals.
- List the properties of unitary transforms, especially useful for image processing.

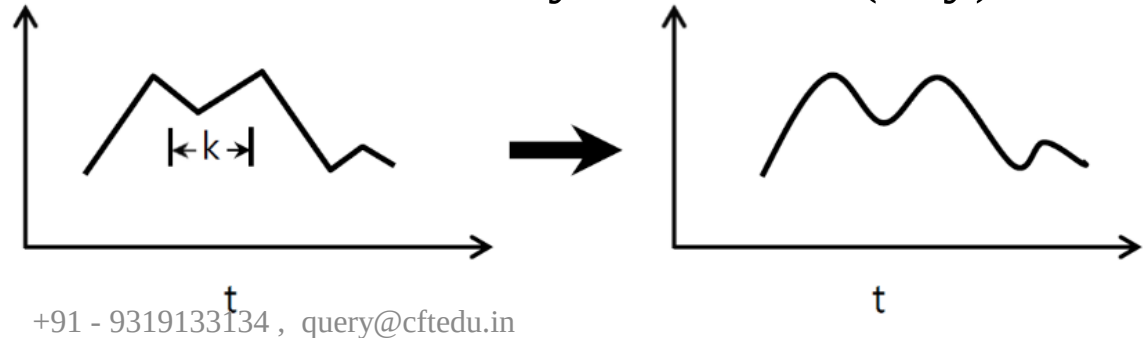


Signals

- A signal is a function of one or more independent variables, where the variables represent some physical meaning. These signals carry information through variables. The dimension of a signal is defined by the number of independent variables that the function is defined on.
- For example, an $E \subset G$ signal is one-dimensional, whereas the intensity of a still image is 2-dimensional.
- Therefore, a way to differentiate the signals is by the dimension of their domain, which is also the number of independent variables that the function/signal is defined on.

Signals

- **Definition(1-D Signal):** A 1-dimesional (1-D) signal is a function of a single variable. A one-dimensional continuous signal is represented as a function, $f(t)$, of the independent variable t , representing the evolution of a physical phenomenon across time 't'.
- **Definition (2-D Signals):** A 2-dimensional (2 – D) signal $f(x, y)$, is a function of two independent variables, x and y , such that (x, y) indicate a point in the 2D space.



Signals

Example 1: Draw the signal of $f(x, y) = \delta(x - 2y)$. Identify the signal impulse type.

Solution: Using the δ -function, we get $\delta(x - 2y) = \begin{cases} 1, & x - 2y = 0 \\ 0, & \text{otherwise} \end{cases}$

We plot these points

X	0	1
Y	0	0.5

on the graph as shown in Fig

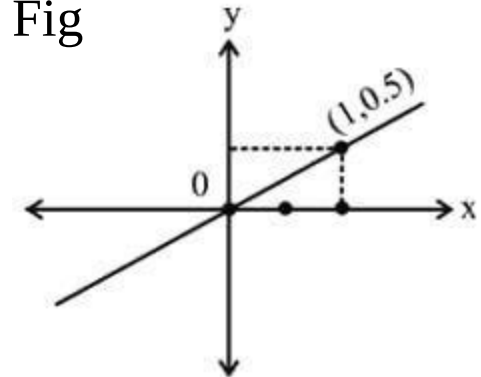


Fig: Representation of $f(x, y) = \delta(x - 2y)$.

The type of signal impulse is diagonal line.

Image as a 2-D signal

- A grayscale image is a discrete 2-D signal $f(x, y)$, having two independent variables, x and y , such that $f(x, y)$ is the value of the signal at a pixel whose location in the image is given by integers x and y . The 2-D signals can be of various types such as separable, periodic, etc.
- Therefore, the 2D signal represents a digital image, where (x, y) is the location of the pixel and $f(x, y)$ is the value at this pixel.

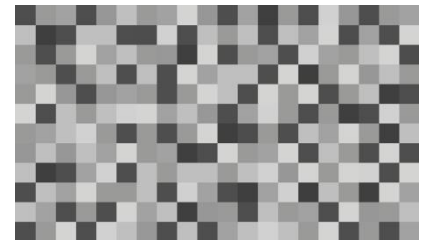


Image as a 2-D signal

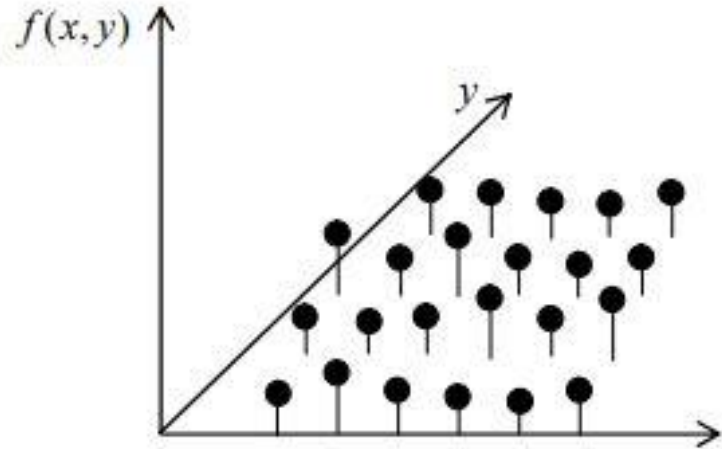
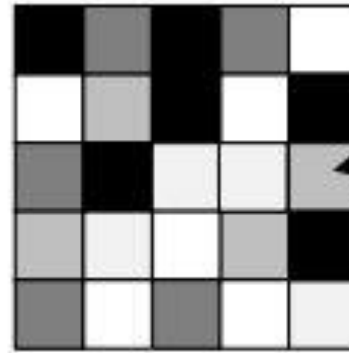


Fig: A Discrete 2-D Signal



**Fig: A Gray-Scale Image,
An Example of a 2-D Signal.**

Image as a 2-D Signal

- In general, brightness or intensity is usually the value of the function in a image. However, in an image, the signal value can represent physical values such as temperature, pressure, depth, etc. also. We are showing some of these examples in Fig



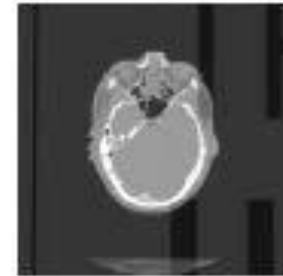
(a) ultrasound



(b) temperature



(c) RGB image



(d) CT scan image

Fig: Examples of different types of images, depending on the physical phenomenon represented by the 2D signal.

Transformations of 2-D Signals

- Transformations are mathematical functions that allow us to convert from one domain to another. Therefore, transformation of a signal $f(x)$ will convert the signal into a new signal, say, $g(y)$, in a different domain. However, transformation does not change the information content present in the signal/image.
- The steps for image transformation from the spatial to the frequency domain are:
- **Step 1:** Transform the image from the spatial domain to the frequency domain.

Transformations of 2-D Signals

- **Step 2:** Carry the required image processing task in the transformed domain.
- **Step 3:** Apply inverse transform to return to the spatial domain.

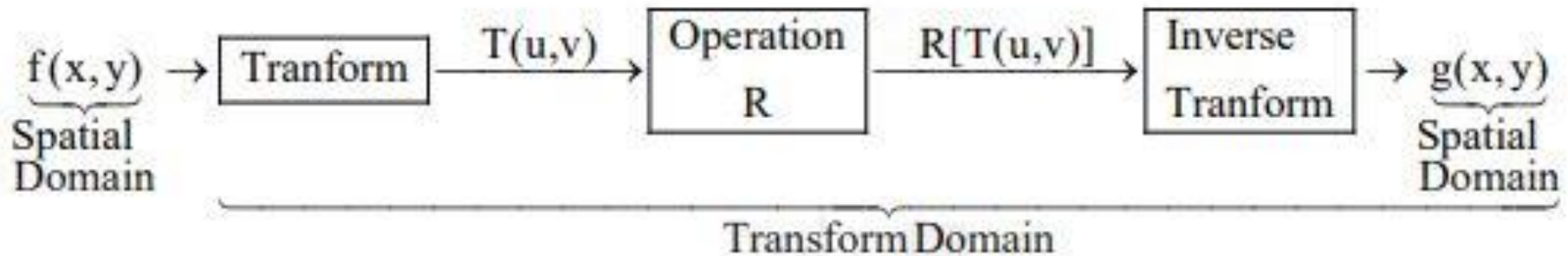


Fig: Steps of 2-D Image Transformation

Orthogonal Transformations of 2-D Signals

- A square matrix $A = [A_1 A_2 \dots A_n]$ where, A_i is the i_{th} column vector of A is real and has the property that $A^{-1} = A^T$, then it is said to be an orthogonal matrix. If, A is not real, it is said to be a unitary matrix if $A^{-1} = A^H$ that is the inverse of A is equal to its conjugate transpose.
- The columns of a unitary and an orthogonal matrix are perpendicular to each other and therefore, are said to be **orthogonal**.
- Also, the columns of these matrices are of unit length, that is, **normalized**.
- Therefore, the columns of these matrices are said to be **orthonormal**.

Orthogonal Transformations of 2-D Signals

- Let us consider an $N \times N$ image $f(x, y)$. The forward and inverse transforms of $f(x, y)$ are given by

$$g(u, v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} T(u, v, x, y) f(x, y), \quad 0 \leq u, v \in N-1.$$

$$f(x, y) = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} I(x, y, u, v) g(u, v), \quad 0 \leq x, y \in N-1$$

- where, $g(u, v)$ is the transformed image, $T(u, v, x, y)$ is called the **forward transformation kernel** and $I(x, y, u, v)$ is called the **inverse transformation kernel**. It is said to be an **orthogonal transform**, when T is an **orthogonal matrix**.

Orthogonal Transformations of 2-D Signals

The forward transformation kernel satisfies the following properties:

i) Orthonormality Property:

$$\sum_{x=0}^{N-1} \sum_{y=0}^{N-1} T(u, v, x, y) I(u', v', x, y) = \delta(u - u', v - v')$$

ii) Completeness Property:

$$\sum_{u=0}^{N-1} \sum_{v=0}^{N-1} T(u, v, x, y) I(u, v, x', y') = \delta(x - x', y - y')$$

Orthogonal Transformations of 2-D Signals

Example 2: Consider the following orthogonal matrix A and image matrix f

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, f = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$$

Apply the orthogonal transform and its inverse.

Solution: The transformed image, obtained according to the Eqn.

$$g(u,v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} T(u,v,x,y) f(x,y), 0 \leq u,v \in N-1 \text{ is}$$

$$g = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 6 & 10 \\ -4 & -4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 8 & -2 \\ -4 & 0 \end{bmatrix}$$

We compute the outer product of the columns of A^{*T} .

Orthogonal Transformations of 2-D Signals

$$A^*(0, 0) = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Similarly, we compute $A^*(0,1)$

$$A^*(0, 1) = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} = A^{*T}(1,0) \text{ and } A^*(1,1) = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Now, we obtain the inverse transformation. We get

$$\begin{aligned} A^{*T}gA^* &= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 8 & -2 \\ -4 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 4 & -2 \\ 12 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \end{aligned}$$

We can verify that the inverse transformation gives us the original image.

Unitary Transformations of 2-D Signals

- Let first learn about unitary transforms for one dimensional signals for easy understanding and will then go to discussing them for 2D signals.
- **Unitary Transforms for 1-D signals**
- A one dimensional sequence $\{f(x), 0 \leq x \leq N-1\}$ can always be represented as a vector of dimension N , as $f = [f(0)f(1)...f(N-1)]$.
- Then, a transformation T of $f(x)$ can be written as:

$$g = T \cdot f \Rightarrow g(u) = \sum_{x=0}^{N-1} T(u,x)f(x), \quad 0 \leq u \leq N-1$$

Unitary Transformations of 2-D Signals

- where, $T(u,x)$ is called the **forward transformation kernel** and $g(u)$ is the transform of $f(x)$, that is $g(u)$ is the result of applying the transformation $T(u,x)$ on $f(x)$.
- There exists an inverse relation that transforms $g(u)$ back to $f(x)$, which is given by $I(x,u)$ such that

$$f(x) = \sum_{u=0}^{N-1} I(x,u)g(u), \quad 0 \leq x \leq N-1$$

- where, $I(x,u)$ is called the **inverse transformation kernel**.

Unitary Transformations of 2-D Signals

Therefore, in the matrix form, we can write

$$\mathbf{f} = \mathbf{I} \cdot \mathbf{g} = \mathbf{T}^{-1} \cdot \mathbf{g}$$

Recall, that a matrix is said to be a unitary matrix, if the inverse of the matrix is also its conjugate transpose. That is, if

$$\mathbf{I} = \mathbf{T}^{-1} = \mathbf{T}^{*T}$$

then the matrix $\mathbf{T}$ is called a **unitary matrix** and therefore, the transformation is also known as the **unitary transformation**.

Unitary Transformations of 2-D Signals

- Like 1D unitary transform, which allows us to represent a 1D signal by a set of orthonormal basis vectors, there exists unitary transforms in the higher dimensions. The 2D unitary transform allows us to represent a 2D signal/ image as set of basis arrays or basis images.
- Given an $N \times N$ image $f(x, y)$ the forward and inverse transforms are given by

$$g(u, v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} T(u, v, x, y) f(x, y)$$

$$f(x, y) = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} I(x, y, u, v) g(u, v)$$

Unitary Transformations of 2-D Signals

- where, $T(u,v,x, y)$ is called the **forward transformation kernel** and $I(x, y, u, v)$ is called the **inverse transformation kernel**. It is said to be a unitary transform when T is a unitary matrix.
- In general, the transformations (both orthonormal and unitary) can have the following properties:
 - i) Separability:** The transformation kernel is said to be **separable** if $T(u,v,x,y) = T_1(u,x)T_2(v, y)$
 - ii) Symmetry:** The transformation kernel is said to be **symmetric** if T_1 is functionally equal to T_2 such that $T(u,v,x,y) = T_1(u, x)T_1(v, y)$

Unitary Transformations of 2-D Signals

Example 3: Check whether the matrix $A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ is unitary or not.

Solution: The members of A are real, therefore the condition for A to be unitary is $A^{-1} = A^T$.

$$A^{-1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \text{ and } A^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

from these equations, it is clear that

$$A^{-1} = A^T$$

Hence, A is unitary matrix.

Fundamental Properties of Unitary Transformations

- Here we discuss the properties of unitary transforms.
- i) **Energy conservation and rotation:** Let $\mathbf{T}$ be a unitary transformation, then
$$\mathbf{g} = \mathbf{T} \cdot \mathbf{f}$$
implies that
$$\|\mathbf{g}\|^2 = \|\mathbf{f}\|^2$$
- This means that the signal energy is preserved after the application of a unitary transformation. This property is called **energy preservation property**. It also implies that every unitary transformation is a rotation of the vector $\mathbf{f}$ in the N – dimensional vector space.

Fundamental Properties of Unitary Transformations

ii) Energy compaction

- A large fraction of the energy in the image is packed into relatively few transform coefficients in most unitary transforms. By the above mentioned property, since the energy is preserved, this implies that most of the transform coefficients have insignificant values and only a few of the transform coefficients that are close to the origin have significant values. This property is very useful for image compression purposes.

Important Topics

Signals

Image as a 2-D signal

Transformations of 2-D Signals

Orthonormal and the Unitary
transformation

Summary

In this session, we discussed:

- 1-D and 2-D signals and that an image is also a 2-D signal.
- Orthonormal and the unitary transformation and the properties of the unitary transformation that makes it very useful for image processing, especially image compression.



*Thank
You*



Digital Image Processing and Computer Vision

Block-1_Unit-3

Image Enhancement In Spatial Domain

Topics to be Covered

Introduction

Image Enhancement

Point Operations

Contrast Stretching

Clipping and Thresholding

Digital Negative

Intensity Levels Slicing

Topics to be Covered

Bit Extraction

Important Topics

Summary

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Introduction

After studying this session, you should be able to:

- Define image enhancement
- Perform image enhancement in spatial domain
- Perform point operations on images
- Perform image enhancement using the following algorithms
 - Point operations
 - Contrast stretching
 - Clipping and thresholding

Introduction

- Digital Negative
- Intensity levels slicing
- Bit plane extraction



Image Enhancement

- Image enhancement techniques improve the quality of image as perceived by human observer/ machine vision system.
- Enhancement techniques improve the perception of information in an image for human viewing and provide ‘better’ input for other automated image processing techniques.
- The main objective is to modify attributes of an image to make it more suitable for a given task or a specific observer.
- Image enhancement techniques are application specific and produce a ‘better’ image.

Image Enhancement

- They are broadly classified into two categories namely
 - i) Spatial domain methods
 - ii) Frequency domain methods
- In spatial (time) domain methods, pixel values are manipulated directly to get an enhanced image, whereas in frequency domain methods, firstly fourier transform of image is taken to convert image into frequency domain. Then the fourier transform is manipulated and the modified spectrum is transformed back to spatial domain to view the enhanced image. Some enhancement techniques operate on combination of these methods to get best results.

Point Operations

- The pixel values (grey values) of an image are directly manipulated. Such operations simply take the grey value of each pixel/neighbouring pixels, map it to a new value and move on to the next pixel.
- Let $f(x, y)$ be an input image.
- An image processing operation in the spatial domain may be expressed as a mathematical function $T[f(x, y)]$ applied to the image $f(x, y)$ to produce a new image $g(v, v)$.
- Therefore, $g(x, y) = T[f(x, y)]$.

Point Operations

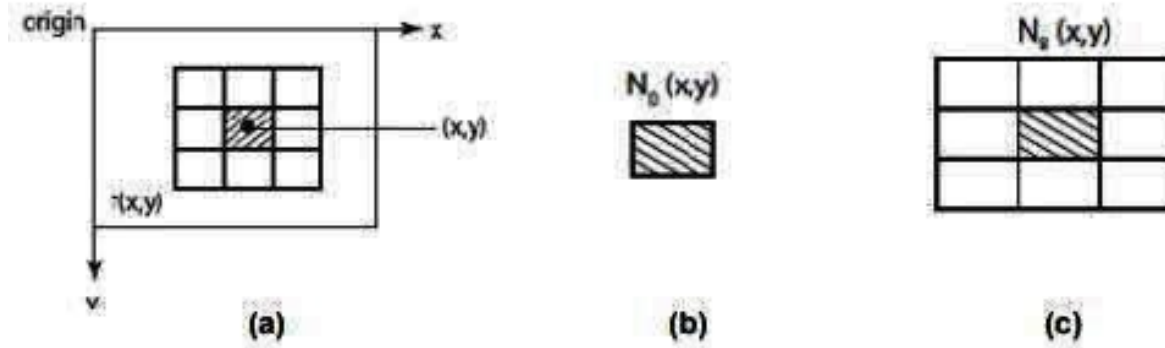


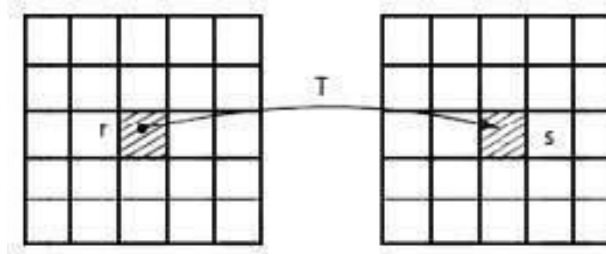
Fig. 1: Point and Neighbourhood Processing

We define the operator T applied on $f(x, y)$ over

- i) a single pixel (x, y) , which is called '**point processing**', as shown in Fig. 1 (b).
- ii) some neighbourhood of (x, y) , which is called '**Neighbourhood processing**', as shown in Fig. 1 (c).
- iii) T may operate on a set of input images instead of a single image.

Point Operations

- Point processing is the simplest case of spatial domain techniques where output at (x, y) only depends on the input intensity at the same point as shown in Fig. 2. It is a memory less operation. In this case, pixels of same intensity get the same transformation.



- $f(x, y)$ $g(u, v)$

Fig. 2: Point operations

Point Operations

Example 1: Perform the transformation $g_1(v) = v + 1$ on the image

$$f(x, y) = \begin{bmatrix} -2 & -1 & 0 \\ 0 & 1 & 2 \end{bmatrix}.$$

Solution: $G_1(x, y) = g_1(f(x, y)) = \begin{bmatrix} -2 + 1 & -1 + 1 & 0 + 1 \\ 0 + 1 & 1 + 1 & 2 + 1 \end{bmatrix}$

$$= \begin{bmatrix} -1 & 0 & 1 \\ 1 & 2 & 3 \end{bmatrix}.$$

Point Operations

Example 2: Perform the transformation $g_2(v) = v^2$ on the image

$$f(x, y) = \begin{bmatrix} -2 & -1 & 0 \\ 0 & 1 & 2 \end{bmatrix}.$$

Solution: $G_1(x, y) = g_2(f(x, y)) = \begin{bmatrix} (-2)^2 & (-1)^2 & 0^2 \\ 0^2 & 1^2 & 2^2 \end{bmatrix}$
 $= \begin{bmatrix} 4 & 1 & 0 \\ 0 & 1 & 4 \end{bmatrix}.$

Point Operations

- Point operations are simplest yet extremely useful and powerful image processing tasks. Point operations are defined as

$$g(u, v) = T[f(x, y)], \text{ also, } s = T(r)$$

Where r = grey level of input image $f(x, y)$, and

s = grey level of output image $g(x, y)$

- Thus, grey levels from the input image are modified with a function T which is independent of image coordinates.
- Examples of point operations are modifying image brightness, histogram processing etc.

Point Operations

Point Processing

- Point operations are zero memory operations where a given gray level values of an individual pixel in the input image $r \in [1, L - 1]$ is mapped into a gray levels $s \in [1, L - 1]$ of the pixels in the output image using the transformation $T()$.

$$s = T(r)$$

Contrast Stretching

- Contrast stretching, also known as contrast enhancement or dynamic range expansion, is a technique used to improve the visibility and clarity of an image by increasing the difference between the darkest and lightest parts of the image.
- The process of contrast stretching involves adjusting the brightness and contrast of an image so that the range of pixel values in the image is expanded. This is done by mapping the original range of pixel values to a new, wider range of pixel values.

Contrast Stretching

- Good contrast in an image is important to distinguish the details else they will merge in background.
- Low contrast images occur due to bad or non uniform illumination conditions or due to nonlinearity of image acquisition devices.



Low contrast image



Enhanced image

Fig: Effect of Contrast Stretching

Clipping and Thresholding

- Clipping and thresholding are two special cases of contrast stretching.

a) Clipping

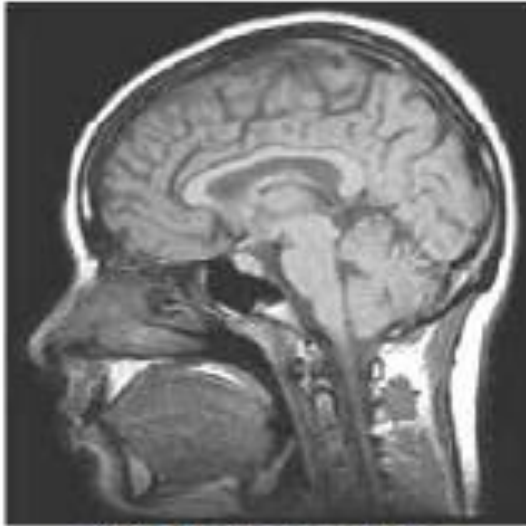
- This operation stretches the contrast to its maximum in a limited range ('window') of original grey level values $\{a, \dots, b\}$. All the grey levels outside this window are either mapped to zero or the maximum values.
- Thus, clipping allows us to focus all the available contrast onto the required range of grey level values.
- This is especially useful when viewing medical images such as CT images. It also helps in viewing under or over exposed images.

Clipping and Thresholding

b) Thresholding

- This operation ‘binarizes’ (only two values present) the image. A suitable threshold value ‘ t ’ is chosen, all the gray levels smaller than ‘ t ’ are mapped to zero whereas all the grey levels greater than or equal to ‘ t ’ are mapped to maximum value.
- This is a very useful operation in image processing applications. Binarization is generally done before segmentation or region extraction.

Clipping and Thresholding



(a) original image



(b) image after thresholding

Fig: Effect of Thresholding

Clipping and Thresholding

Example 3: Perform thresholding on image segment $f(x, y)$ with $t = 128$.

Solution:

0	10	50	100
5	95	150	200
110	150	190	210
175	210	255	100

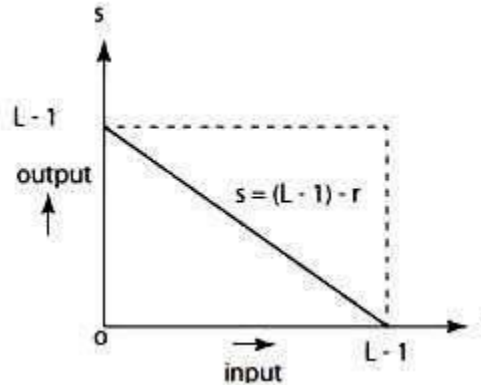
$t = 128$
→

0	0	0	0
0	0	255	255
0	255	255	255
255	255	255	255

Digital Negative

The digital negative of an image with the intensity levels in the range $[0, L - 1]$ is obtained by using negative transformation shown in Fig., which is given by the expression

$$s = (L - 1) - r$$



In this transformation, highest grey level is mapped to lowest and vice versa.

Digital Negative

For an 8-bit image, the transformation is

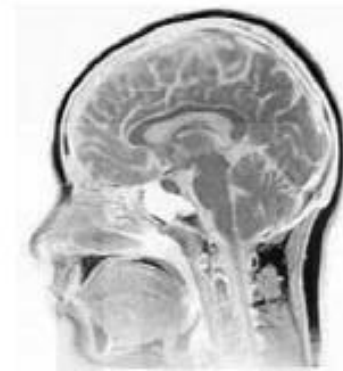
$$s = 255 - r$$

If $r = 20$, (dark pixel) $s = 255 - 20 = 235$ (bright pixel)

If $r = 125$, $s = 255 - 125 = 130$



(a) original image



(b) digital negative of fig (a)

Fig: Effect of digital negative

Digital Negative

Example 4: Find the image negative transformation on an image f given as

$$f(x, y) =$$

0	10	50	100
5	95	150	200
110	150	190	210
175	210	255	100

Solution: The negative of the image is given as

$$g(x, y) =$$

255	245	205	155
250	160	105	55
145	105	165	45
80	45	0	155

Intensity Levels Slicing

- A variant of thresholding is intensity level slicing or double thresholding. It is used to highlight a range of intensity values of interest in an image. In this operation, all the grey levels in a window $\{a, \dots, b\}$ are set to maximum grey level and all the grey levels outside this range are set to zero.

$$s = \begin{cases} M & a \leq r \leq b \\ r & \text{otherwise} \end{cases}$$

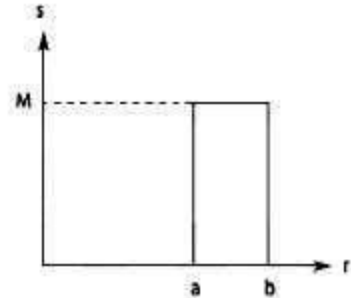


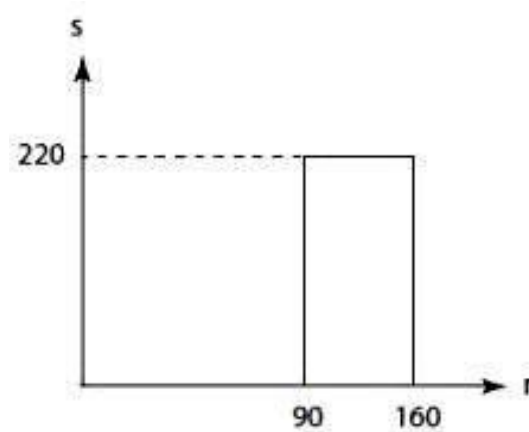
Fig: Transfer Function of Intensity Level Slicing

Intensity Levels Slicing

- Intensity level slicing is a binarization operation as the resulting image has only two grey level values (0 and M). Mostly this technique is used to remove unwanted elements (clutter) from an image so that the useful information becomes prominent.
- Applications are enhancing certain features which are based on grey level values, such as mass of water in satellite images, cancer cells in a monogram images etc.

Intensity Levels Slicing

Example 5: Perform intensity level slicing on $f(x, y)$ as given in Example 4, based on transfer function shown below



Intensity Levels Slicing

Solution: In this case, grey levels less than 90 and greater than 160 are rounded to zero and grey levels between this range are set to 220. This transformation produces a '**binary**' image. In this process, all the background is lost.

0	10	50	100
5	95	150	200
110	150	190	210
175	210	255	100

$f(x,y)$

$$s = T(r)$$



0	0	0	220
0	220	220	0
220	220	0	0
0	0	0	220

$g(x,y)$

Bit Extraction

- Bit extraction, also known as bit plane slicing, is a technique used in digital image processing and computer vision to extract and manipulate specific bits of an image's pixel values.
- In this technique, an image's pixel values are represented in binary format, and each bit of the binary representation is considered as a separate plane or layer of the image.
- The bit planes are ordered from the most significant bit (MSB) to the least significant bit (LSB). The MSB represents the highest power of 2, while the LSB represents the lowest power of 2 in the binary representation of the pixel value.

Bit Extraction

- By extracting specific bit planes, it is possible to perform operations such as noise removal, edge detection, and compression on the image. For example, extracting the MSB bit plane can help in preserving the overall shape of an image, while extracting the LSB bit plane can help in detecting fine details and edges.
- The process of bit extraction involves selecting a specific bit plane and creating a new image from the binary values of that plane. This can be done by masking the original image with a binary value that represents the desired bit plane and then converting the resulting binary image back into a grayscale or color image.

Bit Extraction

Example 6: Find MSB and LSB planes for the given image

255	138	30
65	12	201
183	111	85

Solution: Represent all grey level values in binary format.

		LSB					MSB		
255	→	1	1	1	1	1	1	1	1
138	→	0	1	0	1	0	0	0	1

Bit Extraction

30	→	1	1	1	1	1	1	1	1
65	→	0	1	0	1	0	0	0	1
12	→	0	0	1	1	0	0	0	0
201	→	1	0	0	1	0	0	1	1
183	→	1	1	1	0	1	1	0	1
111	→	1	1	1	1	0	1	1	0
85	→	1	0	1	0	1	0	1	0

Bit Extraction

- MSB plane consists of MSB's of all grey levels as shown in Fig. 14(a) similarly LSB plane consists of LSB's of all grey level values (Fig. 14(b)).

1	1	0
0	0	1
1	0	0

MSB plane

(a)

1	0	1
1	0	1
1	1	1

LSB plane

(b)

Fig. 14

Bit Extraction

Example 7: Compute various bit planes of the following 8-bit image.

0	10	50	100
5	95	150	200
110	150	190	210
175	210	255	100

Solution: To decompose grey level value 100 into 8-bit plane, convert 100 into binary
→ 01100100 .

Then bit for 8th plane is 0, 7th plane is 1, 6th plane is 1 and so on.

Similarly, binary representation of 50 is 00110010. Then bit for 8th plane 0, 7th plane is 0, 6th plane is 1 and so on.

Bit Extraction

Thus, to decompose image grey levels into various bit planes, two steps are followed:

- Convert the grey level value into binary.
- Allocate bits into various planes starting from MSB.

Various bit planes for the given image are:

8th plane (MSB)

0	0	0	0
0	0	1	1
0	1	1	1
1	1	1	0

7th plane

0	0	0	1
0	1	0	1
1	0	0	1
0	1	1	1

6th plane

0	0	1	1
1	0	0	0
1	0	1	0
1	0	1	1

Bit Extraction

5th plane

0	0	0	0
0	0	1	1
0	1	1	1
1	1	1	0

4th plane

0	0	0	1
0	1	0	1
1	0	0	1
0	1	1	1

3rd plane

0	0	1	1
1	0	0	0
1	0	1	0
1	0	1	1

2nd plane

0	1	1	0
1	1	1	0
1	1	1	1
1	1	1	1

1st plane

0	0	0	0
0	1	0	0
0	0	0	0
1	0	1	0

Important Topics

Image Enhancement

Point Operations

Contrast Stretching

Digital Negative

Bit Extraction

Summary

In this session, we discussed:

- Stated Image enhancement in spatial domain.
- Explained point processing and neighbourhood processing
- Explained various point operations such as Contrast stretching, Clipping and thresholding, Digital Negative, Intensity levels slicing, Bit plane extraction.
- Special cases of Contrast stretching: Clipping and thresholding.

*Thank
You*



Digital Image Processing and Computer Vision

Block-1_Unit-4_Part-2

Image Filtering Operations In Spatial Domain

Topics to be Covered

Introduction

Image Sharpening

First Order Filters

Second Order Filters

Histogram Processing

Important Topics

Summary

Introduction

After studying this session, you should be able to:

- Define Image Sharpening
- Define First Order Filters
- Define Second Order Filters
- Define histogram
- Generate histogram from given image

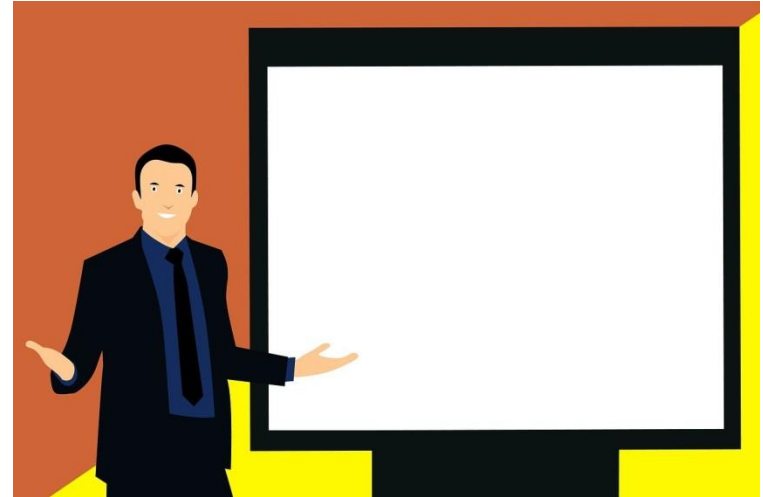


Image Sharpening

- Image sharpening is opposite of image smoothing. This is done to highlight fine details and edges in an image.
- Applications of image sharpening are outlining object boundaries in industrial applications, identifying tumor boundaries, or bone structures in medical imaging etc.
- Smoothing is achieved by pixel averaging which is analogous to integration. Image sharpening is just the reverse process, and is achieved by differentiation. The derivative operation enhances the degree of discontinuity in an image.

Image Sharpening

The advantages of image sharpening are the following:

- It enhances edges and other discontinuities (noise) in an image.
- It de-emphasizes area with slowly varying grey levels (background) in an image.

First order derivative of a one-dimensional signal $f(x)$ is defined as

$$\frac{\partial f}{\partial x} = f(x + 1) - f(x)$$

Similarly 2nd order derivative $f(x, y)$ is defined as

$$\frac{\partial^2 f}{\partial x^2} = f(x + 1) + f(x - 1) - 2f(x)$$

Image Sharpening

Example 5: Find first and second derivative of given data

$$f(x):f(x) = [4 \ 3 \ 2 \ 5 \ 9].$$

Solution: $\frac{\partial f}{\partial x} = [3-4, 2-3, 5-2, 9-5]$

$$= [-1, -1, 3, 4]$$

$$\frac{\partial^2 f}{\partial^2 x} = [4+2-6, 3+5-4, 9+2-10]$$
$$= [0, 4, 1]$$

Image Sharpening

- With respect to the first and second derivatives, following things are observed
 - i) In flat regions (region of constant intensity levels) both derivatives produce zero.
 - ii) In ramp (constant slope), first derivatives gives a non-zero value, whereas second derivative gives non zero values (with different sign) only at the beginning and end of ramp and zero along the ramp.
- First derivative produces a thick edge called '**double edge effect**' along the ramp as the output is non zero. Second derivative produces a double edge one pixel thick separated by zeros.

Image Sharpening

- iii) For a noisy pixel, first derivative produces a positive or negative value leading to one zero crossing. Whereas second derivative produces two zero crossing making it easier to identify.
- iv) In case of an edge (step change in intensity values), first derivative produces a sign change leading to a zero crossing. This makes edge detection easier by second derivative.
- Second derivative enhances the details and edges much better as compared to first derivative. Thus for image sharpening applications, second derivative is most suitable.

Image Sharpening

1. First Derivatives Filters

- **Gradient Operator**
- The gradient operator is often used in first derivative filters to detect edges and boundaries in an image. The first derivative filters use the gradient operator to calculate the rate of change of the image intensity in the x and y directions.
- An edge is the boundary between two regions with distinct grey level properties. Derivative operators are used for most edge detection techniques. The magnitude of first derivative calculated within a neighborhood around the pixel of interest is used to detect presence of edge in an image.

Image Sharpening

First derivatives are directional operators and is defined as

$$\nabla f = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

Magnitude of this vector is given by

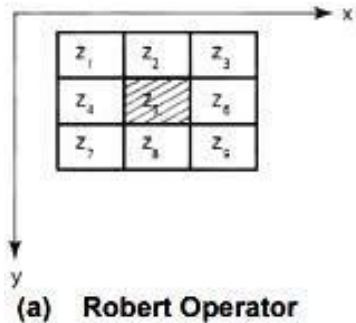
$$\nabla f = \text{mag}(\nabla f(x, y)) = \sqrt{g_x^2 + g_y^2} = \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]$$

In common practice, gradient with absolute values are simpler to implement. Thus

$$\nabla f = |g_x| + |g_y| = \left| \frac{\partial f}{\partial x} \right| + \left| \frac{\partial f}{\partial y} \right|$$

Image Sharpening

- **Robert Operator**
- The Robert operator is a simple edge detection filter used in image processing to detect edges and boundaries in images. It is a first-order derivative filter that uses a 2x2 kernel to calculate the gradient in the x and y directions.
- Consider the a pixel of interest $f(x, y) = z_5$ and a rectangular neighborhood of size 3×3 in Fig. 12 (a).



1	0
-1	0

1	-1
0	0

1	0
0	-1

0	1
-1	0

Fig.: 12

(b)

(c)

Image Sharpening

Equation can be approximated at point z_5 in a number of ways. Simplest is

$$g_x = z_5 - z_8$$

$$g_y = z_5 - z_6$$

$$\nabla f = |z_5 - z_8| + |z_5 - z_6|$$

Another approach is to use cross difference to realized

$$\nabla f \cong |z_5 - z_9| + |z_6 - z_8|$$

Equations can be implemented by the masks in Fig. 12 (a) and Fig. 12 (c).

Image Sharpening

- The original image is convolved with both the masks separately and the absolute values of the two outputs of convolution are added.
- Although Roberts operator illustrates the derivative process, there is a difficulty in implementation. Since only two neighboring pixels are used, it is not clear as to exactly where the derivative value should be stored in the output image.
- This difficulty is taken care of in other filters like Prewitt and Sobel filters which are symmetric in nature.

Image Sharpening

- **Prewitt Operator**

In the Prewitt Operator, the edge value at $f(x, y)$ is calculated by taking the - difference of $f(x+1, y)$ and $f(x-1, y)$. To make sure that the difference is not coming from two arbitrary points, three pairs of points are taken together to result in one edge value. This creates a 3x3 mask.

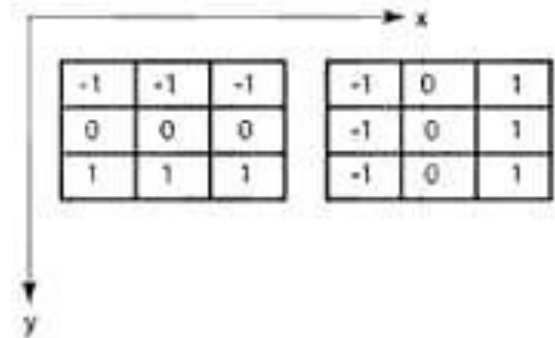


Image Sharpening

The above mask in Fig. 13 can be used to identify the presence of a vertical edge in the following image:

30	30	30	80	80	80
30	30	30	80	80	80
30	30	30	80	80	80
30	30	30	80	80	80
30	30	30	80	80	80
30	30	30	80	80	80

$$\nabla f \cong |(z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)| + |(z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)|$$

Image Sharpening

- **Sobel Operator**
- Another sharpening first order filter is the Sobel Operator. It is slightly different from the Prewitt filter, in that the middle pair of pixels are given higher weight compared to other two pairs of pixels. Both the filters are also called EDGE FILTERS. Sobel is the most popular 3x3 Edge operator. It is described by

$$\Delta f = |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| + |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|$$

- Sobel mask is shown in Fig. 14. Notice that all mask coefficients sum up to zero, thus giving no response in the area of constant intensity.

Image Sharpening

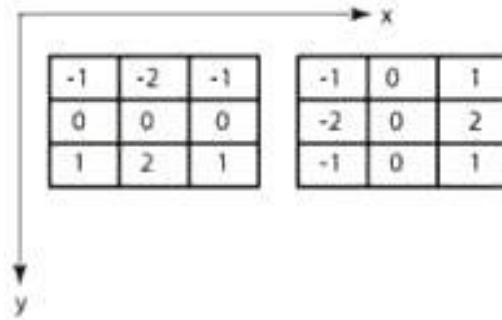


Fig. 14: Sobel Operator

- Sobel operator and Prewitt operator are similar to each other. Sobel operator is also used to detect two kinds of edges in an image; vertical and horizontal.
- You can observe from the masks given in Fig. 14, that first masks has only one difference in the members of first and third rows that is of sign, that is it has “2” and “-2” values in center of first and third rows.

Image Sharpening

- When applied on an image this mask will highlight the horizontal edges. Similarly, the second mask in Fig. 14 shows the vertical edges.
- When we apply this mask on the image it prominent vertical edges. It simply works like as first order derivative and calculates the difference of pixel intensities in an edge region.
- As the center column is of zero so it does not include the original values of an image but rather it calculates the difference of right.

Image Sharpening

2. Second Derivatives Filters

- **Laplacian Operator**

For a 2D function $f(x, y)$, the gradient (first derivative) is defined as

$$\Delta f = \frac{\partial f(x, y)}{\partial x \partial y} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$$

Laplacian (second derivative) is a **rotation invariant** and **linear** operator and it is defined as

$$\Delta^2 f = \frac{\partial^2 f(x, y)}{\partial x^2 \partial y^2} = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Image Sharpening

$$\frac{\partial^2 f}{\partial x^2} = f(x-1, y) + 2f(x, y) + f(x+1, y) = \begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y-1) - 2f(x, y) + f(x, y+1) = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Thus $\Delta^2 f = [f(x+1, y) + f(x-1, y) - 2f(x, y)] + [f(x,$

Image Sharpening

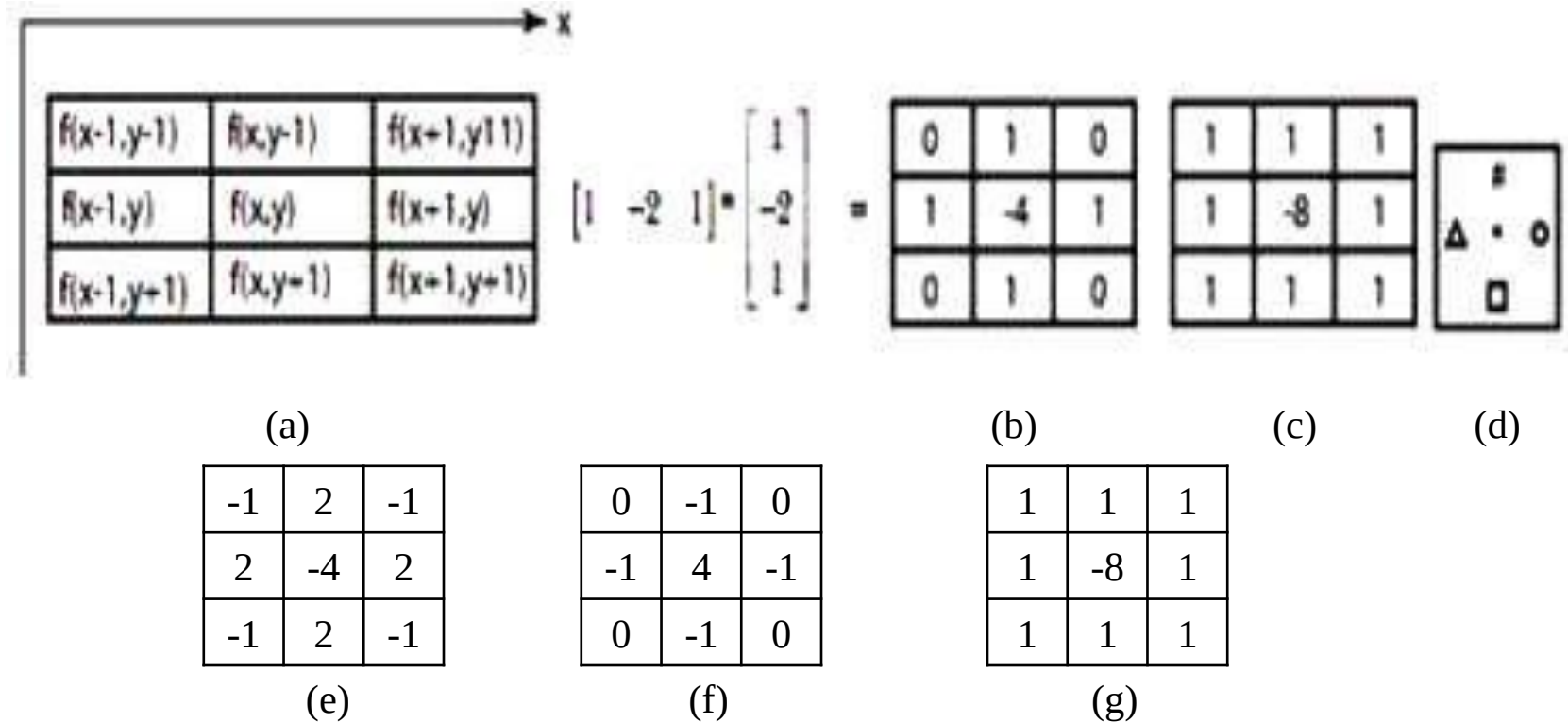


Image Sharpening

- Fig. 15 (b) shows a 3x3 sharpening filter mask. Fig. 15 (a) shows the coordinate notations and Fig. 15 (d) shows the location of non zero elements.
- Notice that addition of all the coefficients of the mask produces zero. In this mask, diagonal elements are not included.
- Fig. 15 (c) shows a mask with diagonal elements also. Fig. 15 (e) shows a weighted Laplacian mask. Notice that the centre element is negative. Fig. 15 (f) and Fig. 15 (g) show two Laplacian masks with centre element positive which gives the same results.

Image Sharpening

- Laplacian operation is a derivative operation which highlights the intensity discontinuities of an image and de-emphasize region with slowly varying intensity values. This tends to produce image that have greyish edge lines with dark featureless background.
- Background features can be '**recovered**' while '**preserving**' the sharpened effect of Laplacian by subtracting Laplacian from the image. Thus, sharpened imaged is generated by the following operation.

Image Sharpening

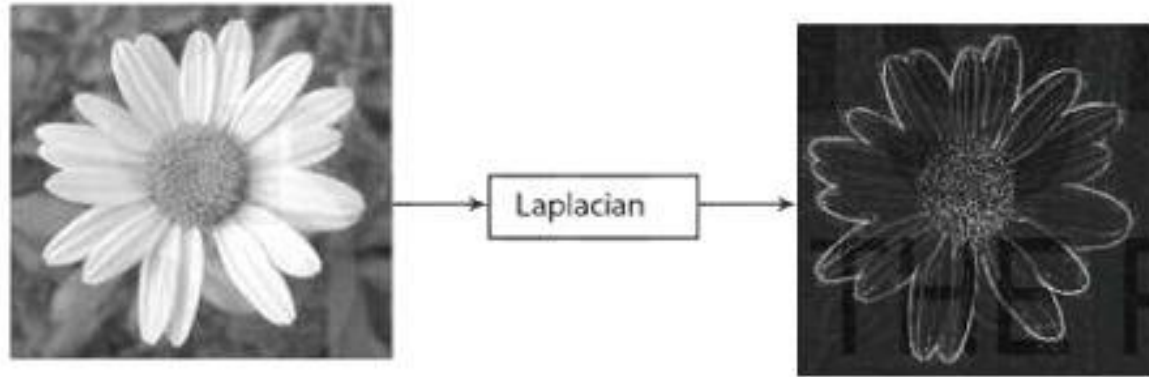


Fig: Output after laplacian operator

$$\begin{aligned} g(x, y) &= f(x, y) - \Delta^2 f(x, y) \\ &= f(x, y) - [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)] \\ &= 5f(x, y) - f(x+1, y) - f(x-1, y) - f(x, y+1) - f(x, y-1) \end{aligned}$$

Image Sharpening

A mask is generated for sharpening of images.

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Similarly

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 9 & -1 \\ -1 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Image Sharpening

- These are high frequency enhancement filters that **pass all frequencies and enhances high frequencies.**
- Notice that all the coefficients in the laplacian mask add up to one, so convolution with this mask will not change the pixel values in the part of the image that has constant grey levels.
- But if the pixels have higher or low grey level values than its neighbours, this contrast will be enlarged by the filter.

Image Sharpening

Example 5: For the image segment $f(x, y)$ given below. Apply laplacian filter of size 3×3 and 5×5 on the centre pixel.

Solution: Laplacian 4-connectivity mask is

0	1	0
1	-4	1
0	1	0

Thus, for Laplacian filter,

$$g_9(2,2) = [1 + 2 + 5 + 6 - 4 \times 7] = -14$$

Laplacian 8-connectivity mask is

1	1	1
1	-8	1
1	1	1

Thus for Laplacian filter

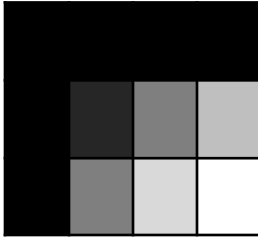
$$g_{10}(2,2) = [3 + 1 + 2 + 2 + 5 + 0 + 6 + 5 - 8 \times 7] = -32$$

Histogram Processing

- Histogram of an image represents the number of times a particular grey level has occurred in an image. It is a graph in which the x-axis represents gray levels, and y-axis represents number of pixels for each grey level.
- Histogram of an image is defined as $h(r_k) = n_k$, where $r_k = k^{\text{th}}$ grey level n_k = number of pixels with grey level r_k , and k takes on the values $0, 1, 2, \dots, L - 1$.

Histogram Processing

Example 1: Find histogram of image in Fig. 1.



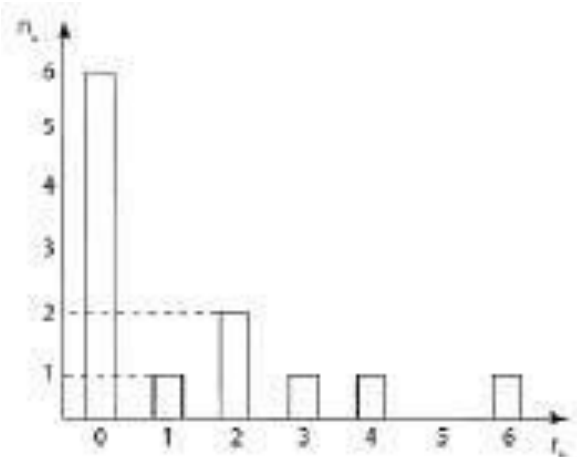
0	0	0	0
0	1	2	3
0	2	4	6

Fig. 1: Image

Solution: A table is generated with r_k and n_k , where r_k contains all possible grey level values 0,1, 2, 3, 4, 5, 6 and n_k consists the number of times that grey value has occurred in the Fig. 1.

0 has occurred 6 times, 1 has occurred 1 time etc. Then histogram is drawn in Fig. 2.

Histogram Processing



r_k	0	1	2	3	4	5	6
n_k	6	1	2	1	1	0	1

Fig. 2: Histogram

Normalized histogram is obtained by dividing the frequency of occurrence of each gray level r_k by the total number of pixels in an image.

Histogram Processing

$p(r_k) = \frac{nk}{n}$, where $k = 0, \dots, L-1$, where n = total number of pixels in the image $p(r_k)$ gives the probability of occurrence of grey level r_k . The sum of all components of a normalized histogram is equal to 1.

Example 2: Find normalized histogram of the image

0	0	0	0
0	1	2	3
0	2	4	6

Solution: Fig. 2 image is similar to Example 1. A table is made with k , n_k , r , and $p(r_k)$ and normalized histogram is drawn in Fig. 3.

Histogram Processing

r_k	1	1	2	3	4	5	6
n_k	6	1	2	1	1	0	1
$P(r_k)$	$\frac{6}{12}$	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	0	$\frac{1}{12}$

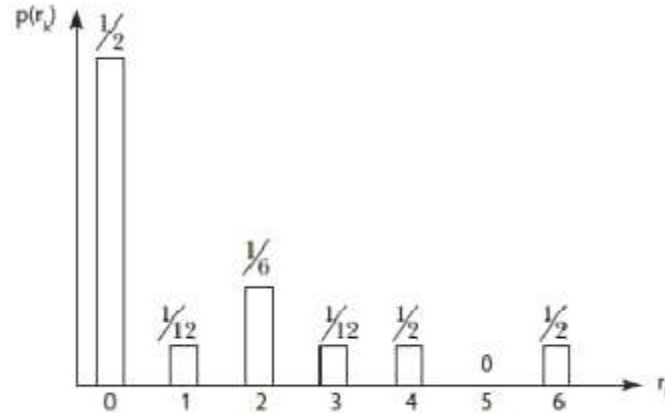


Fig. 3: Normalized Histogram

Histogram Processing

- Histogram gives an insight about the contrast in an image. It tells us the difference between average grey level of an object and that of the surroundings.
- It also provides a useful image statistics which are helpful in various image processing applications, for example, thresholding, intensity level slicing, segmentation etc.
- Besides it is very simple to calculate histogram of an image. Intuitively, it tells how vivid or washed out an image appears.

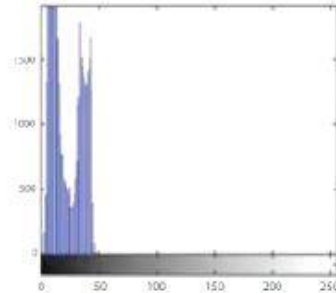
Histogram Processing

Based on histogram, we can categorize images in the four categories, which are defined below.

i) **Under exposed image:** It may be noted that almost all the pixels are concentrated at the lower end of the histogram. The maximum value of grey is less than 50. No details are seen in the image.



(a) under exposed image



(b) histogram

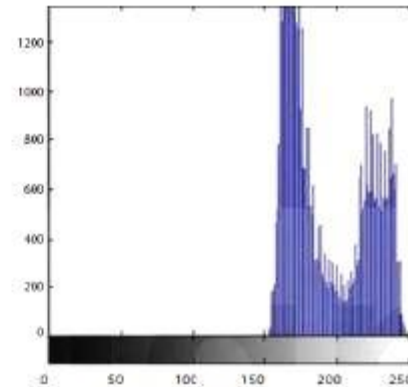
Histogram Processing

ii) **Over-exposed image:** The histogram is prominent towards the higher side of the grey values with the lowest grey level value around 150.

This image has a washed-out look because of the absence of darker shades of grey.



(c) over-exposed image



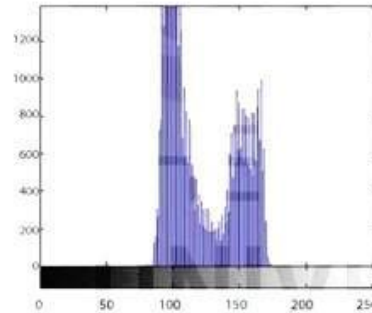
(d) histogram

Histogram Processing

iii) **Low contrast image:** The shape of the histogram is narrow and centered towards the middle of the grey scale. Very few grey levels participate in the image formation as neither the lower grey levels nor the higher grey levels are present. Thus, details are not visible in the image.



(e) low contrast image



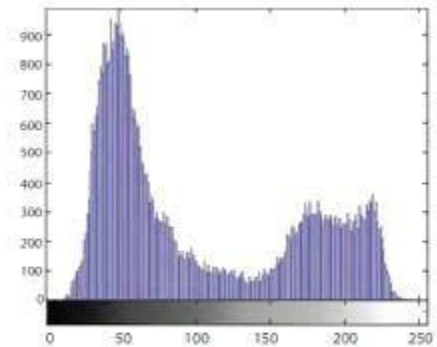
(f) histogram

Histogram Processing

iv) **High contrast image:** This histogram covers a broad range of grey levels, but distribution of pixels is not uniform. The number of some grey levels is very high as compared to the number of other grey levels.



(g) high contrast image



(h) histogram

Histogram Processing

Applications of Histogram

- i) Histogram gives a quick indication as to whether or not we have used the entire dynamic range of digitizer. Information is lost if dynamic range is not used fully.
- ii) Histogram indicates if clipping is a problem in the image.
- iii) It is extensively used in thresholding to separate contrasting objects from background.
- iv) It tells us about image contrast.

Important Topics

Image Sharpening

First and Second Order Filters

Histogram Processing

Summary

In this session, we discussed:

- Image sharpening is opposite of image smoothing. This is done to highlight fine details and edges in an image.
- Defined histogram
- Understood how to draw histogram from the image

*Thank
You*



Digital Image Processing and Computer Vision

Block-2_Unit-5_Part-2

Image Transformations-Frequency Domain

Topics to be Covered

Introduction

Discrete Wavelet Transform

Haar Transform

Important Topics

Summary

Introduction

After studying this session, you should be able to:

- Find the Discrete Wavelet Transform (DWT)
- Find the Haar Transform
- Apply the above mentioned transforms



Discrete Wavelet Transform

- The Discrete Wavelet Transform (DWT) is a mathematical technique used to decompose a signal into different frequency components, called wavelets.
- Unlike the Fourier Transform, which decomposes a signal into a series of sinusoidal functions, the DWT decomposes the signal into wavelets that are localized both in time and frequency.
- The DWT can be applied to both one-dimensional and two-dimensional signals.

Discrete Wavelet Transform

- A wavelet has the important property of not having a fixed-width sampling window.
- “A wavelet is a wave-like oscillation with an amplitude that begins at zero, rises, and then decreases back to zero.” It think of it as a "short oscillation" in the majority of situations, which is analogous to what could be captured by a seismograph or heart monitor.
- A wavelet is a wave-like oscillation that is localised in time.

Discrete Wavelet Transform

- Scale and location are the two fundamental features that wavelets possess. How "stretched" or "squished" a wavelet is can be defined by its scale, which can also be referred to as its dilation.
- This characteristic is connected to frequency in the sense that it is understood for waves. The wavelet's position in time can be determined based on its location (or space).
- We understood that wavelets are functions that are concentrated in time and frequency around a certain location.

Discrete Wavelet Transform

- Generally, got confused for waves and wavelets but they are different the fundamental difference between the two is that a wave is an oscillating function of time or space that is periodic. The wave is an infinite length continuous function in time or space.
- In contrast, wavelets are localised waves. A wavelet is a waveform of an effectively limited duration that has an average value of zero.

Discrete Wavelet Transform

A function () can be called a wavelet if it possesses the following properties:

1. The function integrates to zero, or equivalently its Fourier transform denoted as $\psi(\omega)$ is zero at the origin:

$$\int_{-\infty}^{\infty} \psi(x) dx = 0$$

This implies $|\psi(\omega)|_{\omega=0}=0$ in the frequency domain.

This equation suggests that the function is either oscillatory or has a wavy appearance.

Discrete Wavelet Transform

2. It is square integrable, or equivalently, has finite energy:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty$$

This equation implies that most of the energy in $\psi(x)$ is confined to a finite interval, or in other words, $\psi(x)$ has good space localisation. Ideally, the function is exactly zero outside the finite interval. This implies that the function is a compactly supported function.

Discrete Wavelet Transform

3. The Fourier transform $\psi(\omega)$ must satisfy the admissibility condition given by

$$C_{\psi} \int_{-\infty}^{\infty} \frac{|\psi(\omega)|^2}{|\omega|} d\omega < \infty$$

This equation is useful in formulating the inverse wavelet transform.

From this equation, it is obvious that $\psi(\omega)$ must have a sufficient decay in frequency.

Discrete Wavelet Transform

- This means that the Fourier transform of a wavelet is localized, that is, a wavelet mostly contains frequencies from a certain frequency band.
- Since the Fourier transform is zero at the origin, and the spectrum decays at high frequencies, a wavelet has a band pass characteristic.
- Thus a wavelet is a ‘small wave’ that exhibits good time frequency localisation.

Discrete Wavelet Transform

A family of wavelets can be generated by dilating and translating the mother wavelet in $\psi(x)$ which is given by

$$\psi_{(a, b)}(x) = \frac{1}{\sqrt{a}} \psi\left(\frac{x-b}{a}\right)$$

Here, a is the scale parameter and b is the shift parameter.



Discrete Wavelet Transform

- **Wavelet transforms** can be either continuous or discrete, depending on how they are implemented. The Continuous Wavelet Transform (CWT) uses every wavelet that is feasible across a range of scales and places, meaning that it has an endless number of scales and locations to choose from. This is the primary distinction between these two types.
- While the Discrete Wavelet Transform (DWT) uses a limited number of wavelets, which are defined according to a specific range of sizes and locations, this set of wavelets is not limited in any way.

Discrete Wavelet Transform

Few more comparisons between CWT and DWT are given below:

	CWT- Continuous Wavelet Transform	DWT- Discrete Wavelet Transform
1. Scale	At any scale	Dyadic scales
2. Translation	At any point	Integer point
3. Wavelet	Any wavelet that satisfies minimum criteria	Orthogonal, biorthogonal,

Discrete Wavelet Transform

	CWT- Continuous Wavelet Transform	DWT- Discrete Wavelet Transform
4. Computation	Large	Small
5. Detection	Easily detects direction, orientation	Cannot detect minute object if not finely tuned
6. Application	Object Detection, Pattern recognition Feature extraction	Compression, Denoising, Transmission Characterisation

So, the wavelet transform can be broadly classified into (i) continuous wavelet transform, and (ii) discrete wavelet transform.

Discrete Wavelet Transform

- For long signals, continuous wavelet transform can be time consuming since it needs to integrate over all times. To overcome the time complexity, discrete wavelet transform was introduced.
- Discrete wavelet transforms can be implemented through sub-band coding. The DWT is useful in image processing because it can simultaneously localise signals in time and scale, whereas the DFT or DCT can localise signals only in the frequency domain.

Discrete Wavelet Transform

- Apart from image processing, the DWT is quite promising tool for the Signal processing also. After the suggestion of Mallat's, that signals may be represented at several resolutions using wavelet decomposition, Discrete Wavelet Transform (DWT) emerged as an extremely flexible tool for signal processing.
- The wavelet transformation technique overcomes the limitations of the Fourier method. The Fourier transformation, despite the fact that it deals with frequencies, does not reveal any facts regarding the passage of time.

Discrete Wavelet Transform

The Wavelet Transform offers a number of important benefits, including the following:

- The Wavelet transform has the ability to concurrently extract local spectral and temporal information.
- A selection of different wavelets from a variety to choose

Analyzing electrocardiogram (ECG) readings, which comprise periodic and transient signals of relevance, is an example of a real-world use of the Wavelet Transform.

Haar Transform

- The Haar transform is a wavelet transform. Wavelet transforms are based on small waves called wavelets which are of varying frequencies and limited duration. These are different from the Fourier transform, where the basis functions are sinusoids.
- Haar transform is a transform whose basis functions are orthonormal wavelets. The Haar transform can be expressed as $T = HFH^T$

Haar Transform

- where, F is an $N \times N$ image matrix, H is the $N \times N$ Haar transform matrix and T is the resulting $N \times N$ transform.
- The Haar transform, H , contains the Haar basis functions, $h_k(t)$. They are defined on a continuous interval, $t \in [0, 1]$ for $k = 0, 1, \dots, N-1$, where $N = 2^n$.
- Then, H is generated by uniquely decomposing the integer k as $k = 2^p + q - 1$, where, $0 \leq p \leq n - 1$ and when $p = 0$, $q = 0, 1$; $p \neq 0$ then, $1 \leq q \leq 2^{n-p}$.

Haar Transform

For example, when $N = 4$, k will take the values $k = 0, 1, 2, 3$. For these the corresponding values of p and q have to satisfy that $k = 2^p + q - 1$.

Therefore, we compute the values of k , p and q in Table 1.

Table 1

k	0	1	2	3
p	0	0	1	1
q	0	1	1	2

Haar Transform

Let t take the values from the set $\left\{\frac{0}{N}, \frac{1}{N}, \dots, \frac{N-1}{N}\right\}$.

Then, the Haar basis functions are recursively defined as:

For $k = 0$, the Haar function is defined as a constant

$$h_0(t) = 1/\sqrt{N}$$

When $k > 0$, the Haar function is defined by

$$h_k(t) = \frac{1}{\sqrt{N}} \begin{cases} 2^{p/2}; \frac{(q-1)}{2^p} \leq t \leq \frac{(q-0.5)}{2^p} \\ -2^{p/2}; \frac{(q-0.5)}{2^p} \leq t \leq \frac{q}{2^p} \\ 0; \text{otherwise} \end{cases}$$

Haar Transform

As can be seen from the definition of the Haar basis functions, for the non-zero part of the function, the amplitude and width is determined by p while its position is determined by q .

Example 5: For, $N = 2$, compute the discrete Haar transform of a 2×2 matrix.

Solution: Here, $N = 2$, we know that $N = 2^n$.

Substituting the value of N , we get $2 = 2^n$, which gives $n = 1$.

Since, $0 \leq p \leq n - 1$, we get $0 \leq p \leq 0$.

Haar Transform

Therefore, $p = 0$, and hence $q = 0, 1, 2$.

We determine the value of k using the relation $k = 2^p + q - 1$, we obtain

p	0	1
q	0	1
k	0	1

for $k = 0$, $h_0(t) = \frac{1}{\sqrt{2}}$

for $k = 1$, $h_1(t) = \frac{1}{\sqrt{2}} \begin{cases} 1 & ; t = 0 \\ -1 & ; t = 1/2 \end{cases}$

Thus, Haar transform is $h_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.

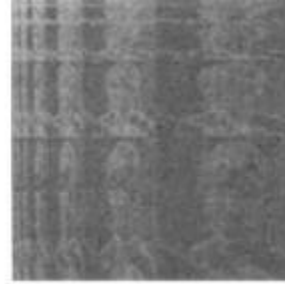
Haar Transform

- An important property of the Haar transform matrix is that it is real and orthogonal, that is, $H = H^*$ and $H^{-1} = H^T$. The orthogonal property of the Haar transform allows the analysis of the frequency components of the input signal. The Haar transform can also be used for analyzing the localized feature of the signal.

Fig. 7: Haar Transform



a) Original Image



b) Output of Haar Transform

Fig. 7 (b) shows the output of Haar Transform of the image in Fig. 7 (a).

Important Topics

Discrete Wavelet Transform

Haar Transform

Summary

In this session, we discussed:

- Two very important image transforms, namely the Discrete Wavelet Transform (DWT) and the Haar transform.
- The properties of each of these transforms, which shall help us in using them for image filtering in the frequency domain.

*Thank
You*



Digital Image Processing and Computer Vision

Block-2_Unit-6_Part-3

Image Enhancement & Filtering In Frequency Domain

Topics to be Covered

Introduction

Image Sharpening in Frequency Domain

Image Degradation

Image Degradation/Restoration Model

Noise Models

Important Topics

Summary

Introduction

After studying this session, you should be able to:

- Apply different types of image sharpening filters
- Describe images degradation models
- State difference between restoration and enhancement
- Apply different noise models



Image Sharpening in Frequency Domain

- Image sharpening in the frequency domain is a technique used to enhance the edges and details of an image by modifying its frequency components. It is based on the fact that high-frequency components of an image correspond to its edges and details.
- High pass filtering is achieved by attenuating low frequency components without disturbing high frequency components. High pass filter (HPF) can also be viewed as reverse operation of low pass filter.
- Transfer function of HPF is given by

$$H_{hp}(u, v) = 1 - H_{lp}(u, v)$$

Image Sharpening in Frequency Domain

Here, $H(u, v)$ is the transfer function of a LPF. Thus

Smoothing $\rightarrow$ LPF $\rightarrow$ attenuates high frequency components

Sharpening $\rightarrow$ HPF $\rightarrow$ attenuates low frequency components

Here, we discuss only real and symmetric filters. Let discuss following sharpening filters:

1. Ideal high pass filter
2. Butterworth high pass filter
3. Gaussian high pass filter

Image Sharpening in Frequency Domain

1. Ideal High Pass Filter (IHPF)

- An Ideal High Pass Filter (IHPF) is a type of frequency domain filter used in image processing to remove low-frequency components from an image while preserving its high-frequency components. As the name suggests, an IHPF allows high-frequency signals to pass through it while blocking low-frequency signals.
- Transfer function of a 2D IHPF is given by

$$H(u, v) = \begin{cases} 1, & \text{if } D(u, v) \geq D_0 \\ 0, & \text{if } D(u, v) < D_0 \end{cases}$$

Image Sharpening in Frequency Domain

Here, D_0 is the cut off frequency and $D(u, v)$ is the distance from the origin of the Fourier transform. Fig. 14 (a) and Fig. 14 (b) is the IHPF and its transfer function respectively. Fig. 14 (c) is plot of IHPF and Fig. 14 (d) is IHPF as an image.

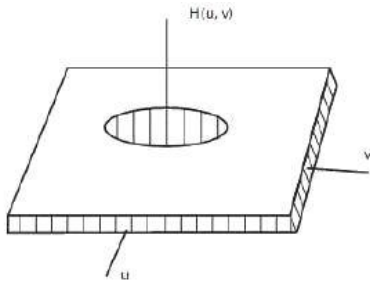


Fig. 14 (a) Plot of IHPF

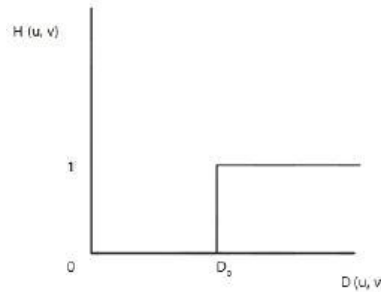


Fig. 14 (b) Transfer function of IHPF

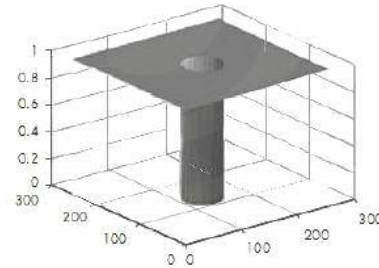


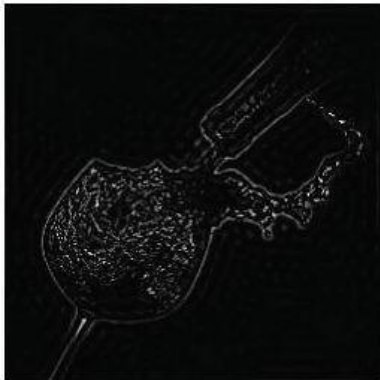
Fig. 14 (c) Plot of IHPF



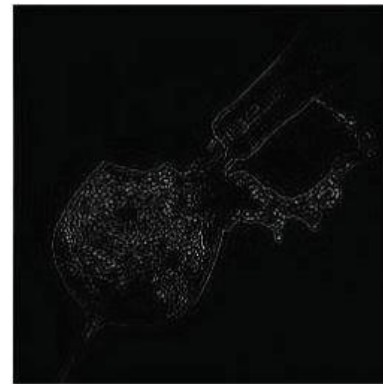
Fig. 14 (d) IHPF displayed as an image

Image Sharpening in Frequency Domain

Note that the origin $(0,0)$ is at the centre and not in the corner of the image. This filter sets to all frequencies inside the circle of radius D and passes all frequencies above D_0 without any attenuation. Output image looks very dark and dull as the high value DC component $G(0, 0)$ is eliminated.



**Fig. 15 (a) Output of
IHPF for $D_0 = 50$**



**Fig. 15 (b) Output of
IHPF for $D_0 = 60$**

Image Sharpening in Frequency Domain

2. Butterworth High Pass Filter (BHPF)

A Butterworth High Pass Filter (BHPF) is a type of frequency domain filter used in image processing to remove low-frequency components from an image while preserving its high-frequency components.

Transfer function of BHPF is $H(u, v) = \frac{1}{1 + \left[\frac{D_0}{D(u, v)} \right]^{2n}}$,

where n is the order of the filter, D_0 is the cut off frequency and $D(u, v)$ is the distance from the origin of Fourier transform.

Frequency response does not have a sharp transition as in the ideal HPF.

Image Sharpening in Frequency Domain

Fig. 16 (a) and Fig. 16 (b) are BHPF transfer function and Fig. 16 (c) and Fig. 16 (d) are plot and image display of BHPF.

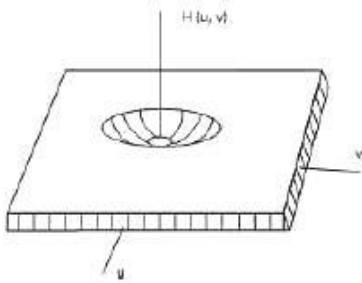


Fig. 16 (a) Transfer function of BHPF

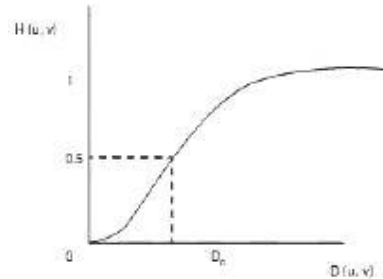


Fig. 16 (b) Transfer function of BHPF

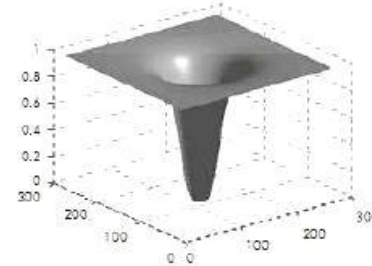


Fig. 16 (c) Plot of BHPF

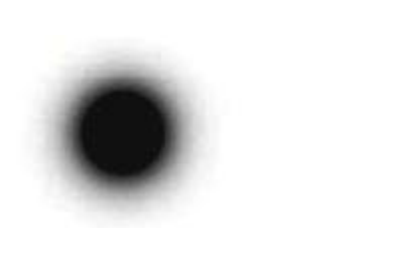


Fig. 16 (d) BHPF displayed as an image

Image Sharpening in Frequency Domain

Fig. 17(a) and Fig.17 (b) are the output of GHPF for $D_0 = 30$ and 130 respectively for $n = 2$. It is clear from the output, as D_0 increases, more and more power is removed from the output image.

Thus, output looks sharper for higher value of D_0 .



Fig. 17 (a) Output of BHPF with $D_0 = 130$, $n = 2$



Fig. 17 (b) Output of BHPF with $D_0 = 30$, $n = 2$

Image Sharpening in Frequency Domain

3. Gaussian High Pass Filter (GHPF)

A Gaussian High Pass Filter (GHPF) is a type of frequency domain filter used in image processing to remove low-frequency components from an image while preserving its high-frequency components. The GHPF is a variation of the Gaussian filter, which is widely used for image smoothing and noise reduction.

Transfer function of GHPF is $H(u, v) = 1 - e^{\frac{-D^2(u,v)}{2D_0^2}}$,

where D_0 is the cut off frequency and $D(u, v)$ is the distance from origin of Fourier transform.

Image Sharpening in Frequency Domain

Fig. 18(a) is GHPF Transfer function filter. Plot and image are displayed in Fig. 18 (b) and Fig. 18 (c).

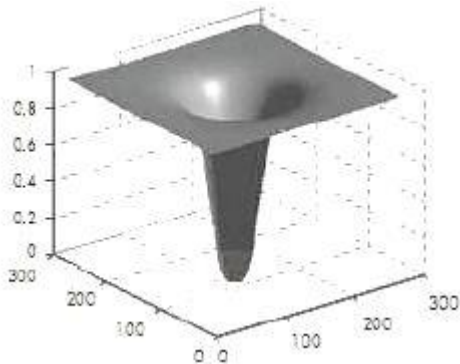


Fig. 18 (a) GHPF transfer function

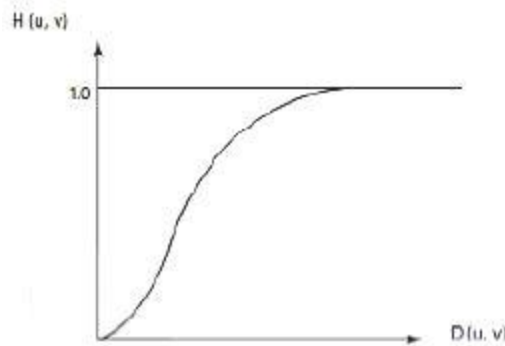


Fig. 18 (b) Plot of GHPF



Fig. 18 (c) GHPF displayed as an image

Image Sharpening in Frequency Domain

Output in Fig. 19 is much smoother than previous two filters with no ringing effect.



Fig. 19 (a) Output of GHPF with $D_0=30$



Fig. 19 (b) Output of GHPF with $D_0=120$

Image Degradation

- Image degradation refers to the loss of quality or information in an image due to various factors such as noise, blurring, distortion, compression, or other forms of damage. Image degradation can occur during image acquisition, transmission, processing, storage, or retrieval.
- Degradations are of three types
 - a) Noise
 - b) Blur
 - c) Artifacts

Image Degradation

a) Noise is a disturbance that causes fluctuations in pixel values. Original image is shown in Fig. 20(a) and noisy image with added Gaussian noise is shown in Fig. 20(b).



Fig. 20 (a) Original



Fig. 20 (b) Noisy Image

Image Degradation

b) Blur is a degradation that makes image less clear. This makes image analysis and interpretation very difficult. Motion blur is a very common cause of blurring where blur occurs due to the movement of object or camera. Fig. 21 (a) shows original image and Fig. 21 (b) shows blurred image.



Fig. 21 (a) Original Image



Fig. 21 (b) Blurred image

Image Degradation

c) Artifacts or distortions are extreme intensity or color fluctuations that can make image meaningless. Distortions involve geometric transformations such as translation, rotation or change in scale.

Image degradation can happen due to

- a) **Sensor Distortions:** Involves quantization, sampling, sensor noise, spectral sensitivity, de-mosaicking, non linearity of sensor etc.
- b) **Optical Distortions:** are geometric distortion, blurring due to camera mis-focus.
- c) **Atmospheric Distortions:** are haze, turbulence etc.

Image Degradation

d) **Other Distortions:** Low illumination, relative motion between object and camera etc.

- The processes that are used to remove degradation are mainly image enhancement and image restoration.
- Restoration is the process of inverting a degradation using knowledge about its nature, whereas enhancement is a process that aims to improve 'bad' quality image so that it looks better.
- Restoration is distinguished from enhancement, as degradation can be considered as an external influence.

Image Degradation

	Enhancement	Restoration
1	It gives better visual representation	It remove effects of sensing environment
2	No model required	Mathematical model of degradation
3	It is a subjective process	It is an objective process
4	Contrast stretching, histogram equalization etc are some enhancement techniques	Inverse filtering, wiener filtering, denoising are some restoration techniques

Image Degradation/Restoration Model

Image degradation and restoration models are used to simulate the degradation of images and to restore them to their original quality.

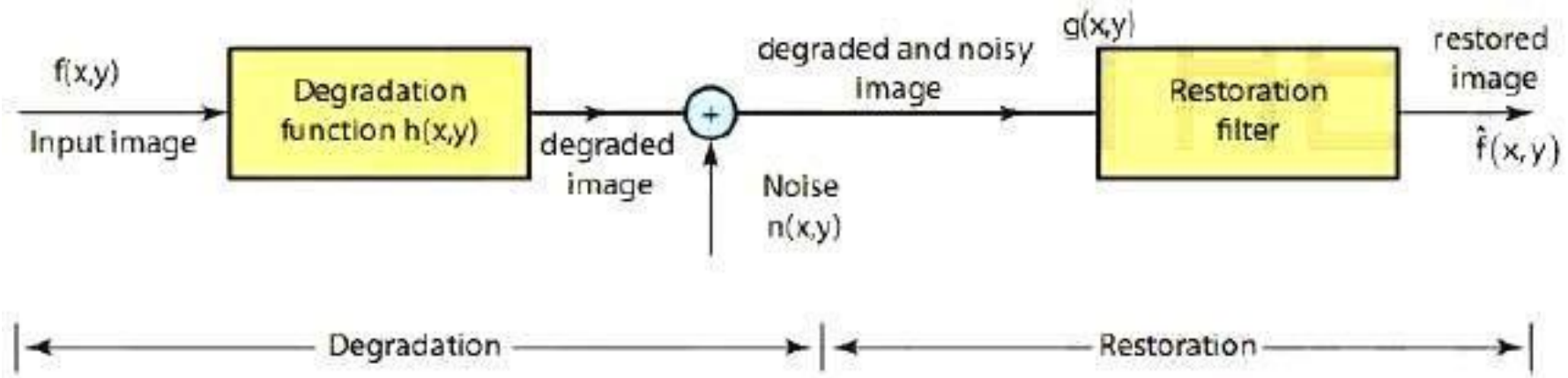


Fig. 23: Block diagram of degradation/restoration model

Image Degradation/Restoration Model

- Consider the block diagram given in Fig. 23 shows the block diagram of degradation/restoration model. Degradation function $h(x, y)$ and noise $n(x, y)$, operate on input image $f(x, y)$ to generate a degraded and noisy image $g(x, y)$.
 - The notations used in the block diagram are
- $f(x, y)$ = original image
- $h(x, y)$ = degradation function
- $n(x, y)$ = additive noise

Image Degradation/Restoration Model

$g(x, y)$ = degraded and noisy image

$\hat{f}(x, y)$ = restored image

- The objective of restoration process is to estimate $\hat{f}(x, y)$ from the degraded version $g(x, y)$, when some knowledge of degradation function H and noise n is available. The degraded image $g(x, y)$ as shown in Fig. 24 can be expressed mathematically as

$$g(x,y) = h(x,y) * f(x,y) + n(x,y)$$

- This equation is in spatial domain and $*$ represents convolution operation.

Image Degradation/Restoration Model

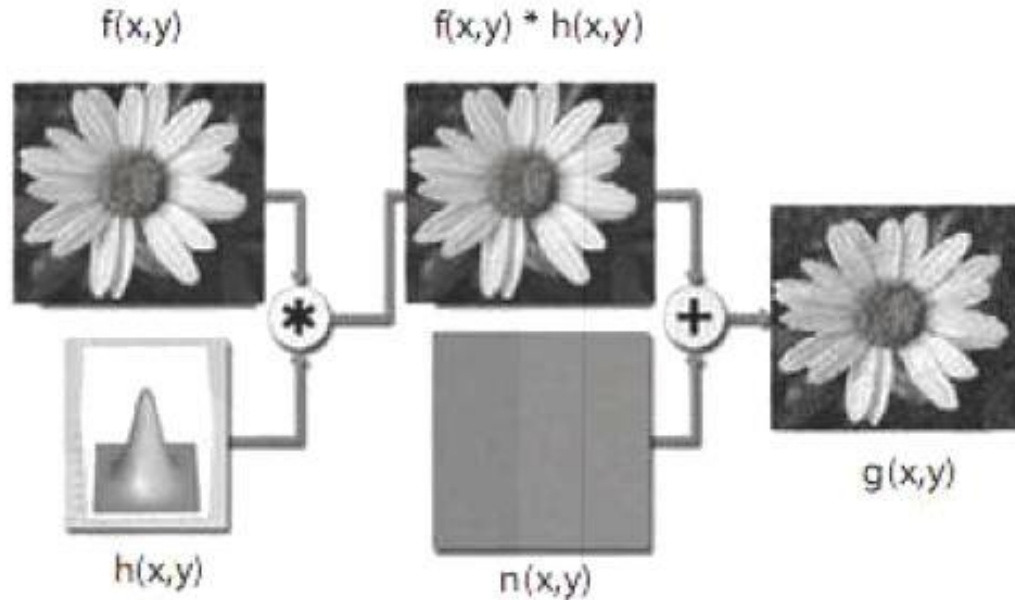


Fig. 24: Image Degradation Model (Spatial Domain)

Image Degradation/Restoration Model

An equivalent frequency domain representation is graphically shown in Fig. 25 and expressed as

$$G(u,v) = H(u,v)F(u,v) + N(u,v)$$

$$\text{Here } G(u,v) = F[g(x,y)]$$

$$H(u,v) = F[h(x,y)]$$

$$F(u,v) = F[g(x,y)]$$

$$N(u,v) = F[n(x,y)]$$

$$F(u,v) = H^4(u,v)[G(u,v) - N(u,v)]$$

Image Degradation/Restoration Model

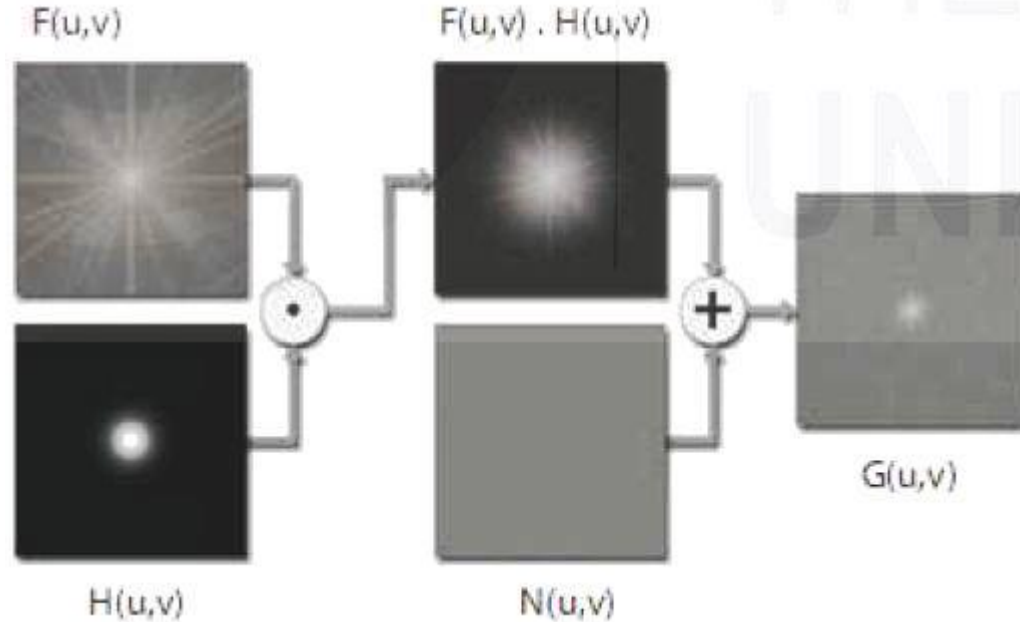


Fig. 25: Image Degradation Model (Frequency Domain)

Noise Models

Noise models are used to simulate or describe the behavior of noise in a system. Here are some common noise models:

Gaussian Noise

Gaussian noise model is most frequently used in practice. The PDF of a Gaussian random variable 'z' is given by

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}},$$

where z = intensity/grey level value

Noise Models

μ = mean (average) value of z

σ = standard deviation

- DFT of **Gaussian (normal) noise is another Gaussian process**. This property of Gaussian noise makes it most often used noise model. Some examples where Gaussian model is the most appropriate model are electronic circuit noise, sensor noise due to low illumination or high temperature, poor illumination.
- Gaussian noise is useful for modeling natural processes which introduce noise

Noise Models

Plot of $p(z)$ with respect to z is shown in Fig. 26.

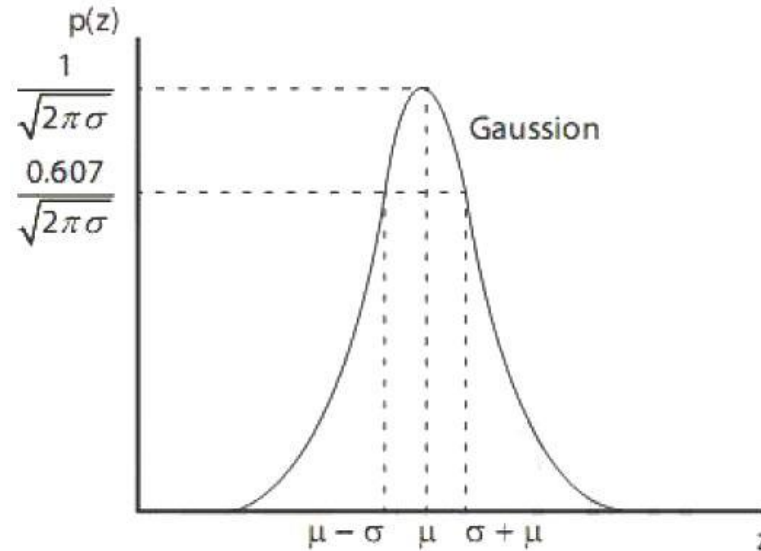


Fig. 26: PDF of Gaussian Noise Model

Noise Models

Rayleigh Noise

- The Rayleigh noise model is a statistical model used to describe the random interference of signals that occurs as they propagate through a medium, such as the atmosphere or a wireless channel. It is based on the Rayleigh distribution, which is a continuous probability distribution that describes the magnitude of the amplitude of the signal.
- Rayleigh distribution is defined by

$$p(z) = \begin{cases} \frac{2}{b}(z - a)e^{-\frac{(z-a)^2}{b}}, & z \geq a \\ 0, & z < a \end{cases}$$

Noise Models

Plot of PDF is shown in Fig 27. Rayleigh density is useful for approximating skewed (nonuniform) histograms. This is mainly used in range imaging.

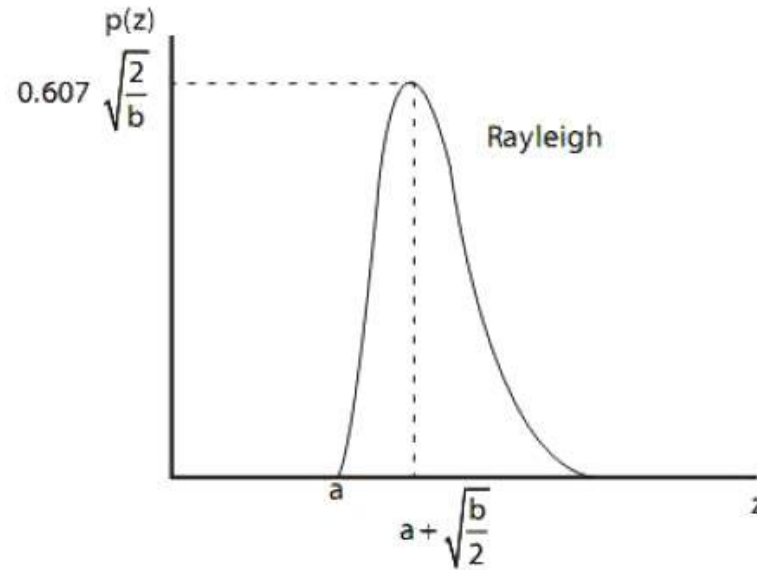


Fig. 27: PDF of Rayleigh Noise

Noise Models

Erlang (Gamma) Noise

- The Erlang noise model, also known as the Gamma noise model, is a statistical model used to describe the random interference of signals that occurs as they propagate through a medium or encounter obstacles in the environment. It is based on the Gamma distribution, which is a continuous probability distribution that describes the magnitude of the amplitude of the signal.
- Erlang noise is given by

$$p(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az}, & z \geq a \\ 0, & z < a \end{cases},$$

Noise Models

Where a and b are positive integers, mean density is given by $\mu = \frac{b}{a}$ and variance is $\sigma^2 = \frac{b}{a^2}$.

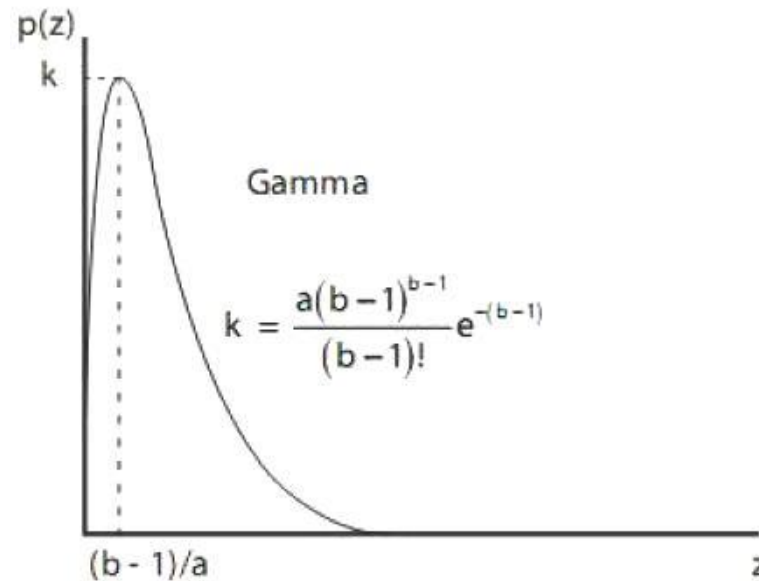


Fig. 28: PDF of Erlang Noise
+91 - 9319133134, query@cftedu.in

Noise Models

Uniform Noise

- Uniform noise, also known as white noise or additive white noise, is a type of noise that has a constant amplitude across all frequencies.
- Uniform noise is specified as

$$p(z) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq z \leq b \\ 0 & \text{otherwise} \end{cases},$$

- Then mean and variance of uniform noise is given by

$$\mu = \frac{a+b}{2},$$

$$\sigma^2 = \frac{(b-a)^2}{12}$$

Noise Models

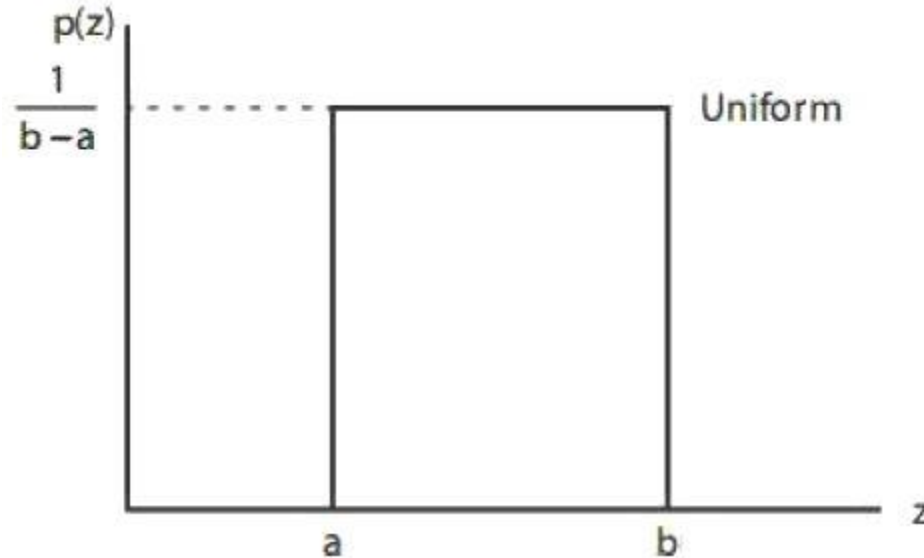


Fig. 29: PDF of Uniform Noise

Fig. 29 shows the plot of PDF of uniform noise. Uniform noise is least used in practice.

Noise Models

Impulse (Salt and Pepper) Noise

- Impulse noise, also known as Salt and Pepper noise, is a type of noise that is characterized by the presence of sudden and sharp disturbances or spikes in the signal.
- Impulse (salt and pepper) noise is specified as

$$p(z) = \begin{cases} P_a & z = a \\ P_b & z = b \end{cases},$$

- Fig. 30 shows the plot of PDF of impulse noise.

Noise Models

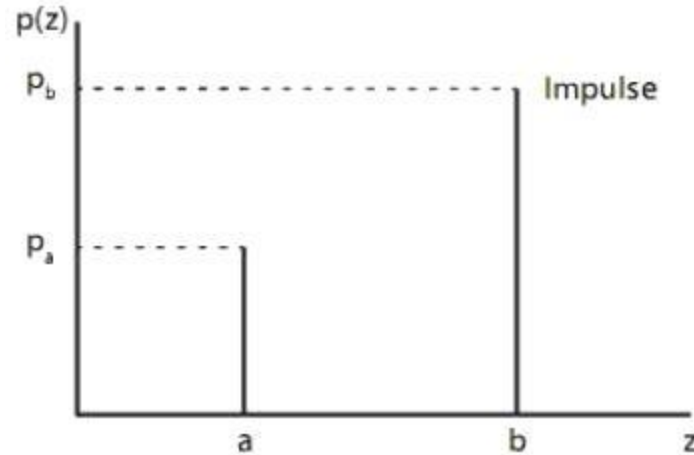


Fig. 30 PDF of Uniform noise

If $b > a$, intensity (grey level) 'b' will appear as a light dot on the image and 'a' appears as a dark dot. This is a '**bipolar**' noise, If $P_a = 0$ or $P_b = 0 \Rightarrow$ **unipolar noise**.

Noise Models

- If $P_a = 0$ and P_b exists, this is called '**pepper noise**' as only black dots are visible as noise. If $P_b = 0$, only P_a exists, this is called '**salt noise**' as only white dots are visible on the image as noise.
- Impulse noise occurs when quick transitions happen, such as faulty switching takes place. Noise parameters are generally estimated based on histogram of small flat area of noisy image.
- The salt & pepper noise is generally caused by malfunctioning of camera's sensor cells, by memory cell failure or by synchronization errors in the image digitizing or transmission.

Noise Models



Fig. 32 (a) Original image



Fig. 32 (b) Image with Gaussian noise



Fig. 32 (c) Image with salt & pepper noise



Fig. 32 (d) Image with uniform noise

Noise Models

Periodic noise is a spatially dependent noise. During Image acquisition, electrical or electromechanical interference may cause such type of periodic noise. A strong periodic noise can be seen in the frequency domain as equi-spaced dots at a particular radius around the centre (origin) of the spectrum.

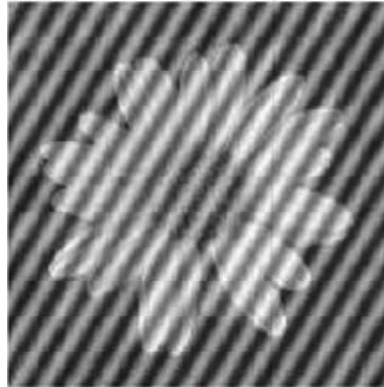


Fig. 33: Image with periodic noise

Important Topics

Image Sharpening in Frequency Domain

Image Degradation

Image Degradation/Restoration Model

Noise Models

Summary

In this session, we discussed:

- Various high pass filters
- Various image sharpening filters in frequency domain
- Sources of degradation
- Difference between enhancement and restoration
- Image degradation/restoration model
- Various types of noises with their pdfs

*Thank
You*



Digital Image Processing and Computer Vision

Block-2_Unit-7

Colour Image Processing

Topics to be Covered

Introduction

Human Vision System

Colour Fundamentals

Colour Models

-RGB Mode

-CMY and CMYK Model

-HSI Mode

Pseudo-colour models

Important Topics

Summary

Introduction

After studying this session, you should be able to:

- Differentiate between thousands of thousands of colours and their shades in the colour images.
- Define the different colour models and use them as per the requirements.
- Apply different pseudo colour models



Human Vision System

- A colour image is a powerful source of information. Human visual system has the ability to differentiate between hundreds of colours and their shades. Therefore, colour images contain a large amount of extra information compared to grey-scale images, that give a better understanding of the contents of the image.
- The human eye is nearly spherical in shape with a 20mm diameter on an average, as shown in Fig. 4.1. The three main parts of the eye are:
 - i) The cornea and sclera outer cover,
 - ii) the choroid and iii) the retina.

Human Vision System

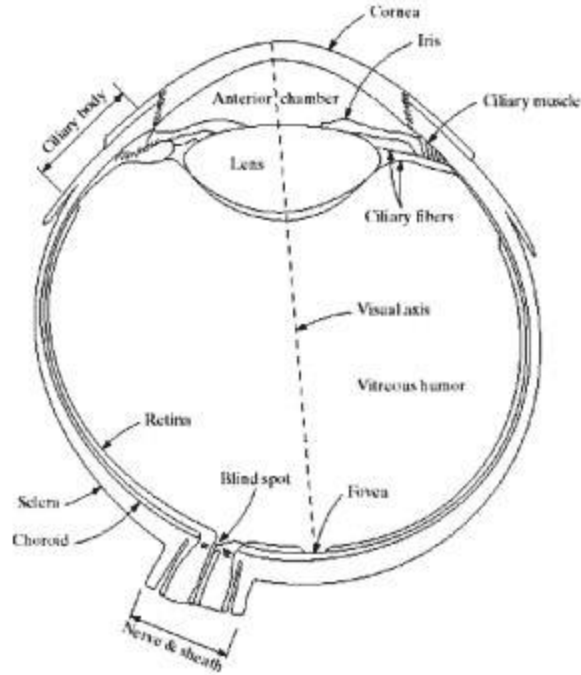


Fig.1: Structure of the human eye source

Human Vision System

- i) The sclera is an opaque member that encloses the optic globe all around, except at the anterior end, which is covered by the cornea. The cornea is a tough, transparent cover of the anterior chamber.
- ii) Choroid is the layer under the sclera. The membrane choroid contains a network of blood vessels. These blood vessels form the major source of nutrition for the eyes. If the choroid is damaged and inflamed, it can restrict blood flow in the eye, resulting in serious damage of the eye cells. The role of the choroid is also to control the amount of light entering the eye as well as reduce the backscatter inside the eye.

Human Vision System

The choroid is divided into two parts:

- a. The ciliary muscles which relax and tighten to enable the lens to focus by changing its shape,
- b. The iris diaphragm, that contracts and expands to control the amount of light that enters the eye.
- iii) The retina is the innermost membrane of the eye. It lines the wall of the complete posterior portion of the eye. The retina can be thought of as the image plane in the eye, since on properly focusing the eye on an object, light from that object is passed through the lens such that the image is formed on the retina.

Human Vision System

- The retina consists of two types of cells called rods and cones. The cones are highly sensitive to colour and are around 6-7 million in a human eye. The cones are located on the fovea which is the central portion of the retina.
- Rods are also responsible for low light vision, also known as SCOTOPIC vision, while cones are responsible for bright light vision, also known as PHOTOPIC vision.
- In the human eye, the distance between the retina and lens, that is the focal length, varies between 14 and 17 mm as the refractive power of the lens increases from min to max.

Human Vision System

- For a nearby object, the lens is most strongly refractive. Moreover, the lens of the eye is very flexible and is flattened by controlling muscles to enable the eye to focus on distant objects.
- To allow the eye to focus on objects close to the eye, the controlling muscles allow the lens to become thicker.
- **Brightness Adaptation and Discrimination:** Human vision system is highly complex and can adapt to an enormous range of light intensity levels-of the order of 10^{10} . The range starts from the scotopic threshold and goes up to the glare limit.

Human Vision System

- The subjective brightness, the perceived intensity by the human eye, has been experimentally found to be a logarithmic function of the light intensity that falls on the eye.
- Since the human eye cannot interpret this dynamic range simultaneously, brightness adaptation is carried out by the eye. The eye can discriminate only a small range of distinct intensity levels simultaneously.
- Brightness adaption level is the current sensitivity level of a human eye for a given set of conditions.

Colour Fundamentals

- Every colour is defined using three quantities that are independent of each other, however, taken together they define a particular shade of a colour. These quantities are:
 - i) **Hue:** Hue component of a colour is defined by the dominant wavelength. The wavelength range on the electromagnetic spectrum that defines the visible colour spectrum lies between 400nm [nm is nanometer] that represents the violet colour and 700nm that represents the red colour.
- This refers to the actual color of an object, such as red, blue, or green.

Colour Fundamentals

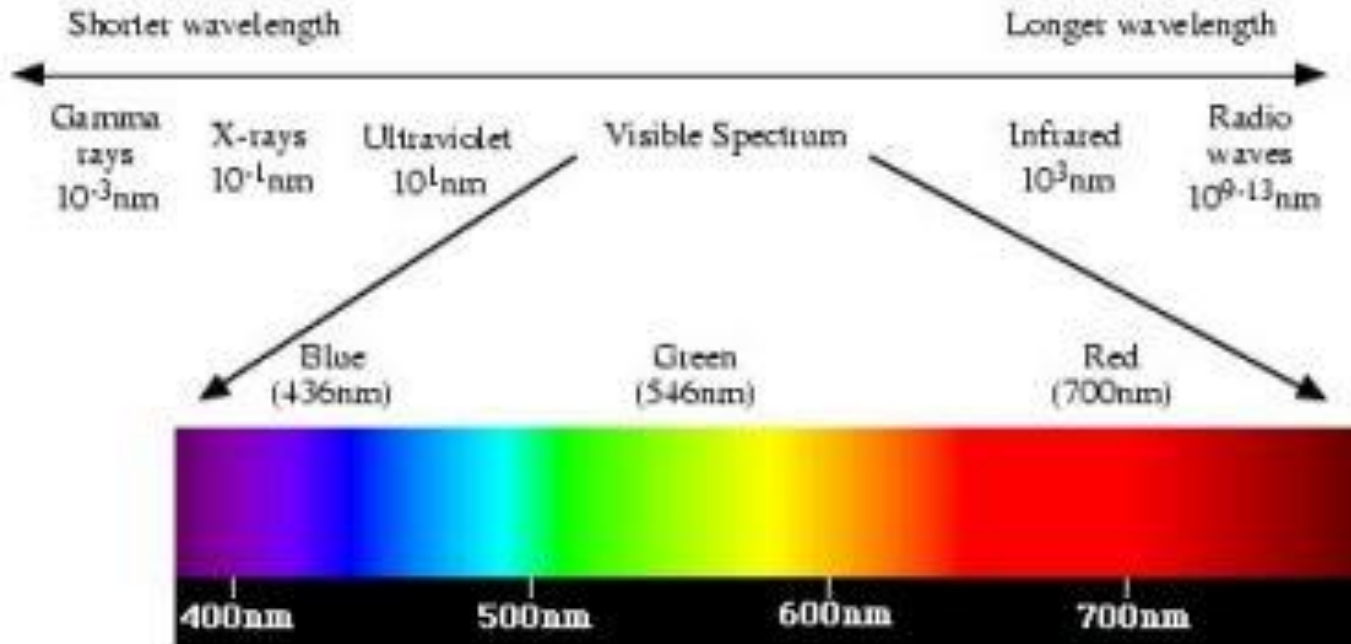


Fig. 2: Part of the electromagnetic spectrum that shows the visible spectrum Source

Colour Fundamentals

- ii) **Saturation:** The excitation purity of the colour is determined by the quantity known as saturation. It is dependent on the amount of white light that is mixed with hue of that colour. A fully saturated colour implies that no white light is mixed with that hue.
- iii) **Chromaticity:** The sum of hue and saturation constitutes the chromaticity of the colour. Therefore, if there is no colour, it is called achromatic light.
- iv) **Intensity:** The amount of light actually present defines the intensity. Therefore, intensity is a physical quantity. Achromatic light has only intensity but no colour.

Colour Fundamentals

v) **Luminance or Brightness:** The perception of colour is the quantity known as luminance or brightness. For example, given two colours of the same intensity, such as blue and green, it is perceived that blue is much darker than green.

vi) **Reflectance:** The ability of an object to reflect light, is the reflectance property of the object. The reflectance property determines the colour of the object, since, we see those colours that are reflected back and not the ones that are absorbed. For example, an object that reflects green, absorbs all other colours in the white light spectrum except green.

Colour Fundamentals

Example 1: Consider the coordinates of warm white (0.45, 0.4) and the coordinates of deep blue (0.15, 0.2). Find the percentage of the three colours red (X), green (Y) and blue (Z).

Solution: We first find the trichromatic coefficients x , y and z . At the point warm white, $x = 0.45$ and $y = 0.4$, therefore

$$z = 1 - (x + y) = 1 - (0.45 + 0.4) = 0.15$$

Now, we shall find the tristimulus values X , Y and Z .

Colour Fundamentals

$$\frac{X}{X+Y+Z} = 0.45$$

$$\frac{Y}{X+Y+Z} = 0.4$$

$$\frac{Z}{X+Y+Z} = 0.15$$

Here, $X : Y : Z = 0.45 : 0.4 : 0.15$

Therefore the percentage of each colour would be as follows:

Percentage of red (X) = 45%

Percentage of green (Y) = 40%

Colour Fundamentals

- Percentage of Blue (Z) =15%
- At the point deep blue, $x = 0.15$, $y = 0.2$, therefore $z = 0.65$.
- We can find the percentage of each colour as we found in case of warm white. We get percentage of red colour as 15%,percentage of green colour as 20% and the percentage of blue colour as 65%.
- We can see the percentage is justified for each colour name.

Colour Fundamentals

Example 2: Find the relative percentage of colours warm white and deep blue which mixes to give the colour which lies on the line joining them. Use the coordinates of these points as given in Example 1.

Solution: Let the colour C lies on the line have the coordinate (x, y).

The distance of C from the warm white colour = $\sqrt{(x - 0.45)^2 + (y - 0.4)^2}$

Similarly, the distance of C from the deep blue colour = $\sqrt{(0.15 - x)^2 + (0.2 - y)^2}$

Colour Fundamentals

The percentage of warm white in

$$C = \frac{\sqrt{(x-0.45)^2+(y-0.4)^2}-\sqrt{(0.15-x)^2+(0.2-y)^2}}{\sqrt{(0.45-0.15)^2+(0.4-0.2)^2}} \times 100$$

This expression can be used to find the percentage of warm white colour at C by substituting the coordinates of the point C as per the situation. Also, the percentage of the deep blue colour would be (100 - percentage of warm white colour).

Colour Models

1. The RGB Model

- This is an additive color model used for electronic displays, such as computer screens and televisions. In this model, colors are created by mixing red, green, and blue light in various intensities. The result is a wide range of colors, including white, black, and all shades in between.
- The geometry of the RGB colour model is a cube as shown in Fig. 3. It is assumed that all colour values are normalised, that is, it is a unit cube and all values of R, G and B lie between 0 and 1. It is clear from Fig. 3, that it is a model of a cube with the three axes for Red, Green and blue.

Colour Models

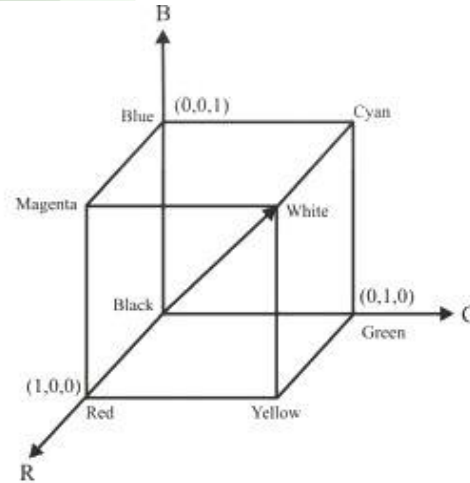


Fig. 3: The RGB colour (Image taken from [1])

A colour image in the RGB model consists of three images corresponding to each of the three colours: Red (R), Green (G) and Blue (B) colours. These three images combine to form one composite colour image on a monitor.

Colour Models

How do we find the composite colour in RGB colour model at any point. For this we follow the following steps:

Step 1: Pixel depth is the number of bits used to represent each pixel. If an image in RGB model has 8-bit image in each of its three colours, then each RGB pixel has a depth of 3 image planes x 8-bit per plane that is 24 bits. This gives rise to 2^{24} colour shades.

Step 2: We fix one of the three colours and let the other two colours to vary. Suppose we fix $R = 127$ and let G and B to vary. Then the colour at any point on the plane parallel to GB plane would be $(127, G, B)$, where, $G, B = 0, 1, \dots, 255$.

Colour Models

Example 3: In a RGB image, the R and B components are at mid and the G component is at 1, then which colour would be seen by a person?

Solution: At the given point, we have

$$\begin{aligned}\frac{R}{2} + \frac{B}{2} + G &= \frac{1}{2}(R + G + B) + \frac{G}{2} \\ &= \text{midgrey} + \frac{1}{2}G \\ &= \text{Pure green with some grey component.}\end{aligned}$$

Colour Models

2. The CMY and CMYK Colour Model

- This is a subtractive color model used in printing. In this model, colors are created by subtracting percentages of cyan, magenta, yellow, and black ink from a white background. The combination of these four inks creates a range of colors that can be printed.
- Fig. 4(a) shows the RGB model, in which the white colour is produced by adding the three primary colours Red, Green and Blue. Fig. 4 (b) shows the CMY model, where black is obtained as the sum of Cyan, Magenta and Yellow. The inverse relation between the RGB and CMY models are also shown by these two images.

Colour Models

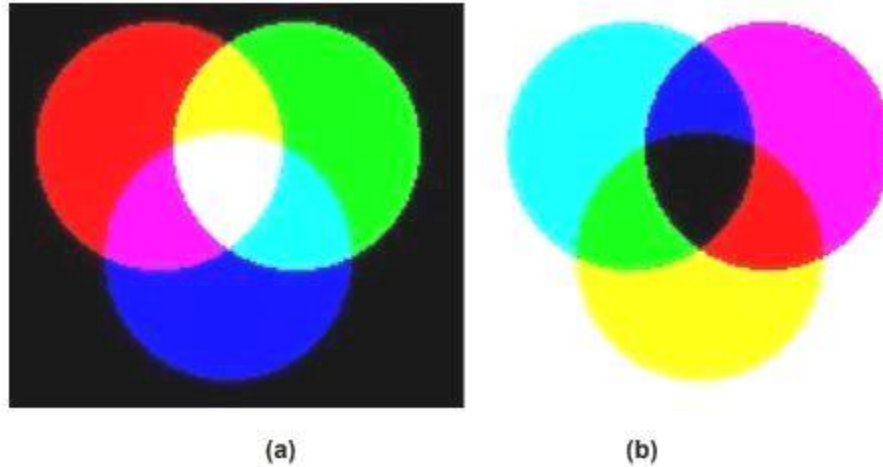


Fig. 4: (a) RGB and (b) CMY

In practice, black is obtained by combining cyan, yellow and magenta, however, this leads to a muddy looking black. In the CMYK colour model, black is added as the fourth colour, where K stands for black.

Colour Models

3. The HSI Model

- This colour model is very close to human colour perception which uses the hue, saturation and intensity components of a colour, when we see a colour, we cannot describe it in terms of the amount of cyan, magenta and yellow that the colour contains.
- Therefore, the HSI colour model was introduced to enable describing a colour by its hue, saturation and intensity/ brightness. Hue describes the pure colour, saturation describes the degree of purity of that colour while intensity describes the brightness or colour, sensation.

Colour Models

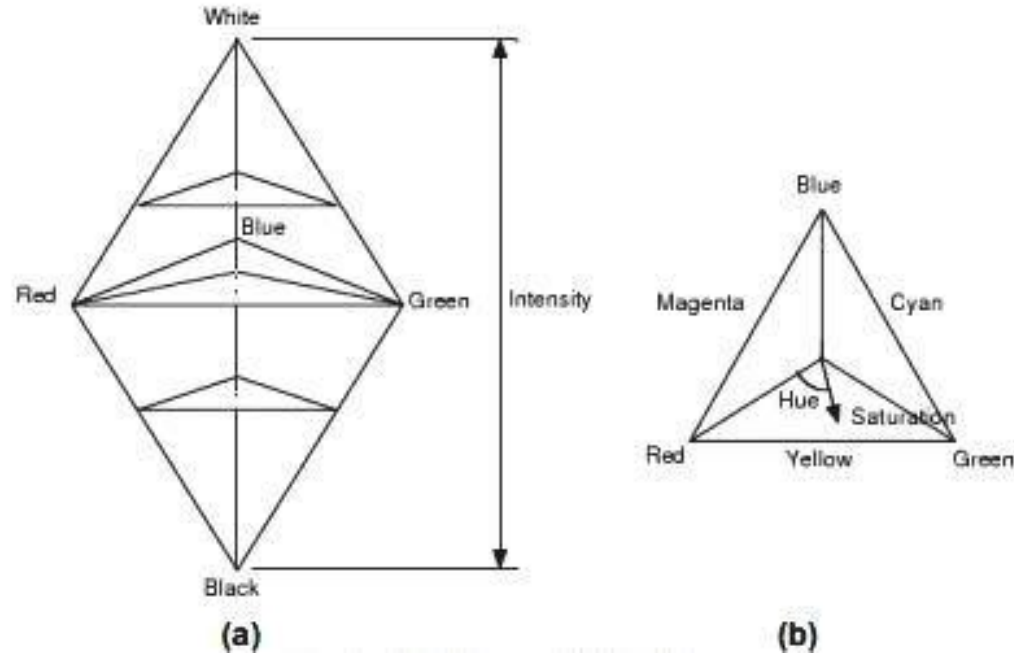


Fig. 5: (a) HSI and (b) RGB

Colour Models

In Fig.5, the HSI colour model is represented and its relation to RGB model is shown in the Fig. 5 (b). The HSI triangle in Fig. 5 (b) shows a slice from the HSI solid at a particular intensity as shown in Fig. 5 (a). You may notice that in Fig. 5, the hue, saturation and intensity values required to form the HSI colour space can be computed using the RGB values.

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Pseudo-Colour Models

- Pseudocolour image processing is the process of assigning each pixel of a grayscale image based on specific conditions.
- Intensity slicing or density slicing is one of the simplest forms of pseudocolour image processing technique. In this technique, the image is interpreted as a 3D function and can be imagined as a set of 2D grid which are parallel to the coordinate planes and placed at each intensity value. Each plane can then be thought of as a slice of the image function in the area of intersection.



Pseudo-Colour Models

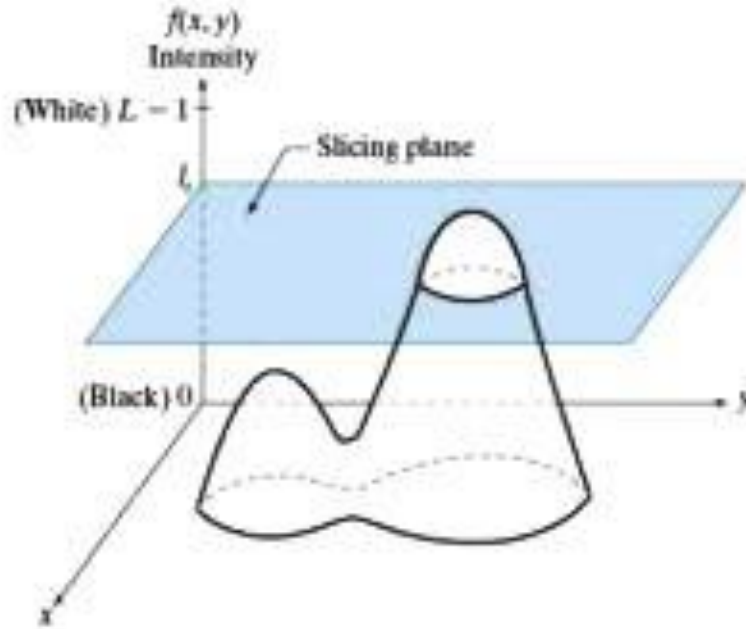


Fig. 9: Intensity slicing technique represented as slicing planes perpendicular to the intensity axis at various intensity levels.

Important Topics

Human Vision System

Colour Fundamentals

Colour Models

Summary

In this session, we discussed:

- The need for colour image processing.
- The structure of the human eye and then the tristimulus theory that connects the perception of colour with the various colour models that exist.
- The main colour models or colour spaces that are mainly used in both TV and print.

*Thank
You*



Digital Image Processing and Computer Vision

Block-3_Unit-8

Introduction To Computer Vision

Topics to be Covered

Introduction

Introduction to Computer Vision

Camera Models

Projections

Transformations

Important Topics

Summary

Introduction

After studying this session, you should be able to:

- Understand about Computer vision.
- Learn various camera models.
- Understand about projections and various geometric transformations.



Introduction to Computer Vision

- Computer vision is the field of study that endeavors to develop techniques to help machines understand the content of images and videos captured by single and/or multiple cameras.
- Applications of computer vision exist in every sphere of life, from medical image understanding to autonomous vehicles and robotics. Therefore, it is an important field of study.



Camera Models

- A camera is a basic tool in computer vision. It is essential to record an image of the scene around us to be able to analyse the scene.
- In general, a camera is a physical device that captures an image by allowing light to pass through a small hole (the aperture) onto a light sensitive surface. The lens in a camera focus the light entering through the aperture onto the imaging surface or the light sensitive surface.
- A camera therefore, maps the 3D world onto a 2D plane (the image). There are two major classes of camera models: camera models with a finite center and camera models with center at infinity.

Camera Models

We shall focus our attention to the camera model with a finite center and discuss the simplest camera model: the pin-hole camera model.

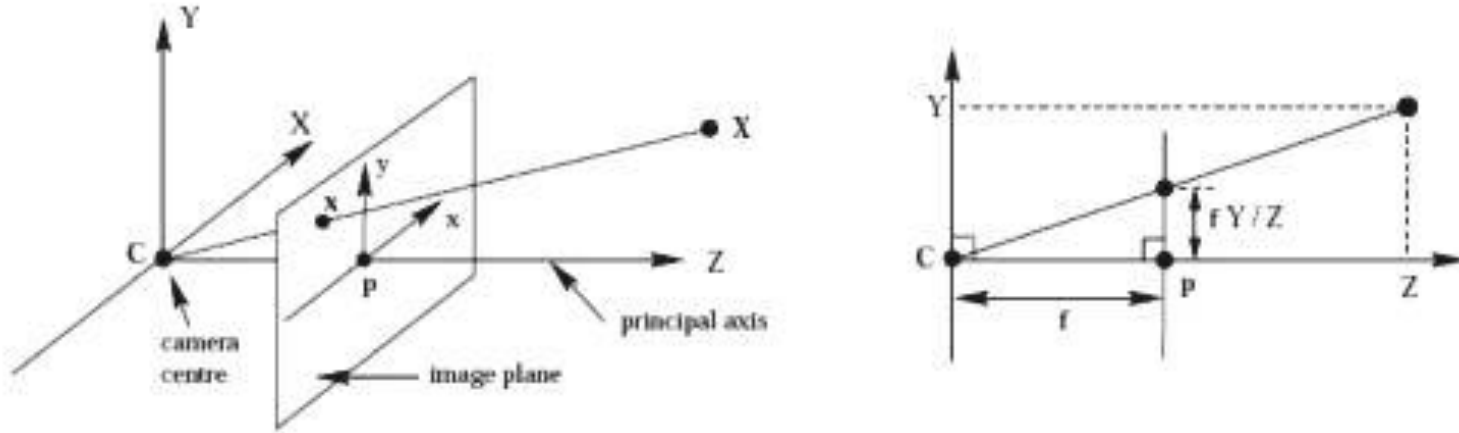


Figure 1. The pin-hole camera geometry. Figure taken from [1]

Camera Models

THE PIN-HOLE CAMERA MODEL

- In a pin-hole camera, a barrier with a small hole (pin-hole) is placed between the 3D object and the 2D image plane (light-sensitive plane). Each point on the 3D object emits/ reflects multiple light rays out of which only one or very few of these light rays fall on the image plane by passing through the pin-hole.
- Thereby, one can see that there exists a mapping between the points of the 3D object and the 2D plane forming an image and such a mapping will be a projection where the pin-hole is called the centre of projection.

Camera Models

HOMOGENEOUS COORDINATES

- A coordinate system where every point having three coordinates, $(x, y, w)^T$, is called a homogeneous coordinate system provided $w \neq 0$ in which for points $X_1 = (x_1, y_1, w_1)^T$ and $X_2 = (x_2, y_2, w_2)^T$, we have $X_1 = X_2 \leftrightarrow \frac{x_1}{w_1} = \frac{x_2}{w_2}$ and $\frac{y_1}{w_1} = \frac{y_2}{w_2}$ will be satisfied. Clearly, the points of a homogeneous coordinate system belong to the 2- dimensional space, $\mathbb{R}^3 \setminus (0,0,0)^T$.
- A two dimensional point in the Euclidean coordinate system can be represented as a point in the homogeneous coordinate system and vice-versa.

Camera Models

In fact, for a 2D point $(x, y)^T$ its corresponding point in the homogeneous coordinate system is taken as $(x, y, 1)^T$. Similarly, for a given a point $(x, y, w)^T$ and $w \neq 0$ in homogeneous coordinates, its corresponding 2D point

is obtained as: Since
$$\begin{pmatrix} x \\ y \\ w \end{pmatrix} = \begin{pmatrix} x/w \\ y/w \\ w/w \end{pmatrix} = \begin{pmatrix} x/w \\ y/w \\ 1 \end{pmatrix}$$

and therefore its corresponding 2D point becomes $(x/w, y/w)$.

Camera Models

- Likewise, as a point in the homogeneous coordinate system having four coordinates $(X, Y, Z, W)^T$ such that $W \neq 0$, the following additional condition holds: For $X_1 = (X_1, Y_1, Z_1, W_1)^T$, $X_2 = (X_2, Y_2, Z_2, W_2)^T$ from the homogenous coordinate system we have $(X_1, Y_1, Z_1, W_1)^T = (X_2, Y_2, Z_2, W_2)^T \leftrightarrow \frac{X_1}{W_1} = \frac{X_2}{W_2}$, $\frac{Y_1}{W_1} = \frac{Y_2}{W_2}$ and $\frac{Z_1}{W_1} = \frac{Z_2}{W_2}$.
- Clearly, these points belong to $\mathbb{R}^4 \setminus (0, 0, 0, 0)^T$.
- An important point to remember is that all scalar multiples of a homogeneous vector represent the same point.

Projections

- Projections are a fundamental concept in computer vision and computer graphics that involve mapping 3D objects onto a 2D image plane. Projections are necessary because most visual data is represented in 2D images or videos, while real-world objects exist in 3D space.
- There are two main types of projections used in computer vision: perspective projections and orthographic projections.

1. Perspective projection

- Perspective projections take into account the effects of perspective, such as foreshortening and the reduction in size of objects as they move further away from the camera.

Projections

- Perspective projections are used to create realistic 3D scenes in computer graphics and to simulate how 3D objects appear in 2D images in computer vision.
- Consider the pin-hole camera model as a perspective projection transformation. Using matrix transformation in homogeneous coordinates
$$\begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & f & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \begin{pmatrix} f X \\ f Y \\ f Z \\ z \end{pmatrix}$$

the projection point in the Euclidean space can be obtained

Projections

$$x = f \frac{x}{Z}, y = f \frac{y}{Z}$$

This way the perspective projection of a 3D point via the center of projection (the pin-hole) onto a point on the image plane $Z = f$ can be obtained in matrix form.

2. Orthographic projection

Orthographic projection is the projection of a 3D object onto a plane by a set of parallel rays that are orthogonal to the image plane, i.e., it is a parallel projection. In this projection, the center of projection is taken at infinity.

Projections

- For any point $(X, Y, Z)^T$ from the object, its orthographic projection on the image plane $Z = f$ is given by $x = X, y = Y$
- The important properties of the orthographic projection are that parallel lines remain parallel and the size of the object does not change.

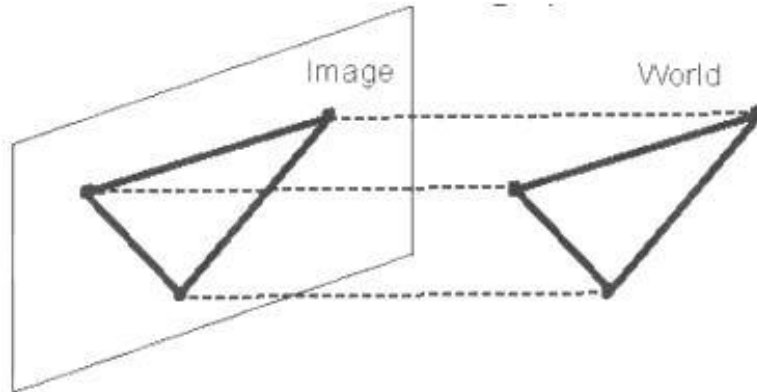


Figure 2. Orthographic projection

Projections

3. Weak perspective projection

- A perspective projection is a non-linear transformation, however, under certain circumstances, we can approximate it by a linear transformation, that is, as a scaled orthographic projection. The important conditions for these are that the object lies close to the optical axis and that the dimension of the object are small compared to its average distance $\bar{Z}$ from the camera. Equations of the weak perspective projection are:

$$x = \frac{fX}{Z} = \frac{fX}{\bar{Z}} ; y = \frac{fY}{Z} = \frac{fY}{\bar{Z}} \text{ in which each point is scaled by } \frac{f}{\bar{Z}} .$$

Transformations

In computer vision, transformations refer to operations that are performed on images or videos to modify their appearance or improve their quality.

Geometric transformations play an important role in Computer Vision.

1. EUCLIDEAN TRANSFORMATIONS

The most important property of Euclidean transformation is that it preserves lengths and angle measures. They are the most commonly used transformations consisting of translation and rotation.

Transformations

(i) Translation:

When a point is moved from one location to another along straight line paths, it is known as a translation.

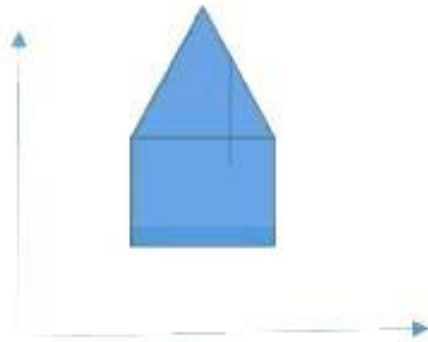
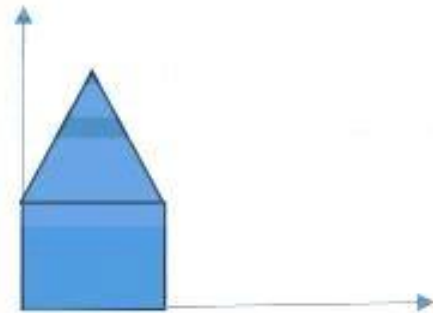


Fig (a) An object



(b) Object translated to the origin

Transformations

Example: If there is a rectangle with coordinates (2, 2), (2, 6), (5,2) and (5, 6) and it is to be translated to the origin. Then the translation vector

will be
$$\begin{bmatrix} t_x \\ t_y \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix}$$

Therefore,
$$\begin{bmatrix} x'1 \\ y'1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

Similarly,
$$\begin{bmatrix} x'2 \\ y'2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} x'3 \\ y'3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} x'4 \\ y'4 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$$

Transformations

The graphical representation is given below:

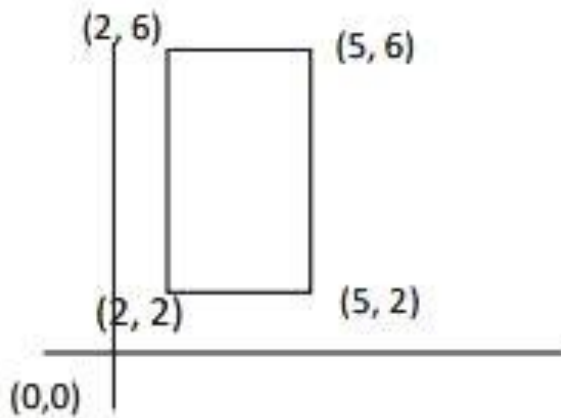
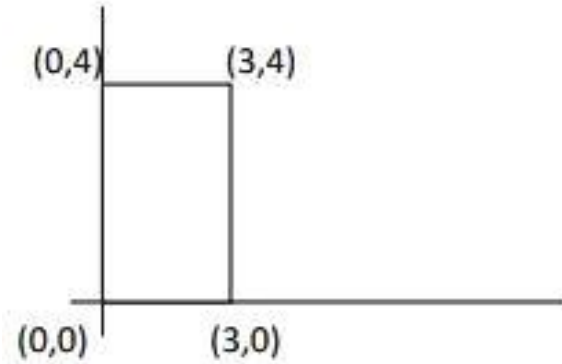


Fig: a) Square before Translation



b) Square after Translation

Transformations

(ii) Rotation

- Rotation is the process of turning an object around an axis. A rotation is specified by an angle θ , and the pivot-point, $(x, y)^T$ about which the object is to be rotated.
- A positive value of θ defines counter-clockwise rotation about the pivot point while a negative value of θ defines clockwise rotation. Rotation about the origin is given as

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Transformations

where, $R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is known as the rotation matrix.

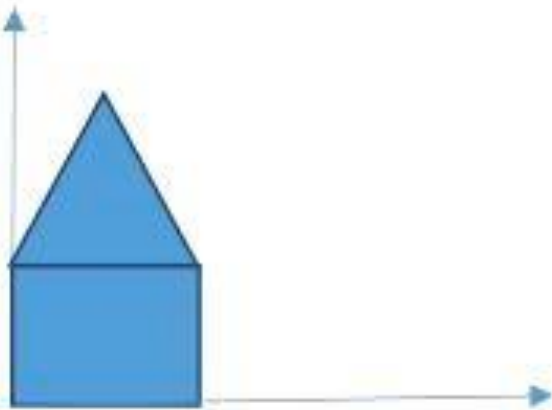
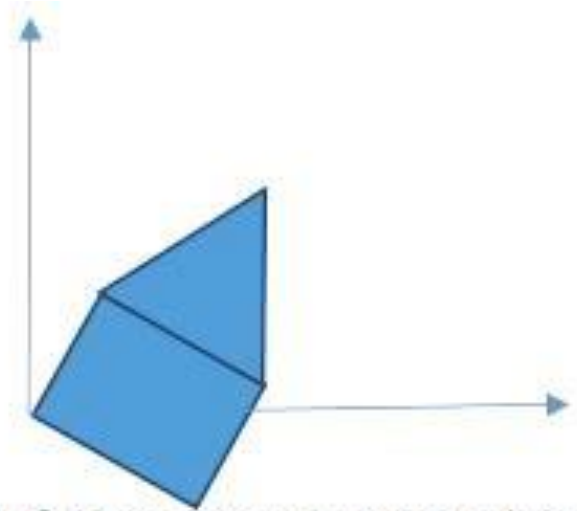


Fig. (c) Object at the origin



(d) Object rotated at the origin

Transformations

Example 2: Perform a 45° rotation of a triangle ABC with coordinates A: (0,0), B: (1,1), C: (5,2) about the origin.

Solution: We can represent the given triangle, in matrix form, using homogeneous coordinates of the vertices:

$$\begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 5 & 2 & 1 \end{bmatrix}$$

The matrix of rotation is: $R_\theta = R_{45}^\circ = \begin{bmatrix} \cos 45 & \sin 45 & 1 \\ -\sin 45 & \cos 45 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Transformations

So the new coordinates A'B'C' of the rotated triangle ABC can be found as:

$$\begin{bmatrix} A' \\ B' \\ C' \end{bmatrix} = \begin{bmatrix} A \\ B \\ C \end{bmatrix} * R_{45}^o = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 5 & 2 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -\sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & \sqrt{2} & 1 \\ 3\sqrt{2}/2 & 7\sqrt{2}/2 & 1 \end{bmatrix}$$

Thus $A' = (0,0)$, $B' = (0,\sqrt{2})$, $C' = (3\sqrt{2}/2, 7\sqrt{2}/2)$

Transformations

The following Figure (a) shows the original, triangle ABC and Figure (b) shows the triangle ABC after the rotation.

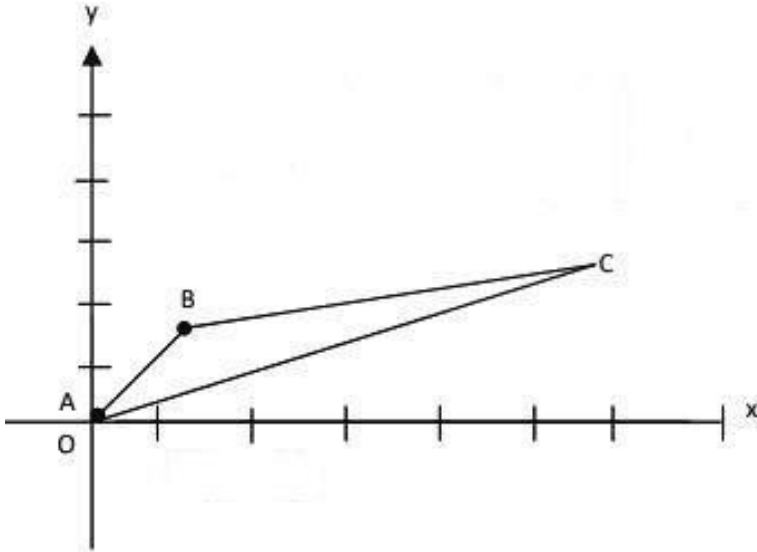


Figure (a)

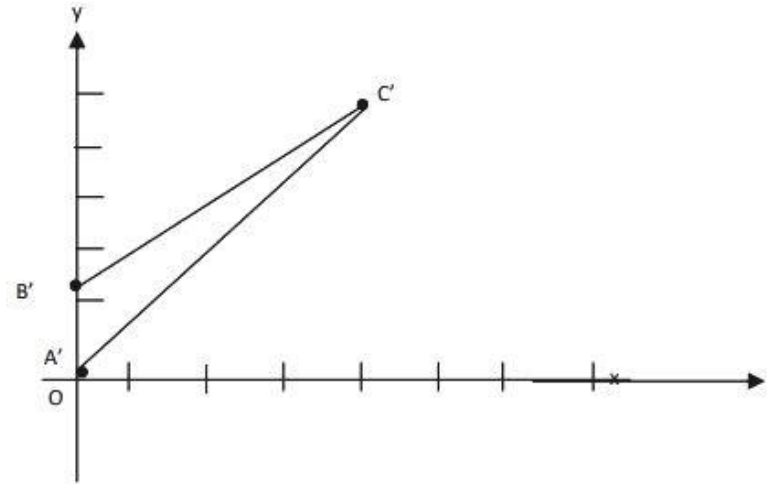


Figure (b)

Transformations

2. AFFINE TRANSFORMATIONS

- Euclidean transformations do not change the shape of the object. Lines transform to lines, planes to planes and circles to circles. However, the lengths and angles are preserved by Euclidean transformation.
- Affine transformations are an extension of the Euclidean transformations which do not preserve lengths and angles. That is, under an affine transformation, circle may transform to an ellipse, however, a line will transform to a line. The important affine transformations are scaling and shear. Moreover, translation and rotation also are affine transformations.

Transformations

(i) Scaling

By scaling, the dimensions of an object are either compressed or expanded. A scaling transformation is carried out by multiplying the coordinate values of each vertex $(x, y)^T$ by scaling factors S_x and S_y , in the x - and y - directions respectively, to produce the transformed coordinates

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Transformations

Example: Consider the triangle ABC with coordinates $A = (0,0)$, $B = (1,0)$ and $C = (1,1)$. If the scaling factor along X-axis, $S_x = 4$ and along Y-axis, $S_y = 1$, then calculate the new coordinates of the sheared triangle ABC.

In matrix form, the shear transformation will be

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Then,

$$A' = \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$B' = \begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

Transformations

$$C' = \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

Therefore, the scaled triangle, $A'B'C'$ will have the coordinates, $A' = (0, 0)$, $B' = (4, 0)$, $C' = (4, 1)$.

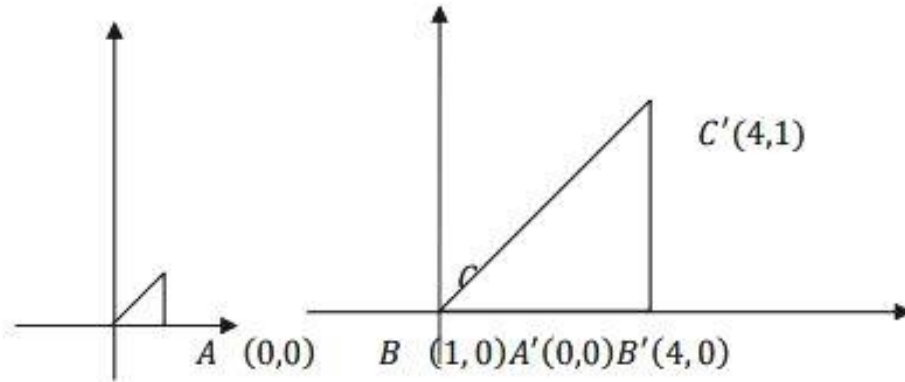


Fig (a) The original triangle ABC Fig(b) The Scaled triangle $A'B'C'$

Transformations

(ii) Shear

- A transformation that slants the shape of an object is called the shear transformation. Shearing transformation can also be carried out in both X, Y directions or only one the directions.
- The new coordinates after shearing in X direction are given by:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} 1 & sh_x \\ 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

and shear in Y-direction is given by:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} 1 & 0 \\ sh_y & 1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Transformations

Here, sh_x and sh_y are the shearing factors along the X- and Y- directions respectively and are given as inputs.

Then, in matrix form,

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} 1 & sh_x \\ sh_y & 1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Example:

Consider the triangle ABC with coordinates A = (0,0), B = (1,0) and C = (1,1). If the shearing factor along X-axis, $sh_x = 4$ and along Y-axis, $sh_y = 1$, then calculate the new coordinates of the sheared triangle ABC.

Transformations

In matrix form, the shear transformation will be

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Then,

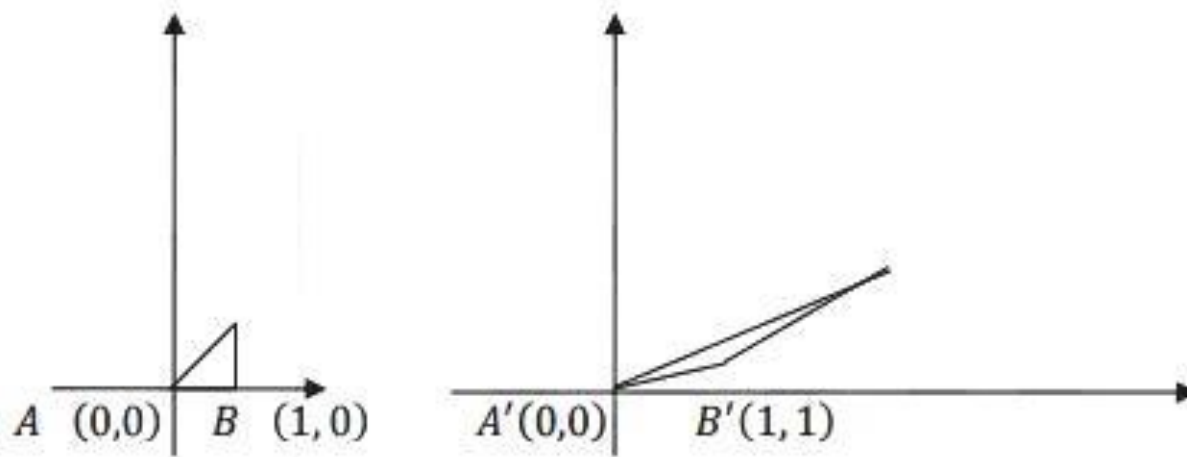
$$A' = \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$B' = \begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$C' = \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

Transformations

Therefore, the sheared triangle, $A'B'C'$ will have the coordinates $A' = (0, 0)$, $B' = (1, 1)$, and $C' = (5, 2)$



Transformations

General Affine Transformation

Using the homogeneous coordinates, a general affine transformation is defined of the form:

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{pmatrix}$$

where the sub-matrix need not be a rotation matrix. An affine transformation preserves parallelism but does not preserve angles.

Transformations

3. PROJECTIVE TRANSFORMATIONS

- Projective transformations are the most ‘general’ forms of linear transformations. It does not preserve angles, distances, therefore, parallelism also. Therefore, under the projective transformation, parallel lines do not remain parallel. The only thing that it preserves is collinearity of points and therefore straight lines remain straight.
- A projective transformation in the most generic form in homogeneous coordinate system is given as:
$$P = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & 1 \end{pmatrix}$$

Important Topics

Camera Models

Projections

Transformations

Summary

In this session, we discussed:

- An introduction to computer vision, the basic pinhole camera model.
- Homogeneous coordinates were introduced, which helps in defining the projection transformations in terms of matrices.
- Perspective projection, orthographic projection and weak perspective projection.
- Geometric transformations and Euclidean, Affine and Projective transformations and their representation in homogeneous coordinates.

*Thank
You*



Digital Image Processing and Computer Vision

Block-3_Unit-9

Single Camera

Topics to be Covered

Introduction

Camera Models

Perspective Projection

Homography

Camera Calibration

Affine motion models

Important Topics

Summary

Introduction

After studying this session, you should be able to:

- To learn about the camera model in detail,
- To understand the concept of camera matrix,
- To understand the process of camera calibration
- To understand the affine motion model.
- To give an overview of the aspects of computer vision related to a single camera and the estimation of 3D parameters from a single camera.

Camera Models

- There are various types of camera models, but here is one common one
- Pinhole camera model: The simplest camera model, which assumes that light travels in straight lines and enters the camera through a small hole or aperture. A pin-hole camera does a perspective projection of a 3D scene onto a 2D plane.

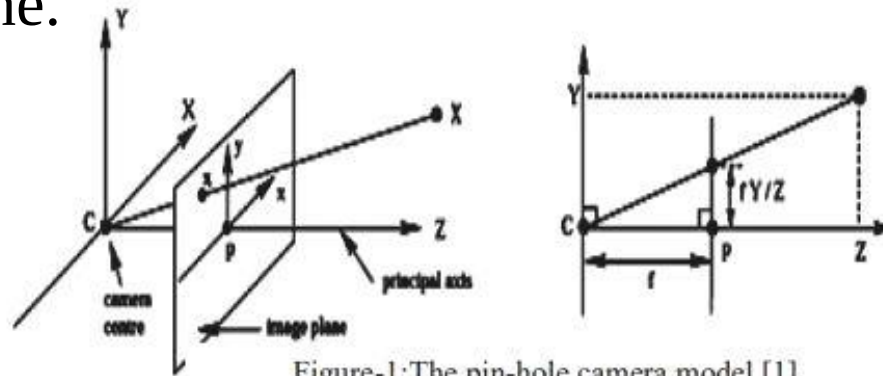


Figure-1: The pin-hole camera model [1].

Camera Models

- Under the pin-hole camera projection, a 3D point $X = (X, Y, Z)$ is mapped ($\sim$) to a 2D point $x = (x, y)$ on the image plane where, the line joining the camera centre C to the 3D point X intersects the image plane at x . Using the homogeneous coordinates, we represent $X = (X, Y, Z, 1)$ and $x = (x, y, 1)$ and therefore, the pin-hole camera model can be represented as

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \sim \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Camera Models

- The **internal or intrinsic parameters** of the camera define the relation between the pixel coordinates of a point on the image with the corresponding camera coordinates that exist in the camera reference frame.
- The **external/ extrinsic parameters** of the camera are the parameters that define the location and orientation of the camera coordinate frame with respect to a known world coordinate frame.
- Both the pixel coordinate system and the world coordinate systems are related by the following physical parameters: (a) size of pixels (b) position of principal point (c) focal length of the lens and (d) position and orientation of the camera.

Perspective Projection

- A camera projects the 3D world onto a 2D image plane. It is a perspective projection and is represented by a 3×4 matrix such that the left 3×3 sub matrix is non-singular.
- Let's see how the camera matrix encodes the camera parameters.

1. External/ Extrinsic parameters of a camera

- The external camera parameters define the relation between the known world coordinate frame and the unknown camera coordinate frame. The two coordinate systems are related by a rotation and translation. Therefore, determining the external parameters of the camera implies:

Perspective Projection

- (a) finding the translation vector between the relative positions of the origins of the camera coordinate frame and the world coordinate frame.
- (b) Finding the elements of the rotation matrix to align the corresponding axes of the two coordinate frames.

Therefore, the extrinsic camera parameters help us in finding the relation between the coordinates of a 3D point in the world coordinate system with its coordinates in the camera coordinate system.



Perspective Projection

2. Internal / Intrinsic parameters of a camera

- The intrinsic parameters of a camera are the internal characteristics of the camera that affect how it captures images.
- In general, the pin-hole camera model is a geometrical model, implying that all points and focal length are measured in millimeters or centimetres or meter. However, the camera coordinates are measured in pixel distance or height and width of pixels (sampling distance).
- Therefore, the internal camera parameters characterize the following:
(a) The perspective projection (focal length)

Perspective Projection

(b) The relation between pixel size and image plane coordinates

(c) The geometric distortions introduced by the optics

The relation between the camera coordinates and the image plane coordinates is given by the perspective projection:

$$x = f \frac{X_c}{Z_c} = f \frac{R_1^T(Aw - T)}{R_3^T(Aw - T)}; \quad y = f \frac{Y_c}{Z_c} = f \frac{R_2^T(Aw - T)}{R_3^T(Aw - T)}$$



Homography

- A planar homography is a transformation that relates two planes. Therefore, a homography relates two images of the same scene, i.e. homography maps an image from one view to another. Given points in one image and the homography matrix, H , we can find the corresponding points in the other image.
- A transformation or a mapping $h: P^2 \rightarrow P^2$ is a homography if and only if there exist a non-singular 3×3 matrix H such that for any point in $x \in P^2$, then $x' = h(x) = Hx$



Homography

An important condition for a transformation to be a homography is that it maps collinear points to collinear points, that is, if x_1 , x_2 and x_3 lie on a line then, $h(x_1)$, $h(x_2)$ and $h(x_3)$ also lie on a line.

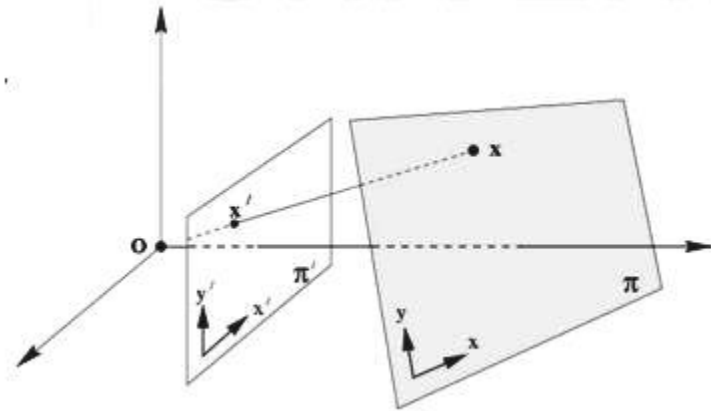


Fig. 2. The projection maps points on one plane to points on another plane. These points are related by a homography, H , such that $x' = Hx$.

Homography

Homography Estimation: Direct Linear Transformation Algorithm

- We can estimate the homography between two images of the same scene under the assumption that the point correspondences are given and that there is planar motion of the camera.
- We assume that $x'_i \leftrightarrow x_i$ are the given set of point correspondences. We need to compute the 3x3 homography matrix H .
- For each point correspondence, we have,

$$\begin{bmatrix} x'_i \\ y'_i \\ w'_i \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ w_i \end{bmatrix}$$

Homography

- Therefore, one pair of correspondences will give two independent equations. There are 8 unknowns, since H has 9 entries but is defined up to scale. In this case, we can divide each entry of H by h_{33} , and fix the last entry as 1. Therefore, H has 8 degrees of freedom. Each point correspondence gives 2 independent equations, therefore, to find the 8 unknowns we shall need at least 4 point correspondences.
- However, in practice, the point correspondences may not be free of noise, therefore a larger number of point correspondences are used to estimate the best solution.

Homography

Planar homography finds a wide range of applications such as in removing perspective distortion, creating panoramas, 3D reconstruction, etc.

For example:

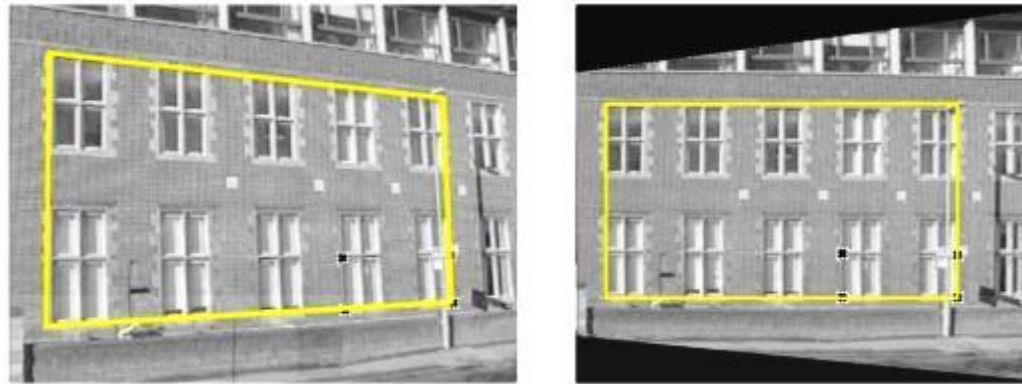


Fig. 3. The figure shows that 4 point correspondences suffice to remove the perspective distortion from a planar building façade.

Camera Calibration

- The aim of camera calibration is to estimate the intrinsic and extrinsic parameters of a camera. To compute the intrinsic and extrinsic camera parameters, we need to know a set of correspondences between the world points (X, Y, Z) and their projections on the image (x, y) .
- Therefore, the first step is to establish the set of correspondences between the world points and their projections on the image plane. To do so, in general, images of a known calibration object are used. The calibration object has a known 3D geometry and location in space. Moreover, it generates image points which can be accurately located.

Camera Calibration



Fig 4. Image of a calibration object

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Camera Calibration

- If we consider the object in Fig. 4, then it can be seen that there are equal spaced black squares on a white background, such that the corner points are clearly visible and can be extracted easily. Also, if we assume a point on this object to be the origin of the world coordinate system, and define the 3 axes, then we can find the coordinate of each corner point in 3D.
- Moreover, the projection of these corner points can be found in the image and therefore, we can establish a set of correspondences between 3D points and their corresponding image points by using a known calibration object.

Affine Motion Models

- An affine transformation is a geometric transformation that preserves lines and parallelism.
- In general, an affine projection is a combination of a linear mapping + translation in homogeneous coordinates. That is,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} T_x \\ T_y \end{bmatrix} = \mathbf{A}\mathbf{X} + \mathbf{b}$$

- where, $\mathbf{b}$ denotes the projection of the world origin.

Affine Motion Models

Affine camera:

- An affine camera is the camera whose projection matrix M has the last row of the form $(0,0,0,1)$. An important property of the affine camera is that it maps the points at infinity to points at infinity. Therefore, an affine camera is a camera at infinity, implying that the camera center lies on the plane at infinity. The affine camera preserves parallelism.
- As we calibrate a projective camera, similarly we can estimate an affine camera also. An affine camera matrix is a 3×4 projection matrix with the last row given as $(0,0,0,1)$. In general, the affine motion model is used for approximating the flow pattern of the camera motion in a video.

Important Topics

Perspective Projection

Homography

Camera Calibration

Summary

In this session, we discussed:

- The single camera geometry, including the camera model, perspective projection, the camera parameters, homography, camera calibration and the affine motion model.
- How image to image homography can be computed with the knowledge of sufficient number of corresponding points.
- The affine transformations, the affine camera, and the affine motion model.

*Thank
You*



Digital Image Processing and Computer Vision

Block-3_Unit-10

Multiple Camera

Topics to be Covered

Introduction

Stereo Vision

Point correspondences

Epipolar Geometry

Motion: Optical Flow

Important Topics

Summary

Introduction

After studying this session, you should be able to:

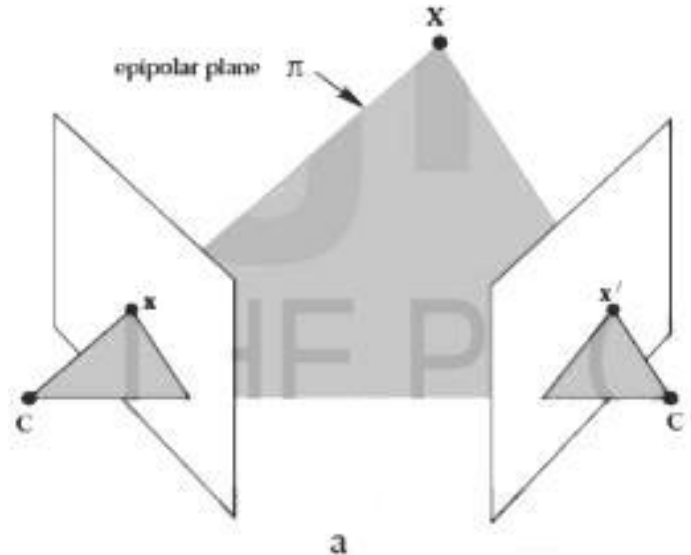
- Learn about the stereo vision system.
- Discuss the concept of point correspondences and epipolar geometry.
- Discuss the concept of motion and optical flow that allows a computer vision system to find the movement pattern in a video.

Stereo Vision

- A stereo vision system consists of two cameras such that both the cameras can capture the same scene, however with some disparity between the views. One stereo vision system that we can easily relate to are the two eyes that we have. The process of using two images of the scene captured by a stereo camera system to extract 3D information is known as stereo vision.
- In stereo vision, the 3D information is obtained by estimating the relative depth of the points in the scene. The corresponding points in the stereo pair are matched and a disparity map is created which helps in estimating the depth of the points.

Point Correspondences

Point correspondences refer to the identification of corresponding points in two or more images of the same scene.

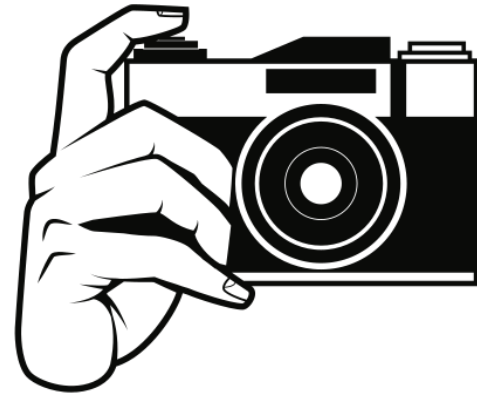


Point Correspondences

- Fig10.1 The figure shows the concept of a stereo pair. C and C' are the two camera centers and x and x' are the images of the 3D point X in the corresponding image planes. x and x' are said to be corresponding points.
- As shown in Figure 10.1, the stereo pair consist of two cameras that are at a certain distance apart. The line joining the camera centers is known as the baseline.
- The view volume of the two cameras is such that they view a common area in the scene. A 3D point that lies in the common view volume of the two cameras will be imaged by both the cameras.

Point Correspondences

- Therefore, as shown in Figure 10.1, the 3D point X is visible to both the cameras and therefore, it has an image x in Camera 1 (with camera center C) and image x' in Camera 2 (with camera center C').
- Therefore, x and x' are called corresponding points.
- Given a point in one image, we can find its corresponding point in the other image because of epipolar geometry.



Epipolar Geometry

Epipolar geometry describes the geometric relation between two views of the same scene captured by a stereo vision system.

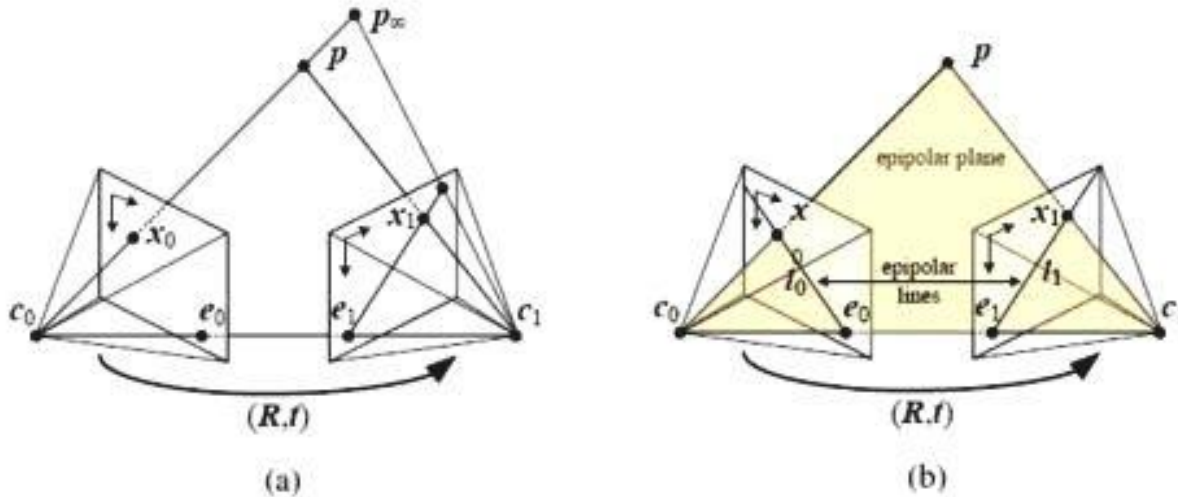


Figure 10.2. (a), (b) Epipolar geometry

Epipolar Geometry

- In Figure 10.2 (a), it can be seen that the two cameras (C_0 and C_1) are related by a rotation (R) and translation (t), implying that the relative pose of the cameras are an inherent part of epipolar geometry. The 3D point $\mathbf{p}$ is imaged in both the image planes I_0 and I_1 . In I_0 , the image of $\mathbf{p}$ is denoted by $\mathbf{x}_0$ and in I_1 , the image of $\mathbf{p}$ is denoted by $\mathbf{x}_1$. It can also be seen that the two camera centres, C_0 and C_1 and the 3D point $\mathbf{p}$ are coplanar. This plane Π is known as the Epipolar plane.
- Moreover, the ray joining C_0 and $\mathbf{x}_0$ extends towards infinity $\mathbf{p}_\infty$ and is intersected by the ray joining C_1 and $\mathbf{x}_1$ at the 3D point $\mathbf{p}$.

Epipolar Geometry

- Therefore, we can see that in the case of stereo vision, it is possible to find the 3D point if it is imaged by both the cameras and the point correspondences are known.
- The line joining the two camera centers C_0 and C_1 is known as the **baseline**. The baseline intersects the two image planes at e_0 and e_1 respectively. The point of intersection of the baseline with an image plane is known as an **epipole**. Therefore, e_0 is the left epipole and e_1 is the right epipole.

Epipolar Geometry

- The left epipole $\mathbf{e}_0$ is the image of the camera center C_0 in the left image plane I_0 while, the right epipole $\mathbf{e}_1$ is the image of the camera center C_1 in the right image plane I_1 . The epipolar line is the intersection of the epipolar plane with the image plane.
- An important point to be noted is that every epipolar line passes through the epipole, therefore, the point of intersection of all the epipolar lines is the epipole. Another point to be noted is that the epipolar line $\mathbf{l}_1$ in I_1 is the image of the back projected ray joining the camera center C_0 with the 3D point $\mathbf{p}$, while the epipolar line $\mathbf{l}_0$ in I_0 is the image of the back projected ray, that is the ray from the camera center C_1 with the 3D point $\mathbf{p}$.

Motion: Optical Flow

- Motion perception is very strong in the human vision system. Motion or perception of motion during the imaging process can be caused by some of the following reasons:
 1. The camera is static but the object is moving.
 2. The camera is moving but the object is static.
 3. Both the camera and objects are moving
 4. The camera is static but light source and objects are moving.

Motion: Optical Flow

- Motion perception plays an important role in applications of computer vision such as activity recognition, surveillance, 3D reconstruction, and many others.
- Therefore, it is important to estimate the motion from images.

Optical Flow

- Estimating motion of the pixels in a sequence of images or videos has a very large number of applications. Optical flow is used to compute where the motion information between sequence of images. It helps to determine a dense point to point correspondence between pixels of an image at time t with the pixels in the image at time $t+1$.

Motion: Optical Flow

- Therefore, optical flow is said to compute the motion in a video or sequence of images.
- Optical flow is based on the assumption that across consecutive frames, the pixel intensities do not rapidly change. It also assumes that neighbouring pixels have similar pattern of motion. Most often, it also assumes that the luminance remains constant.
- There are various methods for solving for optical flow. The Lucas and Kanade optical flow algorithm [3] and Horn and Shunck [4] optical flow methods are two of the most popular methods for estimating the optical flow.

Important Topics

Stereo Vision

Epipolar Geometry

Motion: Optical Flow

Summary

In this session, we discussed:

- About the stereo vision system.
- The concept of point correspondences and epipolar geometry.
- The concept of motion and optical flow that allows a computer vision system to find the movement pattern in a video.

*Thank
You*



Digital Image Processing and Computer Vision

Block-4_Unit-11_Part-3

Object Detection

Topics to be Covered

Introduction

Region Detection

Boundary Detection

Feature Extraction

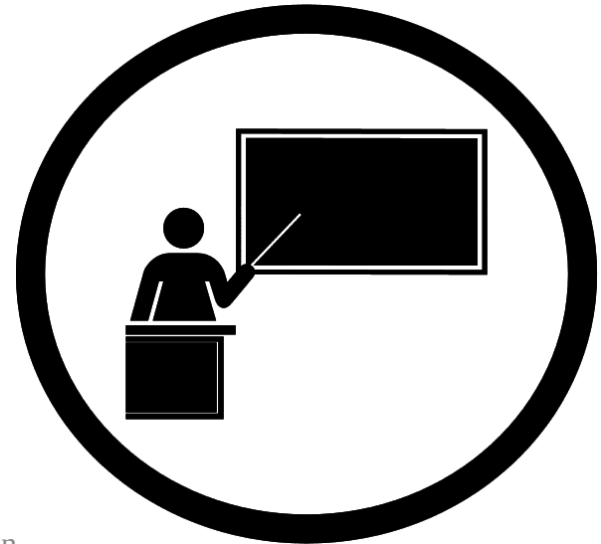
Important Topics

Summary

Introduction

After studying this session, you should be able to:

- Apply region based segmentation and boundary detection.
- Apply Feature Extraction.



Region Detection

- Region detection is an important aspect in image processing tool for semantically significant spatial information in images. Matching establishes similarity between visual entities, which is crucial for recognition.
- In a picture, a region is a collection of connected pixels with comparable features.
- Region-based segmentation and boundary estimation using edge detection are two methods for splitting a picture into areas.

Region Detection

- Further, in boundary detection, semantic boundaries are considered to find different objects or sections of an image. It is different from edge detection because it does not use boundaries between light and dark pixels in an image.
- The brief discussion on Region-based segmentation, boundary estimation, and boundary detection is given below:
 - a) Region based segmentation:** In region-based segmentation, all pixels of an image belonging to a same area are grouped and labeled together.

Region Detection

- Here, pixels are assigned to areas based on unique characteristic feature which is different from other part of image. Value similarity and spatial closeness are two important features of this segmentation process.
- If two pixels are very close to one another and have similar intensity characteristics, they may be allocated to the same region.
- For example, the similar grey values can represent similar pixels and Euclidean distance can represent the closeness of the pixels



Region Detection

b) Boundary estimation method: In this method, segmentation is done by finding the pixels (called edges) lying on a region boundary. The difference between neighboring pixels can be used to determine the region boundary as regions on either side of the boundary may have different grey levels.

- A large number of edge detectors rely on intensity characteristics to detect edges instead of using derived properties such as texture and motion.



Region Detection



Figure 31 Boundary identified by grouping similar pixels

Region Detection

- c) Boundary Detection:** In boundary detection, semantic boundaries are different from edge detection because it does not use boundaries between light and dark pixels in an image. A zebra, for example, has several internal boundaries between black and white stripes that humans would not consider part of the zebra's boundary.
- The focus is more on approximate boundary detection method using training data because a perfect solution requires high-level semantic knowledge about the scene in the image.



Region Detection

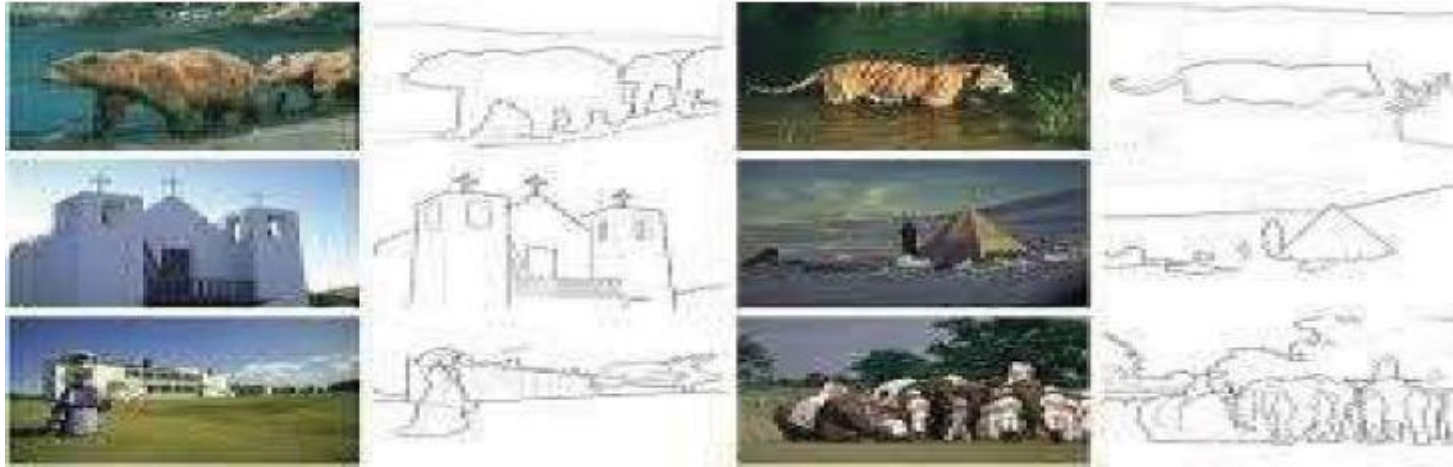


Figure 32

Region Detection

Region based segmentation methods employ two basic steps:

- a) Merging
- b) Splitting

Image segmentation using **merging** has the following steps

Step 1: Obtain the initial segmentation of the image.

Step 2: Merge two adjacent segments to form a single segment if they are similar in same way

Step 3: Repeat step 2 until no segment to be merged remains.

Region Detection

The initial segmentation can be all individual pixels. The basic idea is to combine two pixels (regions) if they are similar. The similarity criteria can be based on grey level similarity, texture of the segment etc.

Image segmentation using **splitting** has following steps

Step 1: Obtain an initial segmentation of an image.

Step 2: Split each segment that is inhomogeneous in some way

Step 3: Repeat step 2 until all segments are homogeneous.

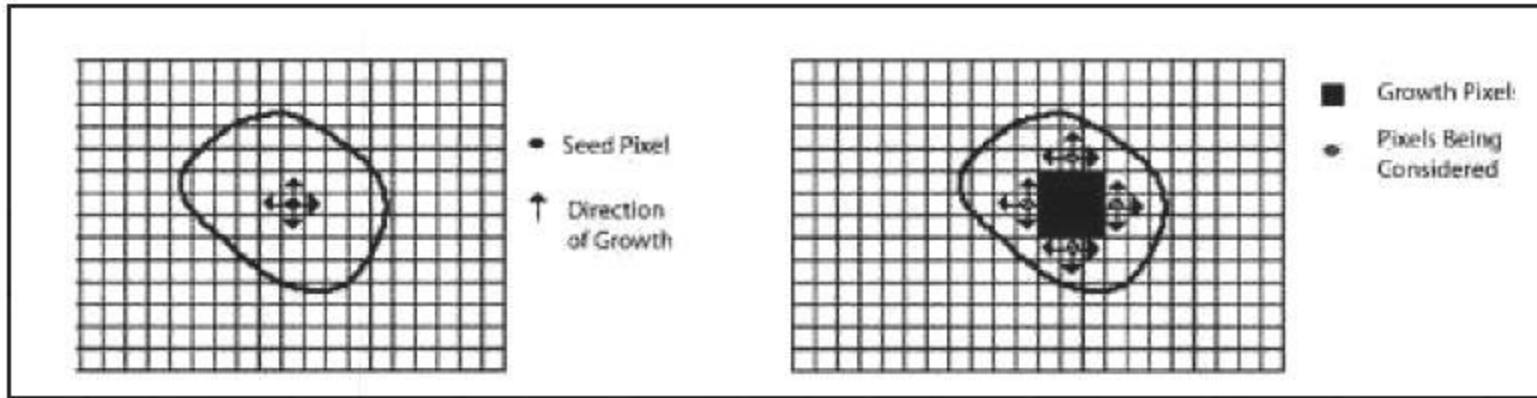
Here, initial segmentation may be the entire image (no segmentation).

Region Detection

Region growing

- Region growing is a process of merging adjacent pixel segments into one segment. It is one of the simplest and very popular method of segmentation which is used in many applications. It needs a set of starting pixels called '**seed**' points.
- The process consists of picking a seed from the set and examining all 4 or 8 connected neighbours of this seed and merging similar neighbours to the seed as shown in Fig.33(a). The seed point is modified based on all merged neighbours Fig.33(b). The algorithm continuous until the seed set is empty.

Region Detection



(a) Start of Growing a Region

(b) Growing Process After a Few Iterations

Fig. 33: Region Growing

Region Detection

- **Region growing algorithm** for a single seed point and same grey level value as similarity measure to merge the pixels is as follows:
- The algorithm uses a data structure entity stack to keep track of seed points. Two operations push and pop are used here. Push: puts the pixel co-ordinates on the top of the stack. Pop: takes a pixel from the top of the stack. Input to the algorithm is an image 'f'. Initial seed has coordinates of (x, y) and grey level value g at (x, y) , $f(x, y) = g$. The goal is to grow a region with all pixels having gray level value of g and assigning them grey level $k=1$, $k \neq 1$, $k \neq g$. Let (x, y) be the coordinates of initial seed, and let (a, b) be the coordinates of pixel under investigation.

Region Detection

The algorithm

Push (x,y)

Do till stack is not empty

Pop (a, b) /(take input point from top of stack)

If $f(a, b) = g$ / if input = desired value g

Set $f(a, b) = 1$ / segmented pixel is assigned as 1/
End

Push (a, b+1) / Test all four neighbors of (a,b)

Push (a, b-1) / Test all four neighbors of (a,b) by

pushing it on the top of the stack/

Push (a+1, b)

Push (a-1,b)

Region Detection

There are several issues about region growing.

- 1) Suitable selection of seed points is important. Seed point selection is user dependent. Histogram and image properties can be taken into account by the user to select the seed point.
- 2) Selection criteria for similarity measure is important. It generally depends on the original image and object to be segmented. Band of intensity values, color, texture, shape are some of the similarity measures generally used.

Region Detection

3) Formulation of a stopping rule is also very important in region growing. Growing of a region should stop when no more pixel satisfy the similarity criteria. Algorithm can be made more powerful, if size and shape of the grown region are also considered.

Main advantage of region growing algorithm is that it correctly separates the regions based on the property defined by the user. The algorithm is very simple to implement. Only input needed are the seed points and selection criterion. Multiple criteria can also be applied. The algorithm works well in noisy environment also.

Boundary Detection

A region may be described by its boundary in terms of features such as its length, the orientation of the straight line joining its extreme points and the number of concavity in the boundary. Many algorithms require the points in the boundary of a region in an ordered clockwise (or anti clockwise) direction. For boundary detection or following or tracking, the following assumptions are made:

- 1) The image is binary where 1=foreground and 0=background
- 2) The image is padded with a border of 0' s so an object cannot merge with the border.

Boundary Detection

3) We limit the discussion to single regions. The extension is straightforward.

Moore Boundary Tracking Algorithm:

Given a binary region R or its boundary.

Step 1: Let the starting point b_0 , be the uppermost, leftmost point in the image labeled 1.

Step 2: Denote by c_0 the west neighbour of b_0 . c_0 is always a background point.

Boundary Detection

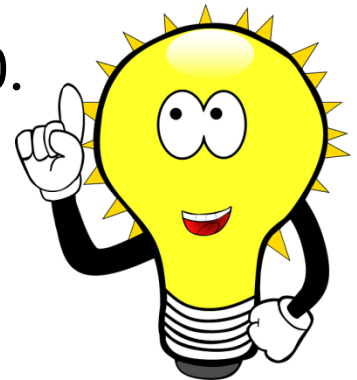
Step 3: Examine the 8-neighbours of b_0 , starting at c_0 and proceeding in a clockwise direction.

Step 4: Let b_1 denote the first neighbour encountered whose value is 1.

Step 5: Let c_1 denote the background point immediately preceding b_1 in the sequence.

Step 6: Store the locations of b_0 and b_1 for use in Step 10.

Step 7: Let $b = b_1$ and $c = c_1$.



Boundary Detection

Step 8: Let the 8-neighbours of b starting at c and proceeding clockwise be denoted as $n_1, n_2, \dots, n_8$. Find the first n_k which is foreground (i.e., a “1”).

Step 9: Let $b = n_k$ and $c = n_{k-1}$

Step 10: Repeat steps 8 and 9 until $b=b_0$, that is, we have reached the first point and the next boundary point found is b_1 .



Boundary Detection

The sequence of b points found when the algorithm stops is the set of ordered boundary points.

		1	1	1	1	
	1			1		
		1		1		
	1			1		
	1	1	1	1		

(a) Image

	c_0	b_0	1	1	1	
	1			1		
		1		1		
	1			1		
	1	1	1	1		

(b) Step 1 and Step 2

Boundary Detection

			c			
			b	1	1	
	1			1		
		1		1		
	1			1		
	1	1	1	1		

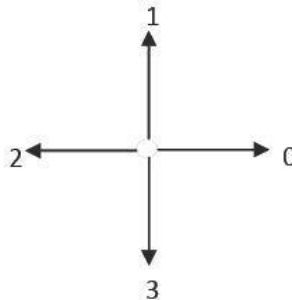
(c) Step 3 to Step 9

(d) Step 10

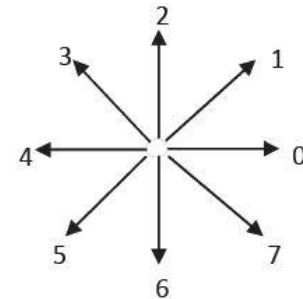
Fig.40: Illustration in Boundary Following Algorithm.

Boundary Detection

Chain codes: Chain codes are used to represent a boundary by a connected sequence of straight line segments of specified length and direction. Freeman codes [1961] represent a boundary by the sequence of straight line segments of specified length and direction. The direction is coded by a numbering scheme (4 or 8- connectivity) as shown in Fig.42



(a) 4-direction chain code



(b) 8-direction chain code

Fig.42: Direction Numbers

Boundary Detection

Example 4: Find chain code and first difference of the following boundary shapes.

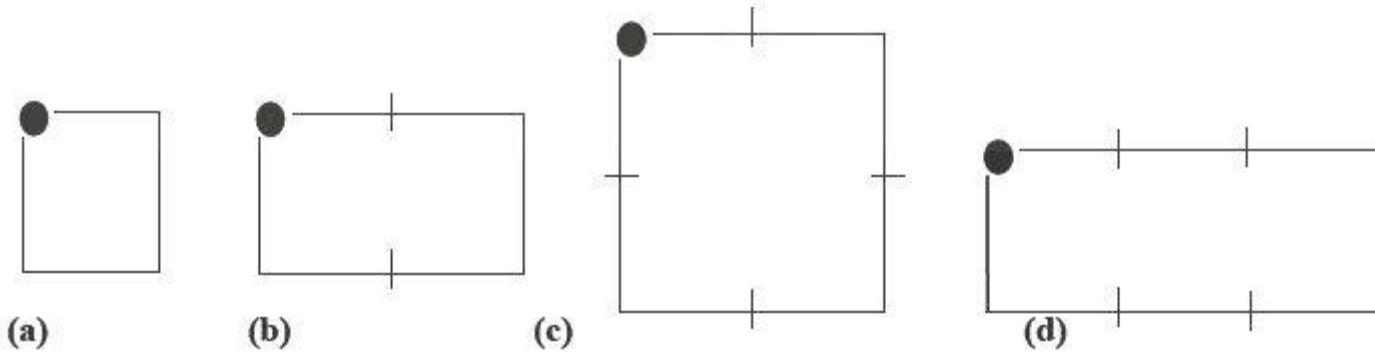
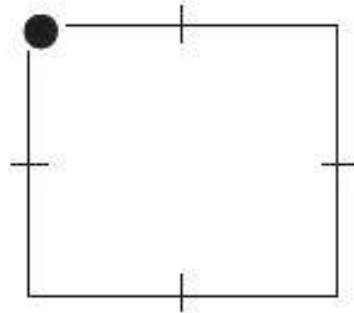
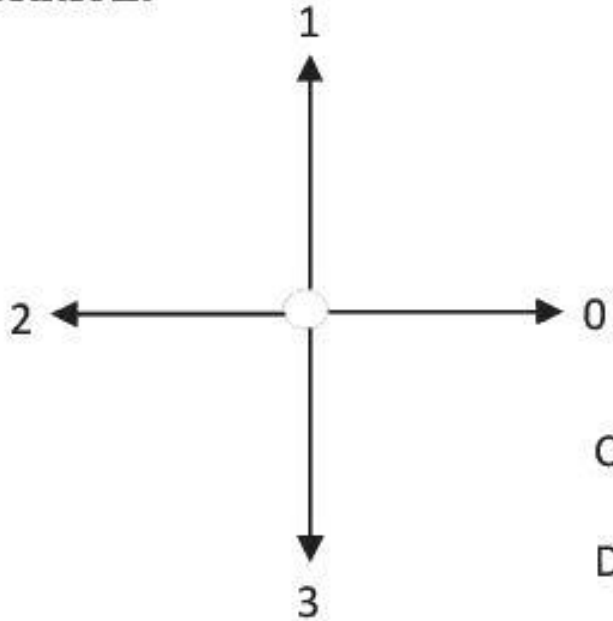


Fig. 44

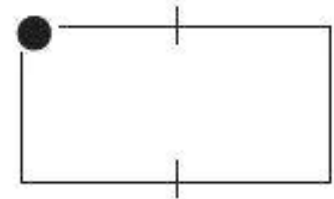
Boundary Detection

Solution:



Chain code: 0 0 3 3 2 2 1 1

Difference: 3 0 3 0 3 0 3 0



0 0 3 2 2 1

3 0 3 3 0 3



Boundary Detection

Chain code: 0 0 3 2 2 1

0 0 0 3 2 2 2 1

Difference: 3 0 3 3 0 3

3 0 0 3 3 0 0 3

4-directional codes are used in this example. First differences are computed by treating the chain as a circular sequence.



Feature Extraction

- Feature extraction divides a large set of data into smaller groups for quick processing. There are a large number of variables in these huge data sets which require a large amount of processing power. Feature extraction extracts the best feature from by selecting and combining variables into features.

Applications of Feature Extraction

- Image Processing, Auto-encoders and Bag of words are some applications of Feature Extraction.

Feature Extraction

1. Image Processing: In image processing, we experiment with images using different techniques to comprehend them better.
2. Auto-encoders: Auto-encoders do efficient unsupervised data coding. As a result, the feature extraction technique may be used to discover significant features from data to code by learning from original data set and generating new ones.
3. Bag of Words: Feature extraction is an important part of this process. This technique is widely used for natural language processing (NLP). Here, words/features are extracted from a document, website or sentence and are classified according to their frequency of use.

Feature Extraction

Traditional methods of feature detection

Traditional methods of feature detection include the following:

1. Harris Corner Detection- It detects corners based on differential corner score with reference to direction directly.
2. SIFT (Scale-Invariant Feature Transform)- Generally used for invariance.
3. SURF (Speeded-Up Robust Features)- This technique is a simplified variant of SIFT.

Feature Extraction

4. FAST (Features from Accelerated Segment Test)- In comparison to SURF, this is a substantially faster corner detecting algorithm.
5. BRIEF (Binary Robust Independent Elementary Features)- This feature descriptor can be used with any other feature detector. By converting floating point integers to binary strings, this approach minimizes memory utilization.
6. Oriented FAST and Rotated BRIEF (ORB) —This Open CV algorithm uses FAST key-point detector and BRIEF descriptor. It is an alternative to SIFT and SURF.

Feature Extraction

Deep Learning Techniques for feature extraction

- Convolutional neural network (CNN) can replace Traditional feature extractors because of their strong ability and efficiency to extract complex features for expressing more detailed part of an image and can learn task specific features.
1. Super Point: It detects points of interest and descriptors using Fully CNN. The extracted features are encoded in VGG style and then using two decoders, it generates inters points and descriptors. The Super Point structure is an example of a fully connected-convolutional neural network architecture.

Feature Extraction

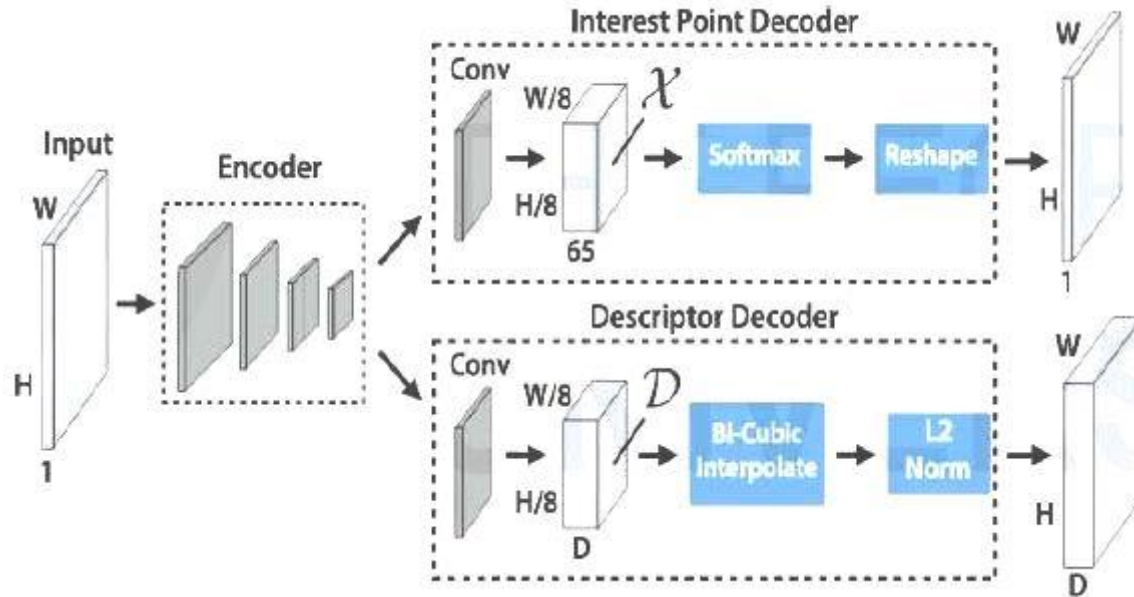


Figure 45 Super Point structure

Feature Extraction

2. D2-Net: It is a trainable CNN based local feature detector and dense feature descriptor(feature descriptor has minimum non zero values)

It's a fully convolutional neural network (FCNN) used for extracting feature maps with a double purpose:

- i) Obtaining the local descriptors d_{ij} at a given spatial position (i,j) is as easy as traversing all the n feature maps D_k ;
- ii) Keypoint detection scores s_{ij} are calculated during training using a soft local-maximum score and a ratio-to-maximum score for each descriptor, and detections are generated by executing a non-local-maximum suppression on a feature map.

Feature Extraction

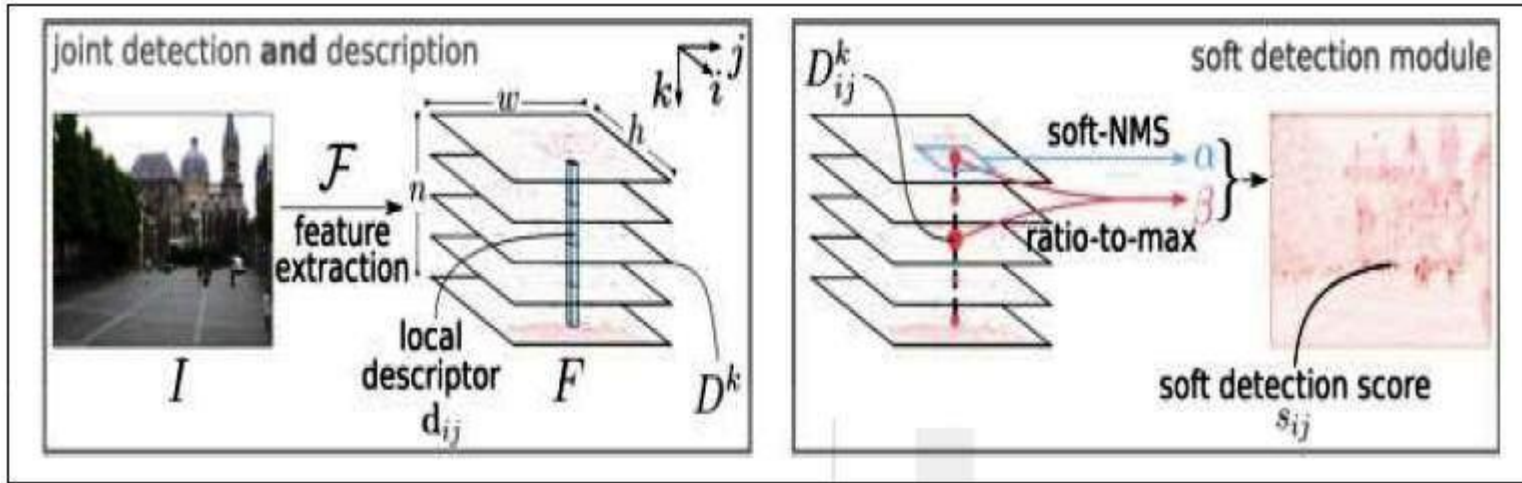


Figure 46 Detect and Describe D2 network

Feature Extraction

3. LF-Net: This approach uses training image pairs with relative pose and depth maps.
- It can be thought of as both a feature detector and a dense feature description at the same time. The key points that are obtained using this method are more stable than their traditional equivalents, which are based on the early detection of low-level structures. This is achieved by delaying the detection until a later stage. We demonstrate that pixel data can be used to train this model.

Feature Extraction

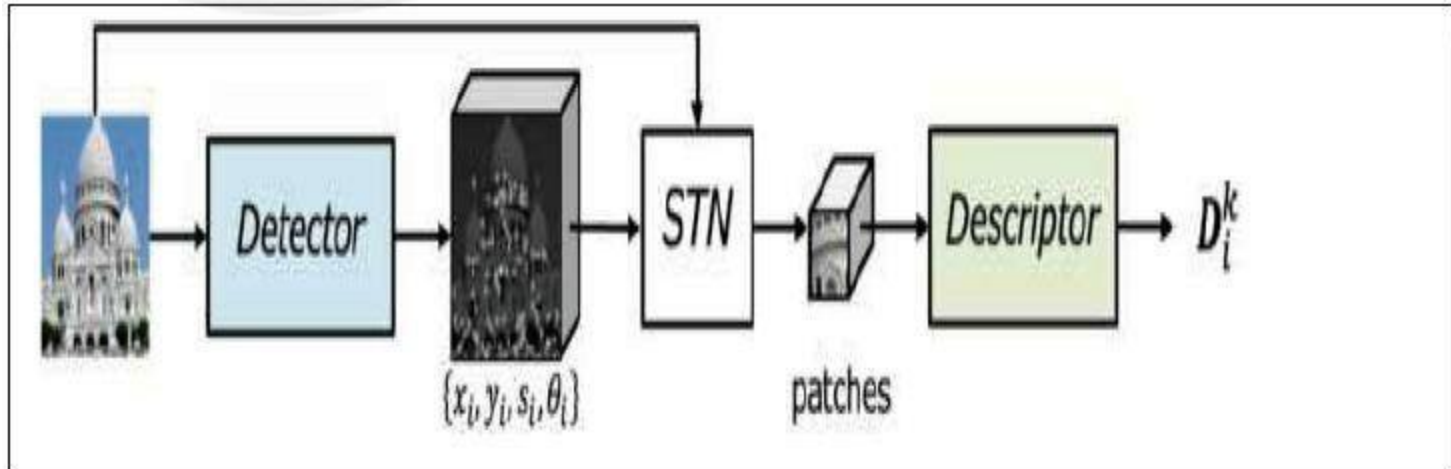


Figure 47: LF-Net

Important Topics

Region Detection

Boundary Detection

Feature Extraction

Summary

In this session, we discussed

- Various region based segmentation techniques.
- Boundary detection algorithm.

*Thank
You*



Digital Image Processing and Computer Vision

Block-4_Unit-12_Part-2

Object Recognition Using Supervised Learning Approaches

Topics to be Covered

Introduction

Minimum Distance Classifiers

Machine Learning Algorithms

Supervised Learning Approach

Important Topics

Summary

Introduction

After studying this session, you should be able to:

- Find minimum distance classifiers.
- Apply machine learning algorithm.
- Describe supervised learning approach.
- Describe unsupervised learning approach.

Minimum Distance Classifiers

- Minimum distance classifier is a pattern classification scheme defined by distance functions. Here pixels which are close to each other in feature space are likely to belong to the same class.
- Distance in feature space is the primary measure of similarity in minimum distance classifier algorithms. Pixels that are "close" in feature space will be grouped in the same class.
- Distance in feature space is the primary measure of similarity in all clustering algorithms.

Minimum Distance Classifiers

1) Euclidean distance:

$$d_i(x) = -D_i(x) = -[(x-z_i)^T (x-z_i)]^{1/2}$$

2) Square of the Euclidean distance:

$$d_i^2(x) = -D_i^2(x) = -[(x-z_i)^T (x-z_i)]$$

3) Square of Euclidean distance after eliminating components that are independent of class:

$$d_i(x) = -[(x^T z_i)^T 1/2(z_i^T z_i)]$$

4) Taxicab distance:

$$d_i(x) = -D_i(x) = -\sum_{k=1}^N |X_k - Z_{ki}|$$

Minimum Distance Classifiers

The **decision boundary** for the single prototype, simple distance discriminant function is the set of planar surfaces perpendicular to and bisecting the lines connecting pairs of prototypes as shown in Fig. 1. This is a minimum-distance classifier. If the prototype is the mean value of the training pixels for each class, it is called a minimum-distance to mean classifier.

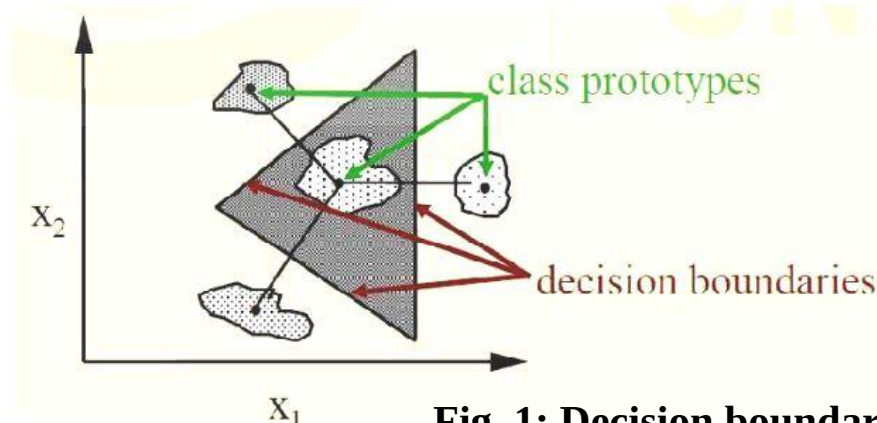


Fig. 1: Decision boundary

Minimum Distance Classifiers

There are two main types of minimum distance classifiers:

The Euclidean Distance Classifier

The Euclidean distance, $D_i(x)$, of a measurement vector, x , from the prototype vector, z_i :

$$D_i(x) = |x - z_i| = [(x - z_i)^T (x - z_i)]^{1/2}$$

The discriminant function is usually defined as the negative of the separation distance:

$$d_i(x) = -D_i(x)$$

Minimum Distance Classifiers

The larger (less negative) $d_i(x)$, the closer the measurement vector lies relative to the prototype vector z_i . The maximum value of $d_i(x)$ is zero and occurs when x matches the prototype vector exactly.

Algorithm

Step 1: Select a threshold, T . T is a representative distance in measurement space. The choice of T in this algorithm is entirely arbitrary; it is also the only input required of the user.

Step 2: Select a pixel with measurement vector, x . The selection scheme is arbitrary. Pixels could be selected at random.

Minimum Distance Classifiers

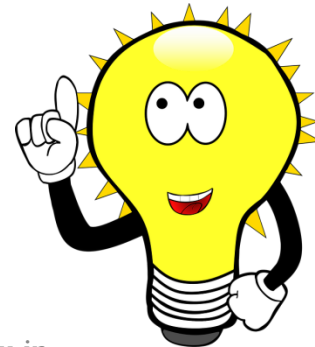
Step 3: Let the first pixel be taken as the first cluster center, z_1 .

Step 4: Select the next pixel from the image.

Step 5: Compute the distance functions, $D_i(x)$. Compute the distance function for each of the classes established at this point, i.e., compute $D_i(x)$, for $i = 1, \dots, N$ where N = the number of classes. ($N = 1$ initially.)

Step 6: Compare the $D_i(x)$ with T .

a) if $D_i(x) < T$, then $x \in w_i$.



Minimum Distance Classifiers

b) if $D_i(x) < T$, for all i , then let x become a new prototype vector: Assign $x \rightarrow Z_{N+1}$. (Do not compute D_{N+1} for pixels already assigned to an existing class.)

Step 7: Return to step #4 until all pixels are assigned to a class.

Step 8: After all pixels have been assigned to a cluster center, re-compute the $D_i(x)$ and reassign pixels according to the minimum $D_i(x)$.



Minimum Distance Classifiers

The Mahalanobis Distance Classifier

- In many applications, the range of all feature value may differ widely. One could be in hundreds while the other could be in decimal fractions.
- If this issue is overlooked some feature values will get neglected. If one relaxes the assumptions required by the Euclidean classifier and removes the last one, the one requiring the covariance matrix to be diagonal and with equal elements, the optimal Bayesian classifier takes the form of minimum Mahalanobis distance classifier.

Minimum Distance Classifiers

That is, given an unknown x , it is assigned to class ω_i if

$$\sqrt{(x - m_i)^T S^{-1} (x - m_i)} < \sqrt{(x - m_j)^T S^{-1} (x - m_j)}, \forall_j \neq i,$$

where S is the common covariance matrix. The presence of the covariance matrix accounts for the shape of the Gaussians distributions of various features.



Machine Learning Algorithms

In order to understand how Machine learning works, we need some essential ML vocabulary first:

- **Parameter:** A value that is learnt during the training of a machine learning program. The inference algorithm uses the parameters to calculate results.
- **Model:** A trained machine learning method. A model is essentially a (sometimes very large) set of parameters and instructions on how to use them – a ready-to-run ML procedure and the result of the training process. The training algorithm generates the model, the inference algorithm uses it to perform a task.

Machine Learning Algorithms

- **Inference:** Running a model using a second algorithm. This algorithm uses the model to create a prediction based on an input, classify an input or create some interesting value based on an input. Hence the term “inference algorithm” or “prediction algorithm”. Inference is a technical term for “execute the ML model and output a result”.
- Machine learning use the combination of a training algorithm and a prediction (or inference) algorithm. The training algorithm uses data to gradually determine parameters.
- The inference algorithm then uses the model and applies it to any given data. Finally, it delivers the desired results.

Machine Learning Algorithms

Execution of a machine learning project:

- We select the machine learning method for which we want to train a model.
- Then we divide the available data into two parts: The training data and the test data. We apply our training data and thus obtain our model. The model is checked on the unknown test data.
- We use the training data to develop the model with the learning algorithm. The more data we have, the “stronger” the model becomes. Using up all available data for the training algorithm is called an “epoch”.

Machine Learning Algorithms

- In order to test it, the trained model is used on the test data unknown to it and makes predictions. If we did everything right, the predictions on unknown data should be as good as on the training data – the model can generalize and solve the problem. Now it is ready for practical use.
- At its most basic, machine learning uses programmed algorithms that receive and analyse input data to predict output values within an acceptable range. As new data is fed to these algorithms, they learn and optimise their operations to improve performance, developing ‘intelligence’ over time.

Machine Learning Algorithms

Machine learning algorithms are organized into taxonomy, based on the desired outcome of the algorithm. Common algorithm type includes:

- **Supervised learning** --- the algorithm generates a function that maps inputs to desired outputs. One standard formulation of the supervised learning task is the classification problem: the learner is required to learn (to approximate the behaviour of) a function which maps a vector into one of several classes by looking at several input-output examples of the function.

Supervised Learning Approach

- Supervised learning is fairly common in classification problems because the goal is often to get the computer to learn a classification system that we have created. Digit recognition, once again, is a common example of classification learning.
- In some cases, it might not even be necessary to give predetermined classifications to every instance of a problem if the agent can work out the classifications for itself. This would be an example of unsupervised learning in a classification context.

Supervised Learning Approach

- Supervised learning often leaves the probability for inputs undefined. This model is not needed as long as the inputs are available, but if some of the input values are missing, it is not possible to infer anything about the outputs.
- Unsupervised learning, all the observations are assumed to be caused by latent variables, that is, the observations is assumed to be at the end of the causal chain. Examples of supervised learning and unsupervised learning are shown in the Fig. 2.

Supervised Learning Approach

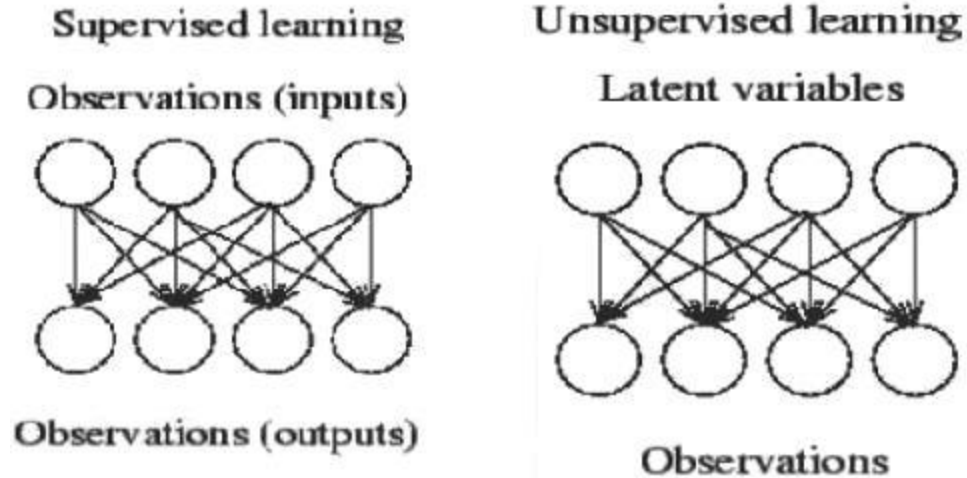


Fig. 2: Examples of Supervised and Unsupervised Learning

Supervised Learning Approach

- Supervised learning is the most common technique for training neural networks and decision trees. Both of these techniques are highly dependent on the information given by the pre-determined classifications.
- In the case of neural networks, the classification is used to determine the error of the network and then adjust the network to minimize it, and in decision trees, the classifications are used to determine what attributes provide the most information that can be used to solve the classification puzzle. Both of these examples have some "supervision" in the form of pre-determined classifications.

Supervised Learning Approach

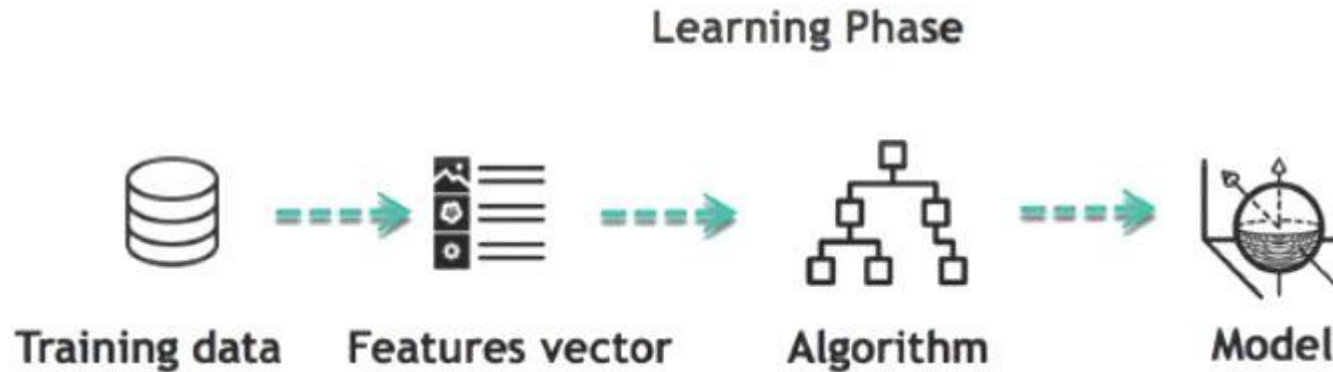


Fig. 3: Learning phase of a supervised learning algorithm

Supervised Learning Approach

Inductive machine learning is the process of learning a set of rules from instances (examples in a training set), or more generally speaking, creating a classifier that can be used to generalize from new instances. The process of applying supervised ML to a real world problem is described below

Step 1 (Collect the dataset): If a requisite expert is available, then s/he could suggest which fields (attributes, features) are the most informative. If not, then the simplest method is that of measuring everything available in the hope that the right (informative, relevant) features can be isolated.

Supervised Learning Approach

Step 2 (Data preparation and data pre-processing): Depending on the circumstances, there are a number of methods to choose from to handle missing data, outlier (noise) detection. There is a variety of procedures for sampling instances from a large dataset.

Step 3 (Define a training set): The goal of the learning algorithm is to minimize the error with respect to the given inputs. These inputs, often called the "training set", are the examples from which the agent tries to learn.

Step 4: Then we need to select a suitable algorithm and perform training and evaluate the results.

Supervised Learning Approach

Classification Models are used to predict the category of the data. Real-life examples include spam detection, sentiment analysis, scorecard prediction of exams, etc.

2) Regression Models — Regression models are used for problems where the output variable is a real value such as a unique number, dollars, salary, weight or pressure, for example. It is most often used to predict numerical values based on previous data observations. Some of the more familiar regression algorithms include linear regression, logistic regression, polynomial regression, and ridge regression.

Supervised Learning Approach

Step 5: If we are satisfied with the results, classifier is trained else parameter tuning is done which includes updating of data preprocessing and training set.

Supervised Machine Learning Categorization

It is important to remember that all supervised learning algorithms are essentially complex algorithms, categorized as either classification or regression models as shown in Fig. 4.

1) Classification Models — Classification models are used for problems where the output variable can be categorized, such as “Yes” or “No”, or “Pass” or “Fail.”

Supervised Learning Approach

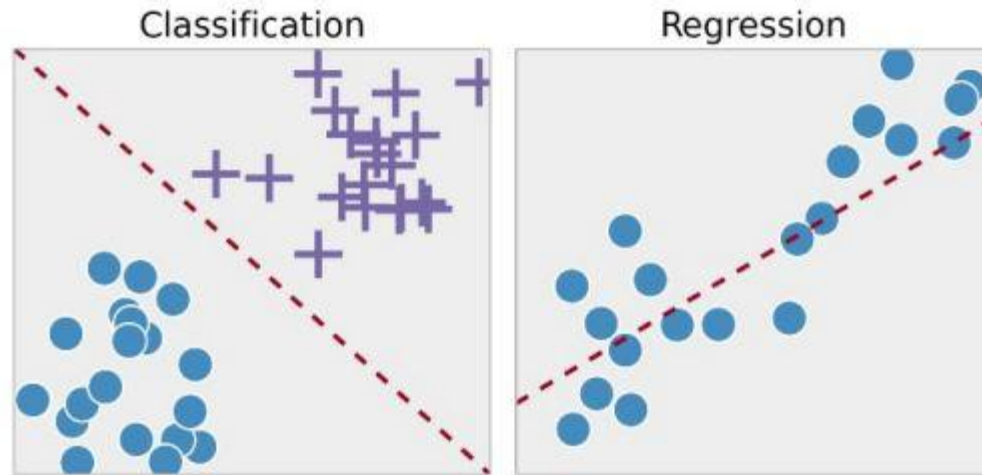


Fig. 4: Categories of Supervised Machine Learning algorithms

Supervised Learning Approach

There are some very practical applications of supervised learning algorithms in real life, including:

- i. Text categorization
- ii. Face Detection
- iii. Signature recognition
- iv. Customer discovery
- v. Spam detection
- vi. Weather forecasting
- vii. Predicting housing prices based on the prevailing market price
- viii. Stock price predictions, among others

Supervised Learning Approach

Parameters	Supervised machine learning technique	Unsupervised machine learning technique
Process	In a supervised learning model, input and output variables will be given.	In unsupervised learning model, only input data will be given.
Input Data	Algorithms are trained using labeled data.	Algorithms are used against data which is not labeled
Algorithms Used	Support vector machine, Neural network, Linear and logistics regression, random forest, and Classification trees.	Unsupervised algorithms can be divided into different categories: like Cluster algorithms, Kmeans, Hierarchical clustering, etc.

Supervised Learning Approach

Parameters	Supervised machine learning technique	Unsupervised machine learning technique
Computational Complexity	Supervised learning is a simpler method.	Unsupervised learning is computationally complex
Use of Data	Supervised learning model uses training data to learn a link between the input and the outputs.	Unsupervised learning does not use output data.
Accuracy of Results	Highly accurate and trustworthy method.	Less accurate and trustworthy method.

Supervised Learning Approach

Parameters	Supervised machine learning technique	Unsupervised machine learning technique
Real Time Learning	Learning method takes place offline.	Learning method takes place in real time.
Number of Classes	Number of classes is Known.	Number of classes is not known.
Main Drawback	Classifying big data can be a real challenge in Supervised Learning.	You cannot get precise information regarding data sorting, and the output as data used in unsupervised learning is labeled and not known.

Supervised Learning Approach

The following are the various situations, where supervised or unsupervised learning approaches are used.

- In Supervised learning, the machine is trained using data which is well "labeled."
- Unsupervised learning is a machine learning technique, where you do not need to supervise the model.
- Supervised learning allows you to collect data or produce a data output from the previous experience.
- Unsupervised machine learning helps you to find all kind of unknown patterns in data.

Supervised Learning Approach

- For example, you will be able to determine the time taken to reach back home based on weather condition, Times of the day and holiday.
- For example, Baby can identify other dogs based on past supervised learning.
- Regression and Classification are two types of supervised machine learning techniques.
- Clustering and Association are two types of Unsupervised learning.
- In a supervised learning model, input and output variables will be given while with unsupervised learning model, only input data will be given.

Supervised Learning Approach

- In manufacturing, a large number of factors affect which machine learning approach is best for any given task. And, since every machine learning problem is different, deciding on which technique to use is a complex process.
- In general, a good strategy for honing in on the right machine learning approach is to:
- **Evaluate the data.** Is it labeled/unlabelled? Is there available expert knowledge to support additional labeling? This will help to determine whether a supervised, unsupervised, semi-supervised or reinforced learning approach should be used

Supervised Learning Approach

- **Define the goal.** Is the problem recurring, defined one? Or, will the algorithm be expected to predict new problems?
- **Review available algorithms** that may suit the problem with regards to dimensionality (number of features, attributes or characteristics). Candidate algorithms should be suited to the overall volume of data and its structure.
- **Study successful applications** of the algorithm type on similar problems.

Important Topics

Minimum Distance
Classifiers

Supervised Learning
Approach

Summary

In this session, we discussed:

- Concept of minimum distance classifier.
- Concept of Classification.
- Supervised Learning.
- Unsupervised Learning.
- Differences between Supervised Learning/ Unsupervised Learning.

*Thank
You*



Digital Image Processing and Computer Vision

Block-4_Unit-13

Object Classification Using Unsupervised Learning

Topics to be Covered

Introduction

Introduction to Clustering

Major Clustering Approaches

Clustering Methods

Hierarchical Clustering

Partitional Clustering

k- Means Clustering

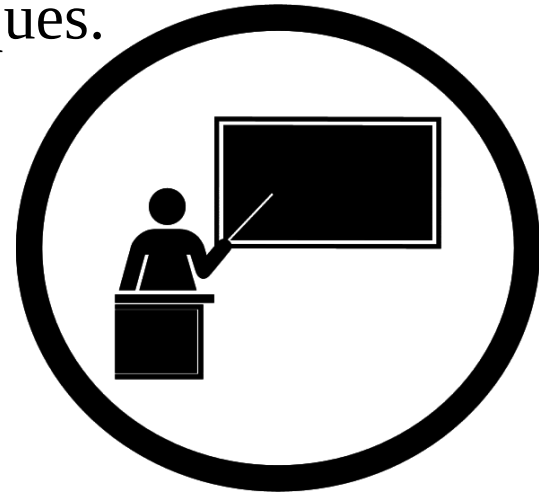
Important Topics

Summary

Introduction

After studying this session, you should be able to:

- Define clustering.
- Define and use different clustering techniques.
- Apply hierarchical clustering.
- Use partition based clustering.
- Apply k - nn clustering.



Introduction to Clustering

- Clustering is a type of unsupervised machine learning technique that is used to group similar data points together based on their features. The objective of clustering is to identify natural groups within a dataset without any prior knowledge of the labels or categories of the data.
- A loose definition of clustering could be “the process of organizing objects into groups whose members are similar in some way”. A cluster is therefore a collection of objects which are “similar to each other and are “dissimilar” to the objects belonging to other clusters.

Introduction to Clustering

- The notion of cluster is not well defined. To better understand the difficulty of deciding what constitutes a cluster, consider Fig. 1, which shows twenty points and three different ways of dividing them into clusters. The shapes of the markers indicate cluster membership.
- Fig. 1(b) and Fig.1 (d) divide the data into two and six parts, respectively. The apparent division of each of the two larger clusters into 3 sub-clusters may simply be an artifact of human visual system. In Fig. 1(c), there are four clusters. The fig illustrates that the definition of a cluster is imprecise and that the best definition depends on the nature of data and the desired result.

Introduction to Clustering

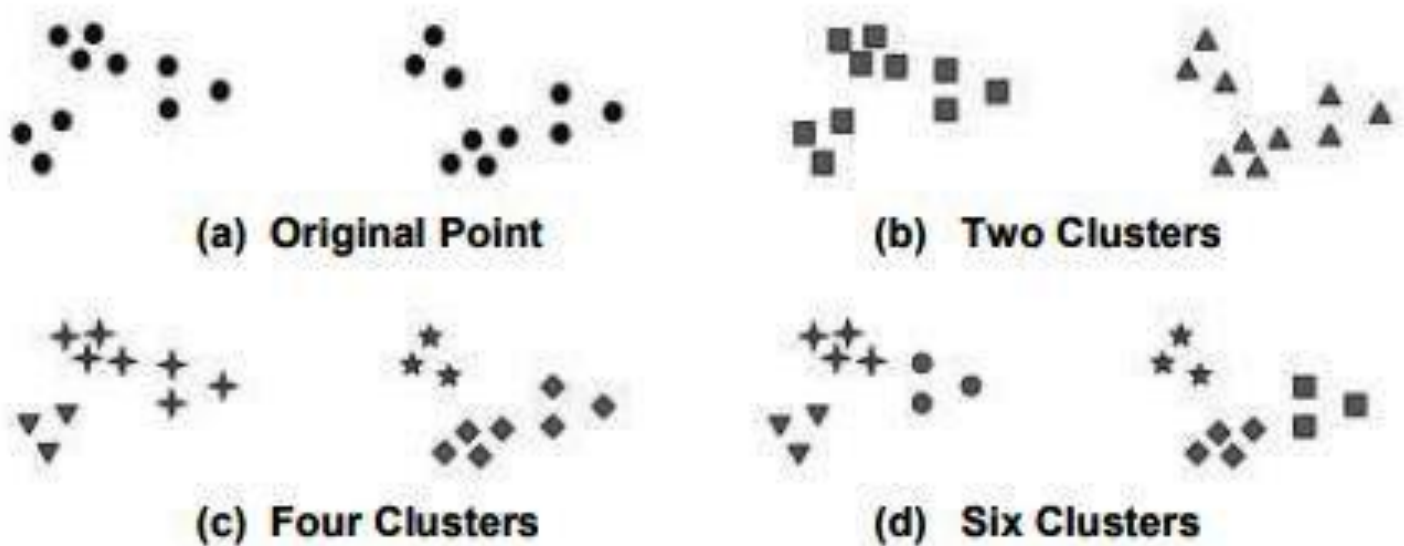


Fig. 1: Different ways of Clustering Same Data Points

Introduction to Clustering

The various examples of clustering applications are as follows:

- 1) **Marketing:** Help marketers discover distinct groups in their customer bases, and then use this knowledge to develop targeted marketing programs.
- 2) **Land use:** Identification of areas of similar land use in an earth observation database.
- 3) **Insurance:** Identifying groups of motor insurance policy holders with a high average claim cost.

Introduction to Clustering

- 4) **City-planning:** Identifying groups of houses according to their house type, value, and geographical location.
- 5) **Earth-quake studies:** Observed earth quake epicenters should be clustered along continent faults.
- 6) **Image processing:** Clustering parts of image having similar RGB values, so that image is clustered into regions such as sky, greenery, road, house, etc.



Introduction to Clustering

Let us see some real-life examples of clustering.

Example 1: Groups people of similar size to gather to make “small”, “medium” and “large” T-Shirts.

- Tailor-made for each person: too expensive
- One-size-fits-all: does not fit all.

Example 2: In marketing, segment customers according to their similarities

- To do targeted marketing.

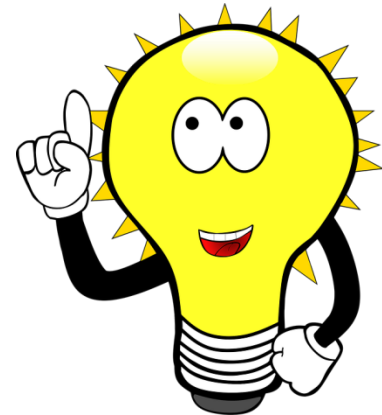
Introduction to Clustering

Example 3: Given a collection of text documents, we want to organize them according to their content similarities,

– To produce a topic hierarchy.

A **good clustering** method will produce high quality clusters with

- a) high intra-class similarity (in the same class)
- b) low inter-class similarity (between two classes)



Introduction to Clustering

- Since clustering is the grouping of similar instances/objects, some sort of measure that can determine whether two objects are similar or dissimilar is required. There are two main type of measures used to estimate this relation: distance measures and similarity measures. Many clustering methods use distance measures to determine the similarity or dissimilarity between any pair of objects.
- Depending on which formula is used to compute the distance between two data points can lead to different classification results. Domain knowledge must be used to guide the formulation of a suitable distance measure for each particular application.

Introduction to Clustering

For high dimensional data, a popular measure is the Minkowski Metric:

$$d(x_i, x_j) = (\sum_{k=1}^d |x_{i,k} - x_{j,k}|^p)^{1/p},$$

where d is the dimensionality of the data.

Special Cases:

If $p = 2$, then the distance is Euclidean distance, and if $p = 1$, then the distance is Manhattan distance.

The commonly used Euclidean distance between two objects is achieved when $p = 2$.

Introduction to Clustering

$$d_{ij} = ((x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{id} - x_{jd})^2)^{1/2}.$$

- Another well-known measure is the Manhattan distance which is defined when $p = 1$.

$$d_{ij} = |((x_{i1} - x_{j1}) + (x_{i2} - x_{j2}) + \dots + (x_{id} - x_{jd}))|.$$

- The Mahalanobis distance is another very important distance measure used in statistics that measures the statistical distance between two populations of Gaussian mixtures having mean μ_i and μ_j and a common covariance matrix Σ_{ij} ... This measure is given by

$$d_{ij} = (\mu_i - \mu_j)^T \Sigma_{ij} (\mu_i - \mu_j).$$

Introduction to Clustering

- In the case of binary attributes, the distance between objects may be calculated based on a contingency table. A binary attribute is symmetric if both of its states are equally valuable. In that case, using the simple matching coefficient can assess dissimilarity between two objects:

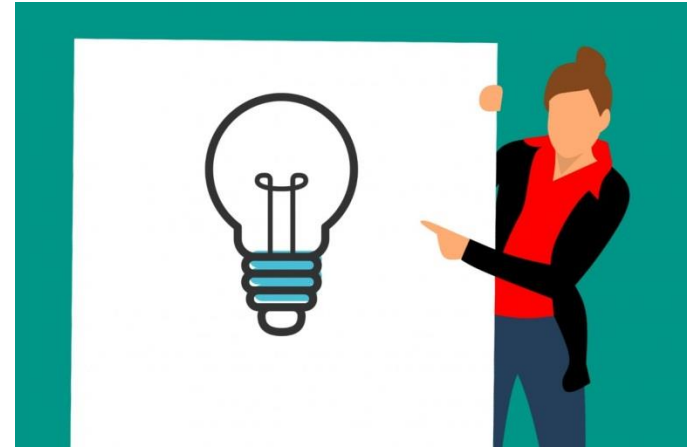
$$d(x_i, x_j) = \frac{r+s}{q+r+s+t} .$$

- where q is the number of attributes that equal 1 for both objects; t is the number of attributes that equal 0 for both objects; and s and r are the number of attributes that are unequal for both objects.

Introduction to Clustering

A binary attribute is asymmetric, if its states are not equally important (usually the positive outcome is considered more important). In this case, the denominator ignores the unimportant negative matches (t). This is called the Jaccard coefficient:

$$d(x_i, x_j) = \frac{r+s}{q+r+s}.$$



Major Clustering Approaches

There are several major clustering approaches:

- 1) **Partitioning algorithms:** Construct various partitions and then evaluate them by some criterion.
- 2) **Hierarchy algorithms:** Create a hierarchical decomposition of the set of data (or objects) using some criterion .
- 3) **Density-based:** Density based clustering approach is based on connectivity and density functions.
- 4) **Grid-based:** It is based on a multiple-level granularity structure.

Major Clustering Approaches

- 5) **Model-based:** A model is hypothesized for each of the clusters and the idea is to find the best fit of that model to each other.
- 6) **Sample based**
- 7) **High-dimensional algorithm**
- 8) **Constraint based**



Major Clustering Approaches

Clustering algorithms may be classified as listed below:

- **Exclusive Clustering**

In exclusive clustering data are grouped in an exclusive way, so that a certain datum belongs to only one definite cluster. K -means clustering is one example of the exclusive clustering algorithms.

- **Overlapping Clustering**

The overlapping clustering uses fuzzy sets to cluster data, so that each point may belong to two or more clusters with different degrees of membership.

Major Clustering Approaches

- **Hierarchical Clustering**

Hierarchical clustering algorithm has two versions: agglomerative clustering and divisive clustering.

- **Agglomerative clustering**

It is based on the union between the two nearest clusters. The beginning condition is realized by setting every datum as a cluster. After a few iterations it reaches the final clusters wanted. Basically, this is a bottom-up version.

Major Clustering Approaches

- **Divisive clustering**

It starts from one cluster containing all data items. At each step, clusters are successively split into smaller clusters according to some dissimilarity. Basically this is a top-down version.

- **Probabilistic Clustering**

Probabilistic clustering, e.g. mixture of Gaussian, uses a completely probabilistic approach.



Major Clustering Approaches

The following requirements should be satisfied by clustering algorithm.

- 1) Scalability
- 2) Dealing with different types of attributes
- 3) Discovering clusters of arbitrary shape
- 4) Ability to deal with noise and outliers
- 5) High dimensionality
- 6) Insensitivity to the order of attributes
- 7) Interpretability and usability

Major Clustering Approaches

Major problems encountered with clustering algorithms are:

- Dealing with large number of dimensions and a large number of objects can be prohibitive due to time complexity.
- The effectiveness of an algorithm depends on the definition of similarity measure.
- The outcome of an algorithm can be interpreted in different ways.



Clustering Methods

1. **Hierarchic versus Non-hierarchic Methods:** This is a major distinction involving both the methods and the classification structures designed with them.
- The hierarchic methods generate clusters as nested structures, in a hierarchical fashion; the clusters of higher levels are aggregations of the clusters of lower levels.
 - Non hierarchic methods result in a set of un-nested clusters. Sometimes, the user, even when he utilizes a hierarchical clustering algorithm, is interested rather in partitioning the set of the entities considered.

Clustering Methods

2. Agglomerative versus Divisive Methods: Agglomerative method is a bottom up approaching and involves merging smaller clusters into larger ones while the Divisive method is a top-down approach where large clusters are split into smaller ones.

- Agglomerative methods have been developed for processing mostly similarity/dissimilarity data while the divisive methods mostly work with attribute-based information, producing attribute-driven subdivisions (conceptual clustering).

Clustering Methods

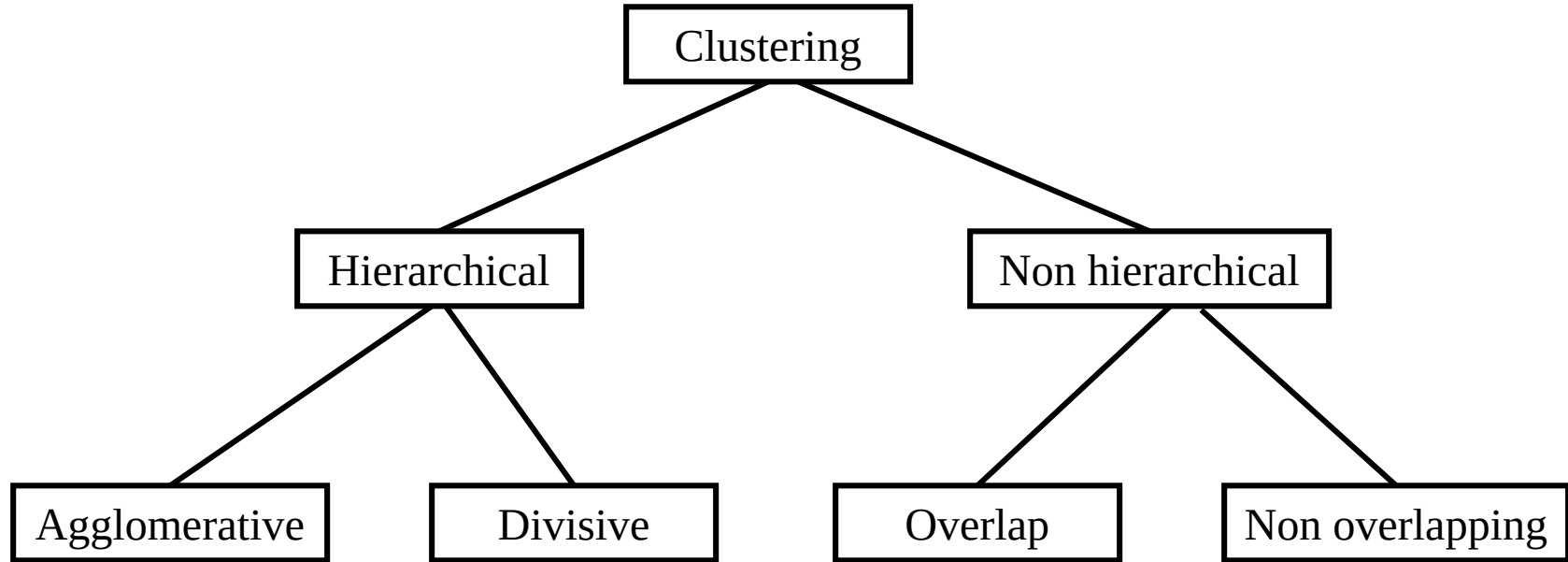


Fig 3: Classical Taxonomy of Clustering Methods

Hierarchical Clustering

These methods construct the clusters by recursively partitioning the instances in either a top-down or bottom-up fashion. A hierarchy can be represented by a tree structure such as the simple one shown in Fig. 4.

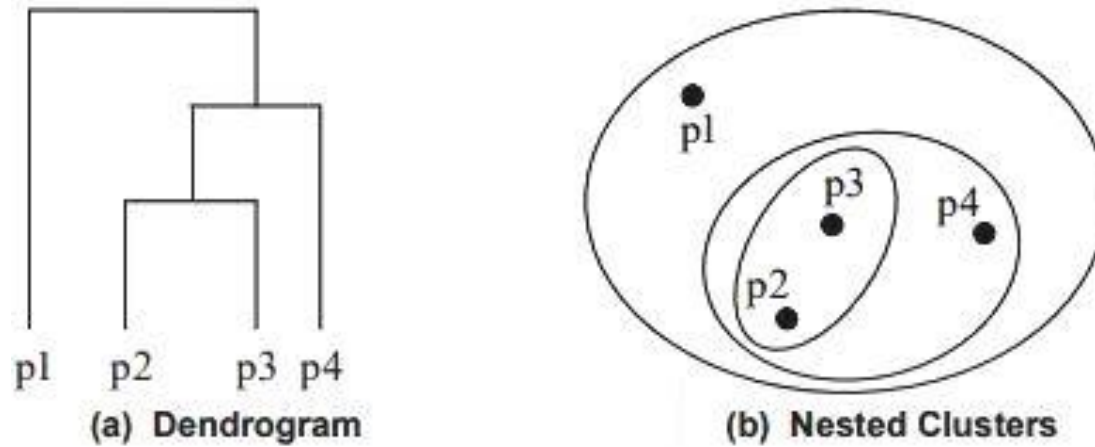


Fig. 4: A Hierarchical Clustering of Four Points

Hierarchical Clustering

- Hierarchical clustering refers to a clustering process that organizes the data into large groups, which contain smaller groups and so on. A hierarchical clustering can be drawn as a tree or dendrogram. The finest grouping is at the bottom of the dendrogram, each sample by itself forms a cluster. The coarsest grouping is at the top of the dendrogram, where all samples are grouped into a cluster.
- Logically, several approaches are possible to find a hierarchy associated with the data. The popular approach is to construct the hierarchy level-by-level, from bottom to top (agglomerative clustering) or from top to bottom (divisive clustering).

Hierarchical Clustering

Agglomerative Hierarchical Clustering

- Agglomerative hierarchical techniques are the more commonly used methods for clustering. Each object initially represents a cluster of its own. Then clusters are successively merged until the desired cluster structure is obtained.
- All objects initially belong to one cluster. Then the cluster is divided into sub-clusters, which are successively divided into their own sub-clusters. This process continues until the desired cluster structure is obtained.

Hierarchical Clustering

- The result of the hierarchical methods is a dendrogram, representing the nested grouping of objects and similarity levels at which groupings change. A clustering of the data objects is obtained by cutting the dendrogram at the desired similarity level. The merging or division of clusters is performed according to some similarity measure, chosen so as to optimize some criterion (such as a sum of squares).

The steps of general agglomerative clustering algorithm are as follows:

Step 1: Begin with N clusters. Each cluster consists of one sample.

Step 2: Repeat Step 2 a total of $N - 1$ times.

Hierarchical Clustering

Step 3: Find the most similar clusters C_i and C_j and merge C_i and C_j into one cluster. If there is a tie, merge the first pair found.

Types of Agglomerative Clustering

- **Nearest neighbor Method (also called Single-link clustering connectedness, the minimum method or the nearest neighbor method)** - methods that consider the distance between two clusters to be equal to the shortest distance between any member of one cluster to any member of the other cluster.
- If the data consist of similarities, the similarity between a pair of clusters is considered to be equal to the greatest similarity from any member of one cluster to any member of the other cluster.

Hierarchical Clustering

- This method has a tendency to cluster together at an early stage objects that are distant from each other in the same cluster because of a chain of intermediate objects in the same cluster. Such clusters have elongated sausage-like shapes when visualized as objects in space.

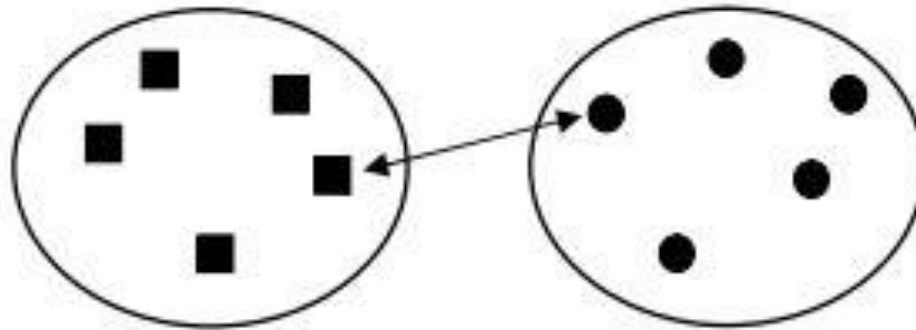


Fig. 5: Cluster Distance in Nearest Neighbour Method

Hierarchical Clustering

- **Complete-link clustering (also called the diameter method, the maximum method or the furthest neighbour method)** - methods that consider the distance between two clusters to be equal to the longest distance from any member of one cluster to any member of the other cluster. The nearest neighbour is not the only method for measuring the distance between clusters.
- Under the furthest neighbor (or complete linkage) method, the distance between two clusters is the distance between their two most distant members.

Hierarchical Clustering

- This method tends to produce clusters at the early stages that have objects that are within a narrow range of distances from each other. If we visualize them as objects in space the objects in such clusters would have a more spherical shape as shown in Fig. 11.

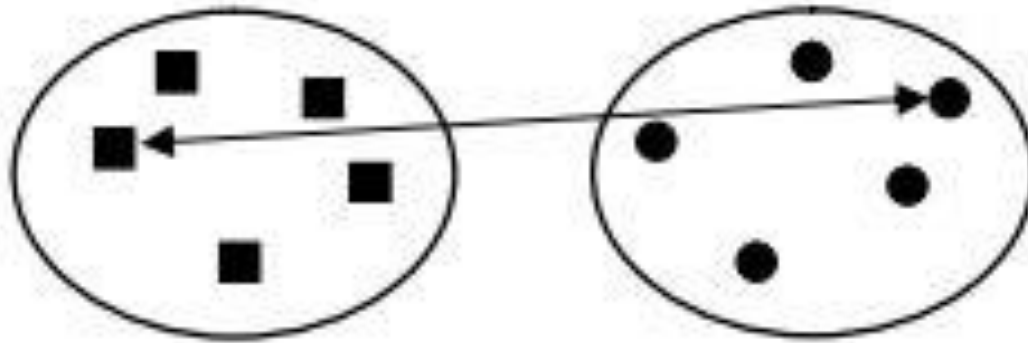


Fig. 11: Cluster Distance (Furthest Neighbour Method)

Partitional Clustering

- Partitioning clustering begins with a starting cluster partition which is iteratively improved until a locally optimal partition is reached. The starting clusters can be either random or the cluster output from some clustering pre-process (e.g. hierarchical clustering).
- Some partitioning methods use multiple passes during which cluster centroids are recalculated and objects are re-evaluated, whereas other methods use a single-pass procedure. Partitioning methods also differ with respect to how they evaluate an object's distance from cluster centroids. Some procedures use simple distance and others use more complex multivariate matrix criteria.

Partitional Clustering

- Let us discuss an important algorithm known as Froggy's algorithm, which is used for partitional clustering.
- **Froggy's algorithm:** One of the simplest partitional clustering algorithm is Froggy's algorithm. Input to the algorithm consists of data, k , number of clusters to be constructed and k samples called seed points. Seed points could be chosen randomly or some knowledge of the desired cluster structure could be the starting point.



Partitional Clustering

The following steps are performed:

Step 1: Initialize the cluster centroid to the seed points.

Step 2: For each sample, find the cluster centroid nearest to it. Put the samples in the cluster identified with the nearest cluster centroid.

Step 3: If no samples changed clusters in Step 2, stop.

Step 4: Compute the centroids of the resulting clusters and go to step 2.



K- Means Clustering

- The K -means clustering technique is simple, and we first choose k initial centroids, where k is a user-specified parameter, namely, the number of clusters desired. Each point is then assigned to the closest centroid, and each collection of points assigned to a centroid is a cluster.
- The centroid of each cluster is then updated based on the points assigned to the cluster. We repeat the assignment and update steps until no point changes clusters, or equivalently, until the centroids remain the same.



K- Means Clustering

The k -means method follows the following steps:

Step 1: Specify the number of clusters and, arbitrarily or deliberately, the members of each cluster.

Step 2: Calculate each cluster's "centroid" (explained below), and the distances between each observation and centroid. If an observation is nearer the centroid of a cluster other than the one to which it currently belongs, re-assign it to the nearer cluster.

Step 3: Repeat Step 2 until all observations are nearest the centroid of the cluster to which they belong.

K- Means Clustering

Step 4: If the number of clusters cannot be specified with confidence in advance, repeat Steps 1 to 3 with a different number of clusters and evaluate the results.

The operation of K -means are shown in Fig. 19, which shows how, starting from three centroids, the final clusters are found in four assignment-update steps. In these and other figures displaying K- means clustering, each subfigure shows (1) the centroid sat the start of the iteration and (2) the assignment of the points to those centroids. The centroids are indicated by the “+” symbol. All points belonging to the same cluster have the same marker shape.

K- Means Clustering

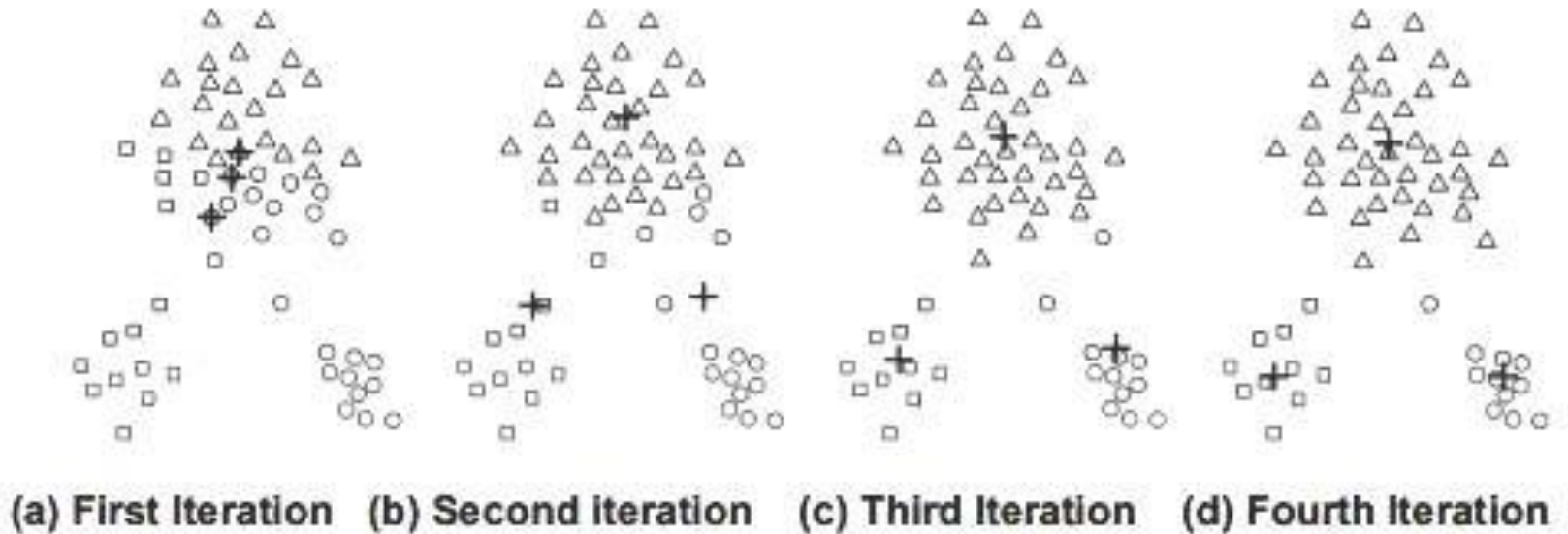


Fig.19: Using the K -Means Algorithm

K- Means Clustering

- In the first step, shown in Fig. 19(a), points are assigned to the initial centroids, which are all in the larger group of points. For this example, we use the mean as the centroid. After points are updated again. In steps 2, 3, and 4, which are shown in Fig. 19(b), (c), and (d), respectively, two of the centroids move to the two small groups of points at the bottom of the figures.
- When the K -means algorithm terminates in Fig. 19(d), because no more changes occur, the centroids have identified the natural groupings of points. Centroid at the beginning of the step and the assignment of points to those centroids. In the second step, points are assigned to the updated centroids, and the centroids.

K- Means Clustering

Benefits and drawbacks of k-means methods:

Benefits:

- 1) Very fast algorithm ($O(k \cdot d \cdot N)$, if we limit the number of iterations)
- 2) Convenient centroid vector for every cluster
- 3) Can be run multiple times to get different results



K- Means Clustering

Limitations:

- 1) Difficult to choose the number of clusters, k
- 2) Cannot be used with arbitrary distances
- 3) Sensitive to scaling – requires careful preprocessing
- 4) Does not produce the same result every time
- 5) Sensitive to outliers (squared errors emphasize outliers)
- 6) Cluster sizes can be quite unbalanced (e.g., one-element outlier clusters)

Important Topics

Introduction to Clustering

Major Clustering Approaches

Clustering Methods

k- MeansClustering

Summary

In this session, we discussed:

- Concept of clustering.
- Various distance measures.
- Various clustering methods.
- Analyzed various Hierarchical clustering algorithms in detail.
- Analyzed various Partitional clustering and k - nn clustering algorithms.

*Thank
You*